ASEAN DIGITAL INTEGRATION INDEX 2.0

Assessing the progress of digital integration measures over time

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TABLE OF CONTENTS

FIGURES

TABLES

ACRONYMS

4IR	The Fourth Industrial Revolution
ACCED	ASEAN Coordinating Committee on E-Commerce and Digital Economy
ADII	ASEAN Digital Integration Index
AITI	Brunei Darussalam Authority for Info-communications Technology Industry
AMS	ASEAN Member States
ASEAN	Association of Southeast Asian Nations
ASW	ASEAN Single Window
BBVA	Banco Bilbao Vizcaya Argentaria
BSP	Bangko Sentral ng Pilipinas
DEPA	Thailand Digital Economy Promotion Agency
DESI	European Union Digital Society and Economy Index
DIF	Digital Integration Framework
DiGiX	BBVA Digitization Index
DII	Tufts University Digital Intelligence Index
e-Phyto	electronic Phytosanitary
Findex	World Bank’s Global Financial Development
GDP	Gross Domestic Product
ICT	Information Communication Technology
ID	Identity Document
IP	Intellectual Property
IT	Information Technology
LaPASS	Lao Payment and Settlement System
MDES	Thailand Ministry of Digital Economy and Society
MIC	Viet Nam Ministry of Information and Communications
MOOC	Massive Open Online Courses
MOU	Memorandum of Understanding
MPT	Ministry of Posts and Telecommunications
MSMEs	Micro, Small, and Medium Enterprises
NDTP	Thailand National Digital Trade Platform
NLE	Indonesia National Logistics Ecosystem
NSW	National Single Window
NTP	Singapore Network Trade Platform
PDPA	Personal Data Protection Act
PhilSys	Philippine Identification System
QR	Quick Response
R&D	Research and development
SBV	The State Bank of Viet Nam
SingPass	Singapore Personal Access
SME	Small and Medium-sized Enterprise
SPBE	Indonesia Sistem Pemerintahan Berbasis Elektronik (e-Government)
STEM	Science, Technology, Engineering, and Mathematics
WIPO	World Intellectual Property Organization

FOREWORD

1 | INTRODUCTION

In 2020-2021, the *ASEAN-USAID Inclusive Growth in ASEAN through Innovation, Trade, and E-Commerce (IGNITE)* project supported the ASEAN Coordinating Committee on E-commerce and Digital Economy (ACCED) as it developed the *ASEAN Digital Integration Index (ADII 1.0)*.[1]

Endorsed by the ASEAN Economic Ministers (AEM), the inaugural ADII 1.0 report represents a major milestone in ASEAN's efforts to strengthen and accelerate its digital transformation, integration, and resilience in the face of regional COVID-19 recovery efforts.

The ADII was developed as a living document that could evolve to best reflect and address ASEAN Member States' (AMS) changing priorities and needs. As such, in 2022-2023, IGNITE was invited to support the development of the ADII's next iteration – *ADII 2.0*.

The first step towards ADII 2.0 consisted of reviewing the third-party datasets to determine which data used in ADII 1.0 could be updated. IGNITE produced a preliminary *Key Highlights Brief* that updated tables and charts for each Pillar and provided analysis on the way the COVID-19 pandemic impacted the region's digital economies since ADII 1.0.

The *Key Highlights Brief* also allowed the identification of quantitative and qualitative indicators for which third-party datasets were no longer updated or available. Working closely with ACCED representatives, AMS representatives, ASEAN sectoral bodies, and the ASEAN Secretariat, IGNITE developed a data completion mechanism that constituted a first step towards AMS and ASEAN systemizing the gathering of quantitative and qualitative datasets for their evidence-based policymaking.

Indeed, the fact that most of the third-party databases used for ADII evolved over time demonstrated the importance of AMS building their own statistical capabilities so that they could track and assess their digital integration efforts without relying on third-party efforts.

To this end, IGNITE conducted a Data Completion Boot Camp (DCBC)—a two-day consultative workshop held in August 2023 in Bangkok, Thailand, and aimed at completing and updating the ADII

[1] ASEAN (2021) ASEAN Digital Integration Index Report, https://asean.org/book/asean-digital-integration-index-report-2021

2.0 dataset. The DCBC served as a data completion capacity-building exercise that will serve multiple purposes beyond ADII 2.0 itself:

1. Build AMS' experience in data gathering on digitalization trends and issues;
2. Support the region's transition towards the goal of independently tracking and assessing digital integration efforts; and
3. Highlight what works and what does not work for AMS and the region on their data- and evidence-gathering journey.

This report is the culmination of these various workstreams. It presents the main ADII 2.0 scores for the ASEAN region, with a focus on the changes observed between ADII 1.0 and ADII 2.0 (Sections 3 and 4). It also contains a number of recommendations to help AMS sustain current digital integration efforts, as well as address foreseeable challenges stemming from the digital divide that persists in the region (Section 5). Lastly, a special chapter envisions the various ways in which ADII 3.0 could evolve to fit emerging technological developments and their corresponding policy priorities (for example, AI), as well as to support the operationalization of upcoming regional engagements such as the *Digital Economy Framework Agreement (DEFA)* (Section 6).

Two Annexes complete this report: one that examines key findings for each individual AMS, tying their ADII 2.0 scores to recent policy and regulatory developments and identifying the areas that may benefit from additional efforts at the national level (Annex A); and one that benchmarks ASEAN's ADII 2.0 performance against that of leading and/or maturing digital economies—namely, Australia, China, Japan, New Zealand, South Korea, and the United States (Annex B).

2 | EXECUTIVE SUMMARY

In 2020-2021, the inaugural ADII 1.0 report offered a starting point for ASEAN to begin reviewing the progress of activities and initiatives driven by the *ASEAN Digital Integration Framework (DIF)*.[2] It also set the foundation to support and strengthen the path towards a single digital community for the digital-driven future.

Today, this ADII 2.0 report provides an update on the status and progress of digital integration efforts in ASEAN, noting the changes observed since the publication of ADII 1.0. It provides an update of the ADII 1.0 database, examining the latest scores, tables, and charts for each of the ADII Indicators and Pillars at the ASEAN level (regional scores).

Where statistically significant changes are observed, the report analyzes and contextualizes the policy and regulatory developments that may have led to this evolution, as well as the potential ways ASEAN can respond to ensure it remains on the path to sustainable and equitable digital integration.

How does ASEAN fare on the ADII 2.0 Pillars?

Table 1 presents an overview of the ADII 1.0 and 2.0 Pillar scores, including the total score for each Pillar, its corresponding position relative to the other Pillars, and some key takeaways.

[2] ASEAN (2019) ASEAN Digital Integration Framework, https://asean.org/storage/2019/01/ASEAN-Digital-Integration-Framework.pdf

Table 1: ADII 1.0 and 2.0 Pillar Scores and Rankings for ASEAN[3]

ADII Pillars	ADII 1.0 Score (/100)	ADII 2.0 Score (/100)	ADII 1.0 Rank	ADII 2.0 Rank	Key Takeaways
Pillar 1 Digital Trade & Logistics	**55.27**	**61.82**	**4**	**4**	▪ Pillar 1 saw its score increase by 6.6 points between ADII 1.0 and ADII 2.0. ▪ This increase is the third highest for the region, demonstrating significant progress has been made in this area since 2020-2021. ▪ Despite this significant improvement, the Pillar remains in 4th position in terms of overall performance for the region.
Pillar 2 Data Protection & Cybersecurity	**62.81**	**69.99**	**2**	**2**	▪ Pillar 2 is among the top three performing ADII 2.0 Pillars for the ASEAN region (2nd behind Pillar 3). ▪ Pillar 2 saw its score increase by 7.2 points between ADII 1.0 and ADII 2.0. ▪ This increase is the second highest for the region, demonstrating significant progress has been made in this area since 2020-2021. ▪ Despite this significant improvement, the Pillar remains in 2nd position in terms of overall performance for the region.
Pillar 3 Digital Payments & Identities	**58.84**	**70.03**	**3**	**1**	▪ Pillar 3 stands out as the strongest ADII 2.0 Pillar for the ASEAN region (1st out of six Pillars, ahead of Pillars 2 and 6). ▪ Pillar 3 saw its score increase by 11.2 points between ADII 1.0 and ADII 2.0. ▪ This increase is the highest for the region, demonstrating significant progress has been made in this area since 2020-2021. ▪ This notable improvement allowed Pillar 3 to advance two positions from 3rd to 1st, pushing Pillar 6 out of the top position.
Pillar 4 Digital Skills & Talent	**48.21**	**50.01**	**6**	**6**	▪ Pillar 4 stands out as the weakest ADII 2.0 Pillar for the ASEAN region (6th out of six Pillars, behind Pillars 5 and 1). ▪ Pillar 4 saw its score increase by 1.8 points between ADII 1.0 and ADII 2.0. ▪ This increase is the lowest for the region, suggesting that very modest progress has been made in this area since 2020-2021.

[3] Each Pillar score has been normalized to a total score of 100 points. The ranking indicates each Pillar's performance relative to the five others.

					▪ This limited improvement is reflected in the Pillar maintaining its 6th position in terms of overall performance for the region.
Pillar 5 Innovation & Entrepreneurship	**49.32**	**51.83**	**5**	**5**	▪ Pillar 5 saw its score increase by 2.5 points between ADII 1.0 and ADII 2.0. ▪ This increase is the second lowest for the region, suggesting that very modest progress has been made in this area since 2020-2021. ▪ This limited improvement is reflected in the Pillar maintaining its 5th position in terms of overall performance for the region.
Pillar 6 Institutional & Infrastructural Readiness	**62.85**	**68.55**	**1**	**3**	▪ Pillar 6 is among the top three performing ADII 2.0 Pillars for the ASEAN region (3rd behind Pillars 3 and 2). ▪ Pillar 6 saw its score increase by 5.7 points between ADII 1.0 and ADII 2.0. ▪ This increase is the fourth highest for the region, suggesting that encouraging progress has been made in this area since 2020-2021. ▪ Despite this encouraging improvement, the Pillar retroceded by two positions in terms of overall performance, pushed from 1st to 3rd and replaced by Pillar 3.

Overall, we find that all six ADII Pillars have seen their scores increase between ADII 1.0 and ADII 2.0, suggesting that digital integration has effectively progressed across all 10 AMS. This trend aligns with the broader global shift towards the digitalization of economies, which has put technological innovation and investment at the top of economic agendas.

It is important to note that Pillar score increases are not necessarily reflected in Pillar rankings; four out of six pillars (Pillars 1, 2, 4, and 5) have retained their previous positions despite scoring higher than in ADII 1.0. This discrepancy may reflect a major dynamic at play in ASEAN: the fact that a broad and concerted approach to digital development is effectively taking place at the regional level, but the improved performance of some AMS is balanced by the slower progress in others.

It is also noteworthy that the score increases have not been uniform across all Pillars. Pillar 3 (Digital Payments & Identities) saw the highest increase, jumping 11.2 points. This remarkable rise may reflect the fact that both AMS and ASEAN have prioritized the implementation of interoperable cross-border digital payments. Meanwhile, Pillar 4 (Digital Skills & Talent) has experienced the smallest growth since ADII 1.0. The modest 1.8-point increase may reflect the fact that access to tertiary education, skill development, lifelong training, and knowledge transfers remains uneven within and across AMS[4]—despite a number of national and regional initiatives aimed at turning things around.[5]

A particularly interesting development is the change in positions between Pillars 3 and 6, with Pillar 3 moving up from 3rd to 1st and Pillar 6 dropping from 1st to 3rd. This switch may indicate a major shift in priorities within the ASEAN digital economy landscape; the rise of Pillar 3 to the top position suggests a growing emphasis on Digital Payments & Identities, possibly driven by the increased adoption of e-commerce and digital financial services. Conversely, the drop in the ranking of Pillar 6, despite an increase in its score, might point to emerging challenges in driving Institutional & Infrastructural Readiness at the same pace as the rapid evolution of digital disruption.

How does ASEAN fare within the broader Asia-Pacific region?

To better assess ASEAN's digital integration efforts, the scores of several leading and/or maturing digital economies—Australia, China, Japan, New Zealand, South Korea, and the United States—were computed using the ADII indicators and methodology. This benchmarking exercise allows a comparable gauge of ASEAN's digital integration efforts in relation to other countries.[6]

Table 2 presents an overview of ASEAN's average Pillar scores for ADII 1.0 and 2.0, as well as for the six benchmarking (non-ASEAN) economies.

Table 2: ADII 1.0 and 2.0 Pillar Scores for ASEAN and Benchmarking Economies

	Pillar 1 Digital Trade & Logistics		Pillar 2 Data Protection & Cybersecurity		Pillar 3 Digital Payments & Identities		Pillar 4 Digital Skills & Talent		Pillar 5 Innovation & Entrepreneurship		Pillar 6 Institutional & Infrastructural Readiness	
	ADII 2.0	**ADII 1.0**	**ADII 2.0**	**ADII 1.0**	**ADII 2.0**	**ADII 1.0**	**ADII 2.0**	**ADII 1.0**	**ADII 2.0**	**ADII 1.0**	**ADII 2.0**	**ADII 1.0**
ASEAN	**61.82**	**55.27**	**69.99**	**62.81**	**70.03**	**58.84**	**50.01**	**48.21**	**51.83**	**49.32**	**68.55**	**62.85**
Australia	92.00	90.72	97.75	90.77	95.50	88.00	53.86	52.99	65.80	66.30	80.87	60.87
China	89.77	86.50	94.60	75.73	83.87	74.73	66.19	64.76	70.56	68.74	72.65	53.63
Japan	93.20	93.36	98.22	90.93	85.00	82.00	47.52	54.77	76.40	77.32	80.53	60.67
New Zealand	90.00	92.04	90.73	85.91	95.75	90.33	58.97	55.70	65.45	65.84	80.94	63.90
South Korea	91.60	89.28	99.09	88.42	88.89	81.42	40.86	53.77	80.87	77.92	83.35	63.59
United States	91.20	91.68	98.33	92.73	89.00	79.00	62.14	58.90	82.93	80.79	88.04	64.90

[4] East Asia Forum (2022) Addressing the digital divide in ASEAN, www.eastasiaforum.org/2022/07/01/addressing-the-digital-divide-in-asean
[5] The ASEAN Digital Innovation Programme (Microsoft), the ASEAN Seeds for the Future program (Huawei), the ASEAN Digital Skills Vision 2020 plan (World Economic Forum), and the Conference on 21st Century Skills and Youth Participation (UNICEF) are just some of the most recent initiatives devoted to improving populations' access to digital skills in ASEAN.
[6] Annex B provides more detailed analysis.

Key findings include:

- **Overall**: Since ADII 1.0, ASEAN has made steady progress across all six ADIII Pillars; the region has experienced a marked upward trajectory, with an average Pillar score that has increased by a low of 1.8 points (Pillar 4) to a high of 11.2 points (Pillar 3). Meanwhile, the scores of the six non-ASEAN economies have greatly fluctuated during the same period. Despite this encouraging upward trend for ASEAN, and despite the mixed progress of the benchmarking economies, ASEAN finds itself scoring the lowest in five of the six ADII Pillars. This suggests that while the region is making great strides in terms of strengthening and integrating its digitalization, much remains to be done when it comes to catching up to more mature digital economies.

- **Pillar 1 (Digital Trade & Logistics)**: Pillar 1 is one of two Pillars (along with Pillar 3) for which ASEAN has experienced the highest point increase since ADII 1.0. Despite this notable progress, ASEAN's Pillar score of 61.82 is the lowest compared to six non-ASEAN economies, whose Pillar 6 scores are comprised between 89 and 93. This discrepancy may reflect the fact that the great strides being made at the regional level (multiple trade agreements that call for a facilitation and dematerialization of cross-border trade) are being weighed down by less conducive measures at the national level (for example, strict data localization measures that hinder the growth of cross-border digital activities).

- **Pillar 2 (Data Protection & Cybersecurity)**: Pillar 2 stands out as the Pillar for which the gap between ASEAN's average score (69.99) and the score of all other benchmarking economies is the widest. With a score of 99.09, South Korea dominates this Pillar and has an advantage of over 30 points over ASEAN. At 90.73, New Zealand is in next-to-last position but is still over 20 points ahead of ASEAN. It is worth noting that Pillar 2 is one of three Pillars (along with Pillars 3 and 6) for which no economies have seen their scores decrease since ADII 1.0, which points to a general strengthening of all the enabling measures that allow digital investment and innovation to thrive.

- **Pillar 3 (Digital Payments & Identities)**: Pillar 3 is one of three Pillars (along with Pillars 2 and 6) for which no economies have seen their scores decrease since ADII 1.0. With a score of 95.75, New Zealand takes top spot and scores almost 25 points more than the ASEAN average (70.03). Pillar 3 is also one of two Pillars (along with Pillar 1) for which ASEAN has experienced the highest point increase since ADII 1.0 (a rise of 11.2 points) but remains at the tail-end of the rankings. This may be because the region started from a much lower point; as such, as remarkable and encouraging as ASEAN's progress has been so far, it is still insufficient to catch up to more mature digital ecosystems.

- **Pillar 4 (Digital Skills & Talent)**: Pillar 4 is not the only Pillar for which ASEAN's average score has increased, but it is the only Pillar that has seen ASEAN move ahead of the benchmarking economies. Indeed, Pillar 4 stands out as the only Pillar for which ASEAN no longer has the lowest score; supported by Japan's and South Korea's respective 7.2- and 12.9-point losses, ASEAN's modest 1.8-point increase allowed the region to move ahead in the benchmarking panel. With a score of 66.19, China takes top position, well ahead of the United States, and New Zealand (interestingly, these same three economies already dominated the top 3 positions in ADII 1.0).

- **Pillar 5 (Innovation & Entrepreneurship)**: Unlike other Pillars, the scores for Pillar 5 have not made much progress across the board. ASEAN's average Pillar score has slightly increased from 49.32 to 51.83 (a 2.5-point increase), while non-ASEAN economies' scores have also gone up or down very moderately (no change above 3 points). This absence of noteworthy progress suggests a significant worldwide stalling of entrepreneurial activity, a trend that is reflected in the fact that the leaderboard remains unchanged since ADII 1.0. Indeed, the United States remains at the top, followed by South Korea and Japan; while ASEAN remains at the bottom, outpaced by New Zealand, Australia, and China.

- **Pillar 6 (Institutional & Infrastructural Readiness)**: With an average score of 68.55, ASEAN saw its score advance by close to 6 points since ADII 1.0. Despite this progress, ASEAN scores the lowest compared to six non-ASEAN economies, whose Pillar 6 scores are comprised between 72 and 88. This is because unlike ASEAN, other economies have seen their Pillar 6 scores increase by 17 points or more, suggesting a faster, stronger, and more wide-reaching maturation of institutional and infrastructural capabilities. This is in stark contrast to ADII 1.0, which placed ASEAN right in the middle of the pack and ahead of China, Japan, and Australia.

Key Recommendations

Table 3 presents an overview of key recommendations, which are more thoroughly developed in Section 5 of this report.

Table 3: Summary of Key Recommendations for ASEAN

Focus Areas	Recommendations
 **Pillar 1:** **Digital Trade & Logistics**	1. **Enhance Digital Infrastructure and Connectivity**: ASEAN should focus on upgrading physical infrastructure critical for digital trade. 2. **Strengthen Digital Authentication and Cybersecurity Frameworks**: ASEAN should intensify its efforts in enhancing digital certificates and signatures. 3. **Adopt Standards and Facilitate Interoperability**: ASEAN should continue to align with international standards and practices to ensure interoperable systems and processes.
 **Pillar 2:** **Data Protection & Cybersecurity**	1. **Make Data Protection and Cybersecurity Laws Interoperable**: Despite the existence of national data protection measures, there is a need for greater regional interoperability to ensure consistency and efficiency in data protection across ASEAN. 2. **Enhance Cybersecurity Cooperation and Integration**: There is an important gap in the convergence of national cybersecurity measures at the regional level. 3. **Strengthen Institutional Cybersecurity Capabilities**: ASEAN should prioritize building and enhancing the capacity of governmental and organizational structures dealing with cybersecurity.

Pillar 3:
Digital Payments & Identities

1. **Expand Access to Digital Finance and Improve Digital Literacy**: ASEAN should focus on increasing digital finance access, particularly in AMS with high rates of unbanked and underbanked populations.

2. **Enhance Security and Interoperability of Digital Payments Systems**: To sustain the rising use of digital platforms for financial transactions, ASEAN should prioritize enhancing the security and interoperability of these systems, as well as developing stronger consumer protection measures to build trust in digital platforms for financial transactions.

3. **Strengthen Digital Identity Frameworks and Enhance Interoperability**: ASEAN should focus on developing digital identity frameworks and making them more interoperable across AMS.

Pillar 4:
Digital Skills & Talent

1. **Improve Access to STEM Education with a Focus on Equity and Inclusivity**: To address the uneven landscape in STEM education and eliminate the gender gap, ASEAN should prioritize laws and policies that provide equitable access to STEM education, particularly in underrepresented and economically challenged regions like CLMV economies.

2. **Strengthen Multi-Stakeholder Collaboration in R&D and Knowledge-Intensive Services**: ASEAN needs to foster renewed focus on cross-sector partnerships and joint initiatives in R&D to stimulate growth in the knowledge-intensive services sector.

3. **Close the Digital Skills Gap through Lifelong Learning and Upskilling Programs**: Given the persistent digital divide and the evolving demands of the digital economy, ASEAN should prioritize policies that promote lifelong learning and upskilling, especially in emerging technologies like AI and data analytics.

Pillar 5: Innovation & Entrepreneurship

1. **Enhance Venture Capital Availability through Incentives**: Despite a conducive environment for start-ups in ASEAN, there is a need to sustain and increase venture capital investments, particularly in light of recent funding slowdowns.

2. **Boost R&D Investment and Collaboration**: Given the relatively low R&D expenditure in the ASEAN region, it is crucial to find alternative ways to raise such investment.

3. **Strengthen Intellectual Property Protection**: With the increasing importance of digital innovation, robust intellectual property (IP) protection plays a significant and vital role in the survival of digital economies. ASEAN should continue enhancing the enforcement of national IP protection measures and enforcing the mechanisms introduced in the *ASEAN IPR Action Plan 2016- 2025 (AIPRAP) v2.0*.

Pillar 6: Institutional & Infrastructural Readiness

1. **Enhance Rural Connectivity**: ASEAN should prioritize closing the rural-urban digital divide by implementing targeted policies for rural internet connectivity.

2. **Improve Internet Quality and Accessibility**: To address the challenge of inconsistent internet quality across AMS, ASEAN should establish regional standards for broadband speed and reliability.

3. **Digitalize Government Services with a Citizen-Centric Approach**: ASEAN should focus on enhancing the interface and delivery of digital government services to ensure they are user-friendly and meet the needs of citizens.

Next Steps for Future ADII Iterations to Remain Relevant and Impactful for Years to Come

1. **Close alignment of DEFA and ADII processes, priorities, and milestones**: Once completed, the *ASEAN Digital Economy Framework Agreement (DEFA)* will significantly influence initiatives like the ADII by providing a consolidated and systematic approach to digital transformation and integration among AMS. As such, we recommend using the *DEFA* as a guiding framework for ADII's metrics and methodologies—If ADII measures and supports the digital economy goals set by *DEFA*, it is likelier to be able to offer a coherent and coordinated direction for digital growth in the region.

2. **Strengthen and consolidate AMS' and ASEC's data gathering, computing, and publishing capabilities**: From unavailable or discontinued datasets to evolving methodologies, third-party databases can be used as a starting point, but should not be the only source of quantitative resources for the ADII. As such, we recommend sustaining the consultative and participative process developed by the data completion mechanism to gradually replace all third-party databases with datasets that are completely produced, selected, and computed by ASEAN, for ASEAN.

3 | OVERVIEW OF ADII 2.0

The ADII is a composite index comprised of six Pillars that align with the six key priority areas identified by the DIF (see Table 4 below).

Table 4: ADII Pillars Mapped Against DIF Priority Areas

DIF PRIORITY AREA	ADII PILLAR
Priority Area 1 Facilitate seamless trade	*Pillar 1* Digitally Enabled Trade & Logistics
Priority Area 2 Protect data while supporting digital trade and innovation	*Pillar 2* Data Protection & Cybersecurity
Priority Area 3 Enable seamless digital payments	*Pillar 3* Digital Payments & Identities
Priority Area 4 Broaden the digital talent base	*Pillar 4* Digital Skills & Talent
Priority Area 5 Foster entrepreneurship	*Pillar 5* Innovation & Entrepreneurship
Priority Area 6 Coordinate actions	*Pillar 6* Institutional & Infrastructural Readiness

Each Pillar comprises five Indicators, with the data retrieved from a range of third-party data sources as well as from national statistical databases. Each Indicator was normalized and scored on a scale of 20 points, and added up to a total Pillar score out of 100 points.

The indicator selection was based on six key criteria:

1. **Relevance** – where data is clear, accurate, and relevant to the DIF's priority areas;
2. **Accessibility** – where data is easily accessible and is freely made available;
3. **Coverage** – where data must cover at least eight of the 10 AMS;
4. **Timeliness** – where updated datasets are used with data published from at least 2017 onwards;
5. **Consistency** – where databases are published consistently and regularly; and
6. **Transparency** – where data comes from reputable sources that openly share their approaches, methodologies, and limitations.

Despite these selection criteria, the process of updating the ADII 1.0 data was limited by the fact that it relied on third-party datasets being readily available, accurately collected, and regularly published by a multitude of organizations.

To counter the reliance on third-party datasets, the ADII 2.0 data collection exercise in 2022-2023 was accompanied by a skills-building exercise aimed at systemizing AMS' ability to gather, normalize, compute, and publish datasets pertaining to the digital economy. Leveraging the expertise of their national Data Completion Committees (DCCs), AMS took completed and updated a wide range of missing and/or outdated third-party data entries. Section 6 provides more details on the way the work of the DCCs can be sustained and expanded in the long term.

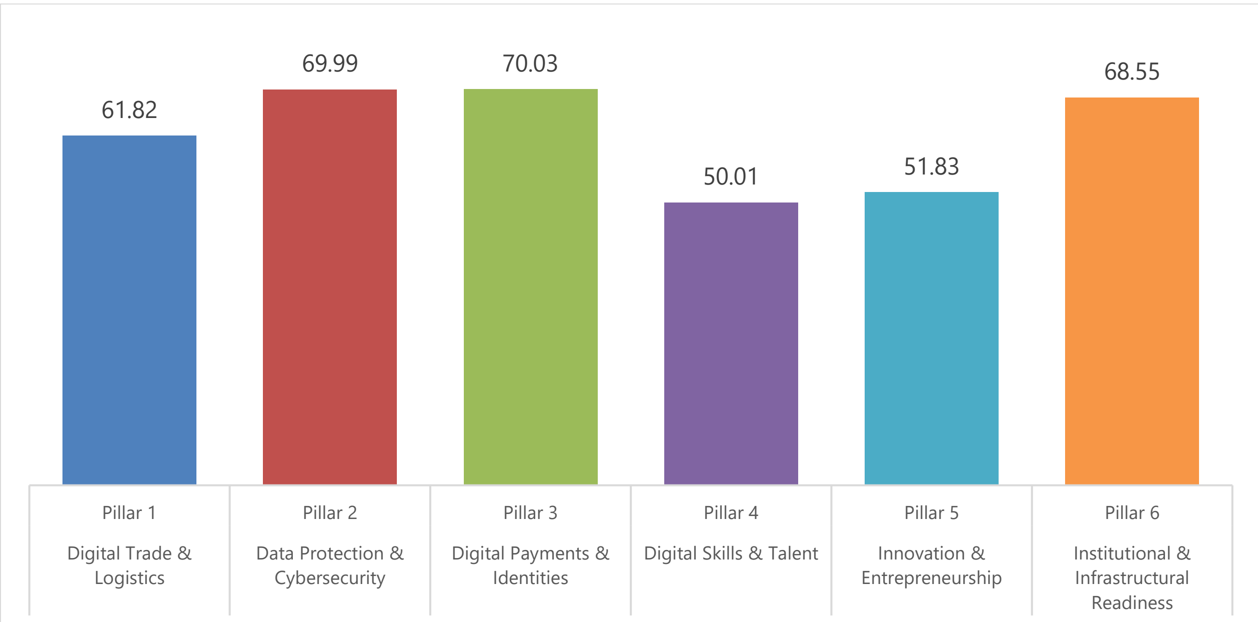
For a closer look at the methodology of both the third-party indicators and the nationally sourced datasets, please refer to Annex C: Methodology

4 | ADII 2.0 FINDINGS

ADII 2.0 SCORES FOR ASEAN

Figure 1 indicates that with a score of 70.03, Pillar 3 is the strongest ADII Pillar for ASEAN, closely followed by Pillar 2 (69.99). With respective scores of 51.83 and 50.01, Pillars 5 and 4 stand out as the weakest ADII Pillars for the region.

Figure 1: ADII 2.0 Pillar Scores for ASEAN



Pillar 3 (Digital Payments & Identities) stands out as the strongest Pillar for ASEAN in ADII 2.0, reflecting the many national and regional efforts that are taking shape and having a tangible impact on the region.

In 2022, the central banks of Indonesia, Malaysia, the Philippines, Singapore, and Thailand to sign an agreement to create an interoperable system for instant digital payments and financial transactions across ASEAN.[7] This system is designed to operate without costly intermediaries and is expected to expand to include all member states.[8]

Additionally, efforts are being made to enhance cross-border financing for businesses through the use of a unique business identification (UBIN) system for micro- and small-sized enterprises within ASEAN.[9] Lastly, the *ASEAN Leaders Declaration on Advancing Regional Payment Connectivity and Promoting Local Currency Transaction* has demonstrated the region's commitment to setting up a regional cross-border Quick Response (QR) code payment network.[10]

Pillar 2 (Data Protection & Cybersecurity) closely tails Pillar 3, reflecting the many initiatives that have strengthened data protection, privacy, and security within and across AMS.

At the national level, a number of laws, policies, and regulations have been launched in Indonesia, Malaysia, the Philippines, Thailand, and Viet Nam to clarify and strengthen existing data protection measures and/or provide further guidance through new tools and mechanisms.[11] At the regional level, the *ASEAN Cybersecurity Cooperation Strategy 2021-2025* was launched in January 2022 to secure the region's cyberspace and promote the digital economy and community.[12]

The overall ADII Pillar scores also show that Pillar 4 (Digital Skills & Talent) is the region's weakest link, closely followed by Pillar 5 (Innovation & Entrepreneurship). It is not surprising that both these Pillars perform very closely; both were already the bottom two Pillars in ADII 1.0, and both share the same set of barriers/challenges that hinder their progress.

Enhancing digital skills and talent was already a major barrier noted in ADII 1.0, and the fact that Pillar 4 remains the region's weakest pillar suggests that the lack of a capable digital workforce will continue to impede digital integration, diversification, and transformation in the region. Indeed, access to tertiary education, skill development, lifelong training, and knowledge transfers remains uneven within and across AMS.[13]

Entrepreneurship in ASEAN faces a similar set of barriers; access to the different elements that spur and sustain digital innovation—capital, skills, knowledge, tools, government support—is severely limited and rather uneven across the region, especially for women and rural populations.

[7] Monetary Authority of Singapore (2022) Central Banks of Indonesia, Malaysia, Philippines, Singapore and Thailand Seal Cooperation in Regional Payment Connectivity, www.mas.gov.sg/news/media-releases/2022/central-banks-of-indonesia-malaysia-philippines-singapore-and-thailand-seal-cooperation-in-regional-payment-connectivity
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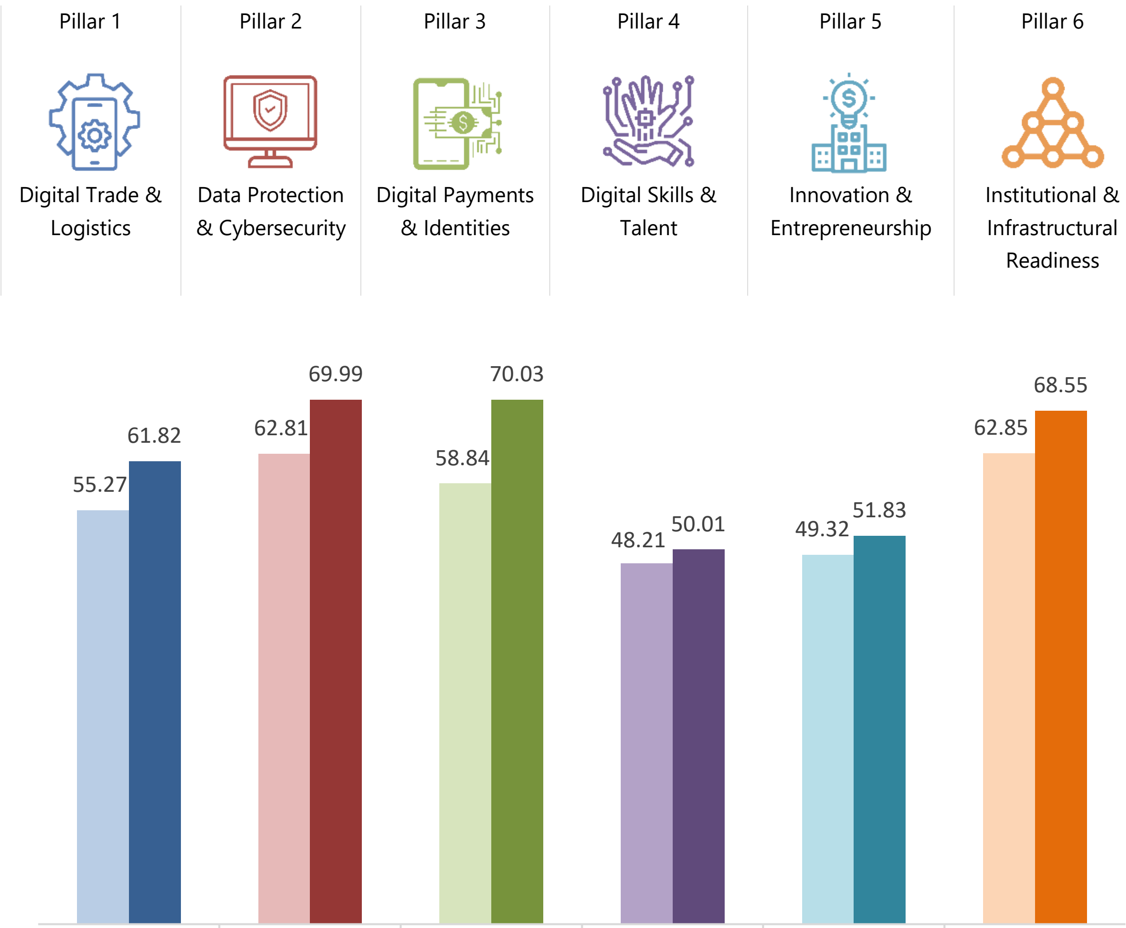
[9] Tech For Good Institute (2022) Promoting the Growth of the Digital Start-up Ecosystem in ASEAN, https://techforgoodinstitute.org/events/past-events/promoting-the-growth-of-the-digital-startup-ecosystem-in-asean
[10] ASEAN (2023) ASEAN Leaders' Declaration on Advancing Regional Payment Connectivity and Promoting Local Currency Transaction, https://asean.org/asean-leaders-declaration-on-advancing-regional-payment-connectivity-and-promoting-local-currency-transaction
[11] Herbert Smith Freehills (2022) Key changes in data privacy and cyber security laws across Southeast Asia in 2022, www.herbertsmithfreehills.com/insights/2022-11/key-changes-in-data-privacy-and-cyber-security-laws-across-southeast-asia-in-2022
[12] ASEAN (2022) ASEAN Cybersecurity Cooperation Strategy 2021-2025, https://asean.org/wp-content/uploads/2022/02/01-ASEAN-Cybersecurity-Cooperation-Paper-2021-2025_final-23-0122.pdf
[13] East Asia Forum (2022) Addressing the digital divide in ASEAN, www.eastasiaforum.org/2022/07/01/addressing-the-digital-divide-in-asean

It seems that the pervasive digital divide that still characterizes the region plays a role in the slow emergence of digital and entrepreneurial vocations in ASEAN, with poor connectivity and inadequate digital infrastructure exacerbating and perpetuating significant gaps between rural/urban, connected/unconnected, and banked/unbanked populations in ASEAN.[14]

Figure 2 shows the progress made by AMS for each Pillar between ADII 1.0 and ADII 2.0. All six ADII Pillar scores have increased between ADII 1.0 and ADII 2.0; a marked upward trajectory that suggests that digital integration has effectively progressed across all 10 AMS.

Figure 2: Pillar Scores for ASEAN – ADII 1.0 & ADII 2.0



ADII 1.0	55.27	62.81	58.84	48.21	49.32	62.85
ADII 2.0	61.82	69.99	70.03	50.01	51.83	68.55

The notable progress made by all six ADII Pillars could be attributed to various macroeconomic factors such as increased digital adoption, sustained policy reforms, and palpable infrastructural improvements across ASEAN.[15] This trend aligns with the broader global shift towards the digitalization of economies, which has put technological innovation and investment at the top of economic agendas.[16]

[14] Tech For Good Institute (2022) Driving Digital Literacy in Southeast Asia, https://techforgoodinstitute.org/blog/articles/driving-digital-literacy-in-southeast-asia
[15] ASEAN (2022) ASEAN Revs Up Digital Transformation, https://asean.org/wp-content/uploads/2022/11/Issue-23-Digital-Transformation-digital-version.pdf
[16] Brookings Institute (2022) How digital transformation is driving economic change, www.brookings.edu/articles/how-digital-transformation-is-driving-economic-change

Looking closer at the numbers, a first noteworthy observation is that score increases have not been uniform across all Pillars. Pillar 3 (Digital Payments & Identities) saw the highest increase, jumping 11.2 points. This remarkable rise may reflect the fact that both AMS and ASEAN have prioritized the implementation of interoperable cross-border digital payments.

Meanwhile, Pillar 4 (Digital Skills & Talent) has experienced the smallest growth since ADII 1.0. The modest 1.8-point increase may reflect the fact that access to tertiary education, skill development, lifelong training, and knowledge transfers remains uneven within and across AMS[17]—despite a number of national and regional initiatives aimed at turning things around.[18]

A particularly interesting development is the change in positions between Pillars 3 and 6; Pillar 3 moved up from 3rd in ADII 1.0 to 1st in ADII 2.0. Conversely, Pillar 6 moved down from 1st in ADII 1.0 to 3rd in ADII 2.0. This switch may indicate a major shift in priorities within the ASEAN digital economy landscape; the rise of Pillar 3 to the top position suggests a growing emphasis on Digital Payments & Identities, possibly driven by the increased adoption of e-commerce and digital financial services. Conversely, the drop in the ranking of Pillar 6, despite an increase in its score, might point to emerging challenges in driving Institutional & Infrastructural Readiness at the same pace as the rapid evolution of digital disruption.

This shifting dynamic may be compounded by the fact that initiatives in the realm of Digital Payments & Identities have long been stuck at various levels of discussions and negotiations and are now (finally) starting to concretize. On the other hand, making institutions nimble and reactive, and modernizing and maintaining digital infrastructure, are two very onerous tasks with no definitive, discernible, or measurable endgames. In any case, this development signals a need for investment in foundational digital infrastructure and focus on governance frameworks to sustain the inclusive and sustainable digitalization of ASEAN.

Overall, the comparative assessment between ADII 1.0 and ADII 2.0 provides a quantitative overview of a major dynamic at play in ASEAN: the fact that a broad and concerted approach to digital development is effectively taking place at the regional level, but the improved performance of some AMS is balanced by the slower progress in others.

Indeed, various ASEAN-driven initiatives are delivering results in the areas of Digital Trade & Logistics, Data Protection & Cybersecurity, Digital Skills & Talent, and Innovation & Entrepreneurship, but the progress being made is not happening in the same manner nor at the same pace for all AMS—underlining the difficulty of driving digital transformation consistently and coherently in a region characterized by varying levels of digital maturity and capacity.

To this end, the upcoming *ASEAN Digital Economy Framework Agreement (DEFA)*, is a region-wide initiative that will have the biggest impact on all facets covered by the ADII. Its potential impact is substantial, promising to reduce the digital divide within ASEAN, improve digitalization processes,

[17] East Asia Forum (2022) Addressing the digital divide in ASEAN, www.eastasiaforum.org/2022/07/01/addressing-the-digital-divide-in-asean

[18] The ASEAN Digital Innovation Programme (Microsoft), the ASEAN Seeds for the Future program (Huawei), the ASEAN Digital Skills Vision 2020 plan (World Economic Forum), and the Conference on 21st Century Skills and Youth Participation (UNICEF) are just some of the most recent initiatives devoted to improving populations' access to digital skills in ASEAN.

increase cross-border interoperability, and better integrate the region's digital economies.[19] Aimed for completion in 2025, the *DEFA* is highly anticipated, as it is poised to address the diverse digital capabilities of member countries, thereby influencing the trajectory of digital trade and economic cooperation in ASEAN and ensuring a more inclusive digital future for the region.

[19] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 1 – DIGITALLY ENABLED TRADE & LOGISTICS

Digitally enabled trade and logistics facilitate seamless trade across ASEAN

The rapid pace of digital transformation is impacting all facets of trade. It enables seamless, digitally enabled trade flows across borders, as well as the flow of new, data-driven digital services across ASEAN.

In this context, it is critical for AMS to rapidly modernize digital trade platforms and processes, including the digitalization of physical infrastructure, the integration of logistical networks, and the alignment of trade policies to fully take advantage of digital trade opportunities in ASEAN. Digital trade is still nascent in ASEAN. It is critical that AMS agree on digital trade priority areas to focus on and how to enable and coordinate trade processes and related sectors. As a key component of digital economies, seamless and digitally enabled trade is crucial for the region to effectively grow and expand.

To this end, Pillar 1 of the ADII measures the extent of digital technologies (and complementary infrastructure) that are being used to facilitate trade. This includes digital technologies used by traders and customs authorities, the acceptance and use of digital signatures, the use of international standards for trade documents and procedures, infrastructure connecting and facilitating trade across borders, and the competence and quality of logistics services which support trade including multimodal transport operators and customs brokers.

Figure 3: Pillar 1 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0

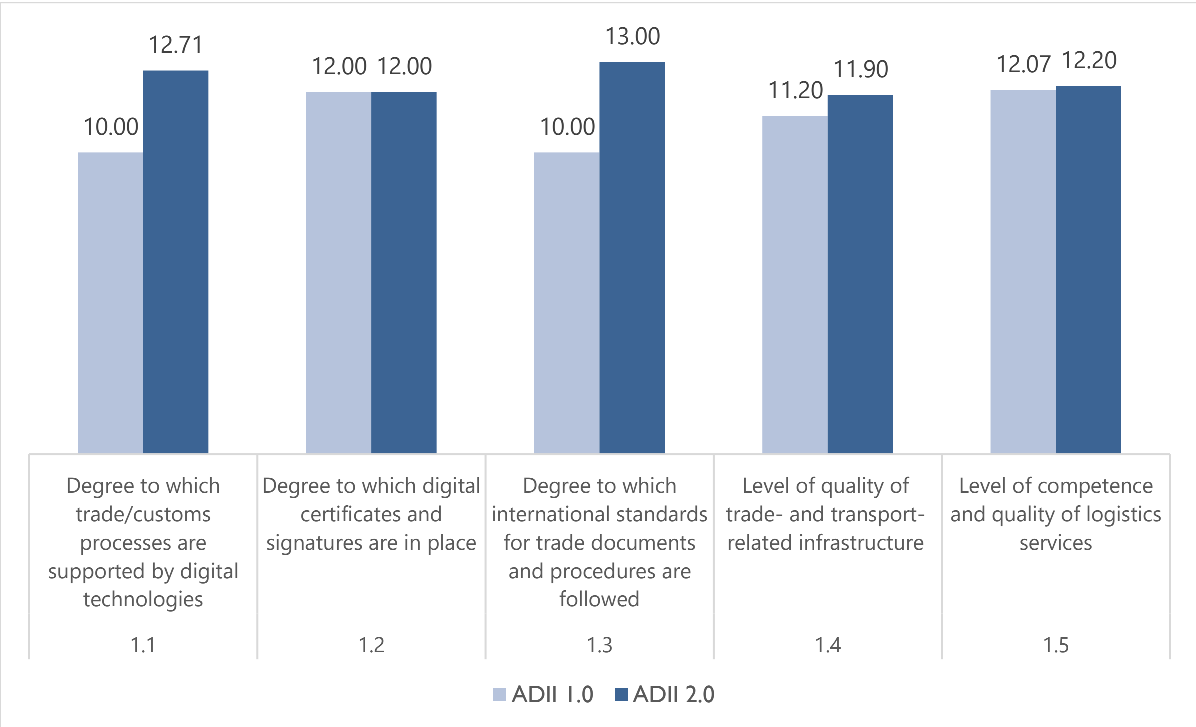


Figure 3 shows that ASEAN has made commendable progress in several key areas of digital trade since ADII 1.0. All Indicator scores have remained stable or improved in ADII 2.0, reflecting a positive trajectory for ASEAN's digital trade environment. The biggest improvements were seen in processes supported by digital technologies (Indicator 1.1) and the use of international standards (Indicator 1.3), denoting targeted policy efforts in both digital and physical trade facilitation. Despite this encouraging trend, the increases remain modest compared to the ones observed in other Pillars, indicating that the great strides being made at the regional level may be weighed down by less conducive measures at the national level.

Indicator 1.1: Indicator 1.1 (12.71) is the second strongest Indicator in this Pillar, reflecting the high levels of digitalization that trade and customs processes are undergoing in the region. This relatively high score reflects investments in information and communications technology (ICT) infrastructure and the push for digitalization in the wake of the COVID-19 pandemic. It also indicates that there have been concerted policy efforts to modernize customs through digital platforms, potentially leading to more efficient cross-border transactions and reduced trade barriers.

Indicator 1.1 is the second most improved Indicator with a 2.71-point increase. This increase allowed it to hoist itself to 2nd position out of five indicators, whereas it was tied with Indicator 1.3 for last position in ADIII 1.0. This evolution demonstrates significant progress in the adoption of digital technologies in trade and customs processes since ADII 1.0. This trajectory likely reflects regional commitments to streamline trade processes, including the *ASEAN Single Window (ASW)*, which streamlines customs processes among member states, and policies encouraging the adoption of digital solutions to manage trade documentation and procedures.

Indicator 1.2: With a score of 12.00, this Indicator stands in 4th position, just ahead of Indicator 1.4. The score points to an adequate level of adoption and implementation of digital certificates and signatures, which are key components of secure digital transactions. While this denotes the presence of policies aimed at enhancing cybersecurity and trust in digital trade, its relatively low position within the Pillar suggests that more could be done to make these policies both more effective and more impactful.

The score for Indicator 1.2 remained stable at 12.00, possibly indicating a maturity in the region's digital authentication infrastructure, which is foundational for secure digital trade. It suggests that ASEAN had already established robust digital frameworks by ADII 1.0, and these have presumably remained effective into ADII 2.0. This score will likely increase in the next iteration of ADII to reflect the progress currently being made in the development of legal and regulatory frameworks that recognize and enforce digital certifications and agreements.

Indicator 1.3: Indicator 1.3 scored the highest (13.00) among the five Pillar 1 Indicators, suggesting a robust alignment with global best practices across AMS. This high level of adoption indicates regional efforts to integrate with global trade systems, reduce trade frictions, and increase the interoperability of trade documentation. It may also signal investments in capacity-building initiatives to train stakeholders on international best practices.

Indicator 1.3 also made the most notable improvement over time (a 3-point increase), lifting itself from last position in ADII 1.0 to 1st position in ADII 2.0. This progress translates a number of decisive and concerted efforts that have been made since ADII 1.0. From the implementation of policies like the *ASEAN Trade in Goods Agreement (ATIGA)* and other international agreements aimed at making standards and procedures for trade documentation more interoperable, these efforts contribute to the efficiency and reliability of cross-border trade, making ASEAN more competitive globally. For example, the *Bandar Seri Begawan Roadmap* lays out a comprehensive plan for ASEAN's digital transformation and integration and the *ASEAN-Australia Digital Trade Standards Initiative* aims to expand bilateral political and economic linkages between ASEAN and Australia.

Indicator 1.4: With a score of 11.90, Indicator 1.4 scored the lowest out of all Indicators in Pillar 1, revealing that physical infrastructure remains a major hurdle for the region. This suggests that there is room for improvement in the quality and coverage of infrastructure related to trade and transport, and that further policy efforts are needed to effectively improve port facilities, road networks, and other critical logistics corridors to boost trade efficiency.

Indicator 1.4 saw a modest increase from 11.20 to 11.90. In addition, Indicator 1.4 was in 3rd position in ADII 1.0 and tumbled to last position in ADII 2.0. Together, this trend may indicate limited and uneven improvements in infrastructure, as well as room for further enhancements. There are in fact several ongoing infrastructure projects; for example, the *Master Plan on ASEAN Connectivity 2025* aims to improve physical, institutional, and people-to-people linkages. But it may be too soon for such broad and long-term initiatives to deliver both tangible and measurable results.

Indicator 1.5: At 12.20, Indicator 1.5 is right in the middle of all Pillar 1 Indicators. This score reflects an adequate quality of logistics services, indicating that there is a reasonably high standard of operational performance and reliability in logistics within the region. It also suggests that ASEAN may have been investing in training, technology, and systems to enhance the quality of logistics services, which is critical for competitive international trade.

The slight increase experienced from 12.07 to 12.20 could be attributed to the various initiatives that are facilitating the movement of goods and supplies across the region. In addition to various transport-facilitation agreements (the *ASEAN Framework Agreement on the Facilitation of Goods in Transit*,[20] the *ASEAN Framework Agreement on Multimodal Transport*,[21] and the *ASEAN Framework Agreement on Facilitation of Inter-State Transport)*,[22] ASEAN's logistics sector has benefitted from advances in trade and competition policy. On trade, ASEAN is working on widening the electronic exchange of intra-ASEAN certificate of origin and ASEAN Customs Declaration Document (ACDD).[23] On competition, the mid-term review of the *ASEAN Competition Action Plan 2016-2025 (ACAP 2016-2025)*[24] has allowed ASEAN to develop new priorities more in line with the rapid digitalization of AMS' economies.[25]

It is worth noting that much like the other ADII Pillars, Pillar 1 should greatly benefit from the finalization of the much-anticipated *ASEAN Digital Economy Framework Agreement (DEFA)*.[26] Indeed, the first (Digital Trade) and second (Cross-border E-Commerce) of its nine priority areas cover the establishment of a seamless and interoperable digital trading environment across the region, with the aim to facilitate cross-border trade and commerce through electronic documents and interoperable processes.[27]

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[20] ASEAN (2019) ASEAN Framework Agreement on The Facilitation of Goods in Transit, https://acts.asean.org/Legal_Framework/asean-framework-agreement-facilitation-goods-transit-afafgit
[21] ASEAN (2019) ASEAN Framework Agreement on Multimodal Transport, https://asean.org/wp-content/uploads/2019/11/Implementation-Framework-AFAMT_FINAL.pdf
[22] ASEAN (2012) ASEAN Framework Agreement on Facilitation of Inter-State Transport, https://asean.org/wp-content/uploads/images/archive/documents/Inter-State%20Transport%20Agreement.pdf
[23] Singapore Customs (2022) Electronic exchange of the ASEAN Customs Declaration Document (ACDD) under the ASEAN Single Window (ASW), www.customs.gov.sg/businesses/international-connectivity/acdd/
[24] ASEAN (2020) ASEAN Competition Action Plan (ACAP) 2016-2025, www.asean-competition.org/about-aegc-asean-competition-action-plan-acap-2016-2025
[25] ASEAN (2021) New ACAP 2025 and Implementation Schedule 2021-2025, www.asean-competition.org/read-publication-new-acap-2025-and-implementation-schedule-2021-2025
[26] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf
[27] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 2 – DATA PROTECTION & CYBERSECURITY

Robust data protection and cybersecurity support digital trade and innovation

Globalization and digitalization have led to greater connectivity, which in turn has rapidly increased the quantities of data being accessed, transferred, and exchanged both within and between countries. As such, digital integration requires consumer and business data to flow safely, securely, and responsibly across territories and jurisdictions.

ASEAN has recognized this as a priority under the *ASEAN Framework on Digital Data Governance* which strives to promote intra-ASEAN data flows and drive digital economy growth. From manufacturing and services to agriculture and retail, all sectors increasingly rely on data to plug in to global value chains and contribute to the global economy. It is thus essential for AMS to have cybersecurity, privacy, and other frameworks in place that protect and secure data. This is not only in the interest of digital businesses and their business models, but also for citizens, whose personal data (everything from biometrics to Internet surfing habits) powers data-driven platforms and services.

In this context, Pillar 2 aligns with DIF Priority Area 2 on the importance of protecting data while supporting digital trade and innovation. It measures the degree to which data protection measures are in place among AMS, the levels of legislative and regulatory cybersecurity capabilities, institutional cybersecurity capabilities, technical cybersecurity capabilities, and the level of international cooperation on cybersecurity.

Figure 4: Pillar 2 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0

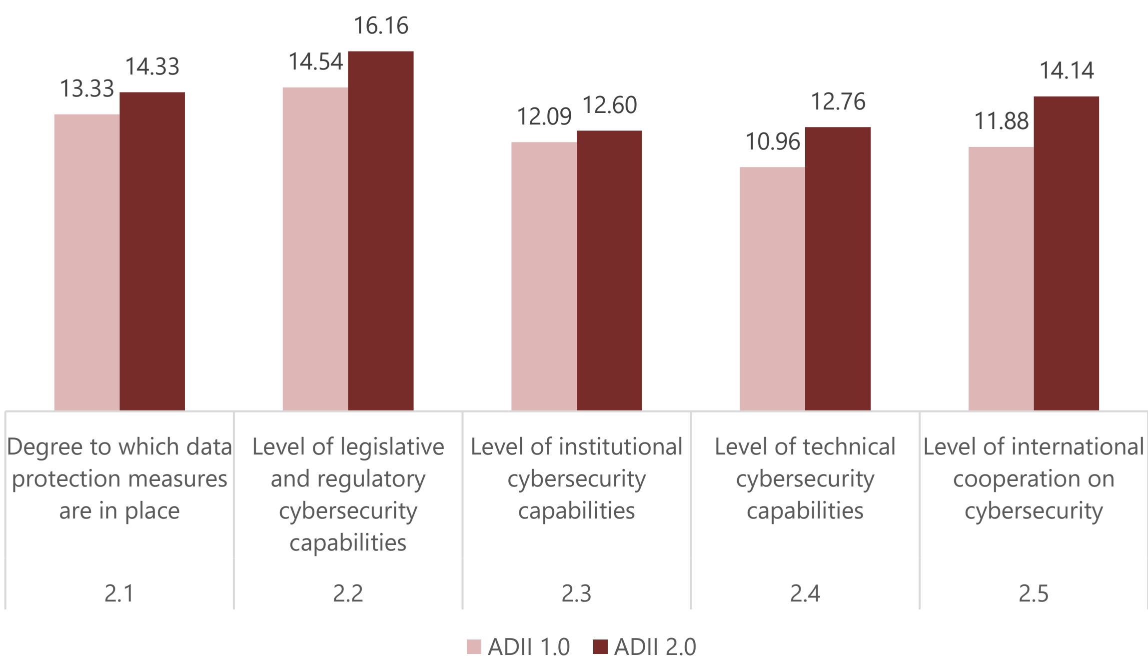

Figure 4 shows a clear upward trajectory across all indicators, translating a consistent improvement of the region's data protection and cybersecurity capabilities. Pillar 2 is indeed one of three Pillars (along with Pillars 3 and 6) for which no AMS has seen its score decrease since ADII 1.0, which points to a general strengthening of all the enabling measures that allow data-driven innovations to thrive. The most significant leap is observed in the level of technical cybersecurity capabilities (Indicator 2.4) and the level of international cooperation on cybersecurity (Indicator 2.5), indicating a strong national and regional policy response to global data governance challenges. Collectively, these trends demonstrate ASEAN's proactive stance in fortifying its digital domain, which is crucial for the region's economic integration and competitiveness in the digital economy.

Indicator 2.1: A score of 14.33 (the second highest score within this Pillar) suggests a robust level of data protection measures across ASEAN countries. Indeed, a number of laws, policies, and regulations have been launched since ADII 1.0: In Indonesia, *Law No. 27 of 2022 on Personal Data* is Indonesia's first comprehensive data protection legislation, applicable to both electronic and non-electronic systems and to all types of businesses, industries, and organizations.[28] In Malaysia, the *General Code of Practice of Personal Data Protection* introduces new legal requirements and providing best practice recommendations with respect to the implementation of principles under the *Personal Data Protection Act 2010* and its subsidiary legislation.[29] In the Philippines, its National Privacy Commission (NPC) issued *Circular No. 2022-01* on the *Guidelines on Administrative Fines* which intensified its drive to ensure compliance by introducing a new way to quickly determine liability for violations of the *Data Privacy Act 2012*, its IRR, and other issuances.[30] In Thailand, multiple guidelines with various effective dates were introduced since the main data protection provisions under the *Personal Data Protection Act (PDPA)* came into full effect in June 2021.[31]

There is, however, a discrepancy between the number of national data protection measures that have been enacted and the modest 1-point increase from 13.33 to 14.33 that we observe. This may be explained by the fact that while these foundational measures are essential for national digital economies, much remains to be done in making regional data protection initiatives more interoperable; AMS are at different stages when it comes to establishing, implementing, and enforcing comprehensive data protection regulations. We expect this score to evolve in the next iteration of the ADII, as several regional measures should start impacting data protection at the regional level. Notably, the *ASEAN Data Management Framework (DMF)*[32] and the *ASEAN Model Contractual Clauses for Cross-Border Data Flows (MCCs)*[33] should have a visible effect on Indicator scores in the next couple of years.

Indicator 2.2: At 16.16, Indicator 2.2 scores the highest within Pillar 2, demonstrating a prioritization of cybersecurity in AMS' digital policy agendas. Indeed, all AMS recognize the importance of strong legislative and regulatory cybersecurity foundations for digital business models to emerge and thrive. A key question for the region will be the extent to which national measures will effectively converge and work together at the regional level. To this end, the *ASEAN Cybersecurity Cooperation Strategy 2021-2025* was launched in January 2022 to consolidate and promote a regional approach to cyber-

[28] Future of Privacy Forum (2022) Indonesia's Personal Data Protection Bill: Overview, Key Takeaways, and Context, https://fpf.org/blog/indonesias-personal-data-protection-bill-overview-key-takeaways-and-context

[29] Global Compliance News (2023) Malaysia: New legal requirements under the General Code of Practice of Personal Data Protection, www.globalcompliancenews.com/2023/02/06/https-insightplus-bakermckenzie-com-bm-technology-media-telecommunications_1-malaysia-new-legal-requirements-under-the-general-code-of-practice-of-personal-data-protection_02022023

[30] National Privacy Commission (2022) NPC Circular No. 2022-01, www.privacy.gov.ph/wp-content/uploads/2022/08/NPC-CIRCULAR-NO.-2022-01-GUIDELINES-ON-ADMINISTRATIVE-FINES-dated-08-AUGUST-2022-w-SGD.pdf

[31] Zico Law (2022) Thailand issues PDPA guidelines on consent and personal data collection, www.zicolaw.com/resources/alerts/thailand-issues-pdpa-guidelines-on-consent-and-personal-data-collection

[32] PDPC (2021) ASEAN Data Management Framework and Model Contractual Clauses on Cross Border Data Flows, www.pdpc.gov.sg/Help-and-Resources/2021/01/ASEAN-Data-Management-Framework-and-Model-Contractual-Clauses-on-Cross-Border-Data-Flows

[33] Baker McKenzie (2021) ASEAN: Adopting the ASEAN Model Contractual Clauses for cross-border data transfers, https://insightplus.bakermckenzie.com/bm/data-technology/asean-adopting-the-asean-model-contractual-clauses-for-cross-border-data-transfers_1

threat mitigation. It updates the previous strategy, addressing newer cyber-developments and building upon AMS' collective efforts.[34]

Looking at progress over time, the score has increased from 14.54 to 16.16, indicating an improvement of AMS' legal and regulatory frameworks. This may be a result of AMS actively taking measures to ensure their approaches to cybersecurity evolve to remain useful and impactful in a context of rapidly changing cyber-threats. In Vietnam, *Decree No. 53/2022/ND-CP* provides much-needed clarity on important aspects of the *Law on Cyber Security No. 24/2018/QH14* (most notably on the requirements for local data storage and local presence requirements).[35] Likewise, Thailand's National Cyber Security Agency (NCSA) has announced plans to expand the enforcement of its standard framework of security requirements.[36] In a similar vein, Singapore has announced that it will review its *Cybersecurity Act 2018* to "improve awareness of threats over Singapore's cyberspace, protect virtual assets (e.g., systems hosted on the cloud) as critical information infrastructure (CII) if they support essential services".[37]

Indicator 2.3: With a score of 12.60, institutional cybersecurity capabilities score the lowest out of all Pillar 2 Indicators. This could signal operational and implementational gaps in cybersecurity regulation, with institutional capacity-building (training, staffing, and inter-agency coordination) needing to be strengthened across the board. A major institutional obstacle within AMS is the fact that there is little to no coordination between the different bodies and organizations in charge of operationalizing cybersecurity policies.[38] As such, collaboration and breaking down silos among organizations and sectors are needed to match the scope and pace of ever-evolving cybersecurity challenges more effectively.[39]

The slight increase from 12.09 to 12.60 points to modest improvements in institutional capabilities; while they have improved, they have done so at a slow and uneven pace across AMS, implying a need for further strengthening and capacity building within governmental and organizational structures. Initiatives such as the *ASEAN CERT Information Exchange Mechanism*,[40] the *ASEAN Regional Computer Emergency Response Team (CERT)*,[41] the *ASEAN-Singapore Cybersecurity Centre of Excellence (ASCCE)*,[42] and the *ASEAN-Japan Cybersecurity Capacity Building Centre*[43] will undoubtedly lead to better cybersecurity preparedness in the region by enhancing the transmission and exchange of operational knowledge, skills, and information.

Indicator 2.4: The score of 12.76 indicates that the technical measures and tools in place to counteract cyber-threats are present but possibly not as advanced nor as widespread as they should be. This highlights a possible focus area for AMS to develop policies that encourage the adoption of state-of-

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[34] ASEAN (2022) ASEAN Cybersecurity Cooperation Strategy 2021-2025, https://asean.org/wp-content/uploads/2022/02/01-ASEAN-Cybersecurity-Cooperation-Paper-2021-2025_final-23-0122.pdf
[35] Allen & Gledhill (2022) Data localisation requirements for specific data and entities in Vietnam, effective 1 October 2022, www.allenandgledhill.com/perspectives/articles/22527/vnkh-data-localisation-requirement-for-specific-data-and-entities-in-effective-1-october-2022?agreed=cookiepolicy
[36] Bangkok Post (2022) NCSA set to boost cyber security laws, www.bangkokpost.com/thailand/general/2369148/ncsa-set-to-boost-cyber-security-laws
[37] Data Guidance (2022) Singapore: MCI announces review of Cybersecurity Act, www.dataguidance.com/news/singapore-mci-announces-review-cybersecurity-act
[38] Voice of International Affairs (2023) Emerging Concerns of Cyber-Security in Southeast Asia, https://internationalaffairsbd.com/cyber-security-in-southeast-asia
[39] OpenGov Asia (2023) ASEAN Cybersecurity: The Need for Public-Private Partnerships, https://opengovasia.com/asean-cybersecurity-the-need-for-public-private-partnerships
[40] ZDNet (2022) Singapore champions Asean CERT as region's cyber armour, www.zdnet.com/article/singapore-champions-asean-cert-as-regions-cyber-armour
[41] CSA (2022) Establishment of ASEAN Regional Computer Emergency Response Team (CERT), www.csa.gov.sg/News-Events/Press-Releases/2022/establishment-of-asean-regional-computer-emergency-response-team
[42] CSA (2021) ASEAN-Singapore Cybersecurity Centre of Excellence, www.csa.gov.sg/News-Events/Press-Releases/2021/asean-singapore-cybersecurity-centre-of-excellence
[43] Ministry of Foreign Affairs of Japan (2023) Opening Ceremony of the Training Program for the ASEAN-Japan Cybersecurity Capacity Building Centre in Thailand, www.mofa.go.jp/press/release/press5e_000016.html

the-art cybersecurity technologies, promote R&D in cybersecurity, and foster a skilled workforce to manage these technologies.

Encouragingly, the score's improvement from 10.96 to 12.76 demonstrates that advancements have been made in this area since ADII 1.0. According to Palo Alto Networks' *State of Cybersecurity in ASEAN* report, over 90% of all businesses in ASEAN declare putting cybersecurity at the top of their agendas, resulting in 68% of all organizations planning to increase the budget allocated for cybersecurity.[44] At the same time, the survey finds that small businesses feel relatively less confident in coping with cybersecurity challenges, mainly due to constrained cybersecurity budgets and relatively weaker in-house cybersecurity talent.

Indicator 2.5: At 14.14, the Indicator score translates the many regional and international linkages that allow AMS to communicate and cooperate on cybersecurity issues. This cooperation is crucial for managing the transnational nature of cyber-threats and for ensuring uninterrupted growth of the region's digital economies. While an encouraging score, there remains room for more robust policy initiatives aimed at enhancing collaborative efforts, such as information sharing, joint exercises, and alignment of cybersecurity standards.

The Indicator score increased from 11.88 to 14.14, the biggest improvement seen within Pillar 2. This shows a significant enhancement in international cooperation since ADII 1.0, reflecting ASEAN's strategic focus on regional and global partnerships to address the complexities of cross-jurisdictional cyber-criminality. Within ASEAN, the *ASEAN Cybersecurity Cooperation Strategy (2021-2025)* clearly aims to help AMS develop their understanding and capacity to implement cybersecurity measures.[45] Looking at partnerships beyond the region, the *ASEAN-Australia Cyber Policy Dialogue*[46] and the *EU-ASEAN Cooperation on Cybersecurity and Emerging Technologies*[47] will play an important role in building a cohesive regional response to cross-border cyber-challenges.

Here too, Pillar 2 should greatly benefit from the finalization of the *ASEAN Digital Economy Framework Agreement (DEFA)*. Out of nine main priority areas, the fifth (Online Safety and Cybersecurity) and the sixth (Cross-border Data Flows and Data Protection) are devoted to improving cooperation and establishing frameworks to better protect data privacy and better secure data movements across the region.[48]

[44] Palo Alto Networks (2022) Cybersecurity leads boardroom agendas among ASEAN organisations in light of pandemic-induced shifts, www.paloaltonetworks.sg/company/press/2022/cybersecurity-leads-boardroom-agendas-among-asean-organisations-in-light-of-pandemic-induced-shifts-malaysia
[45] ASEAN (2022) ASEAN Cybersecurity Cooperation Strategy (2021-2025), https://asean.org/wp-content/uploads/2022/02/01-ASEAN-Cybersecurity-Cooperation-Paper-2021-2025_final-23-0122.pdf
[46] ASEAN (2023) Chairman's Statement of the 3rd Annual ASEAN-Australia Summit, https://asean.org/wp-content/uploads/2023/09/FINAL-Chairmans-Statement-of-the-3rd-ASEAN-Australia-Summit.pdf
[47] Carnegie Europe (2023) EU-ASEAN Cooperation on Cybersecurity and Emerging Technologies, https://carnegieeurope.eu/2023/07/04/eu-asean-cooperation-on-cybersecurity-and-emerging-technologies-pub-90082
[48] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 3 – DIGITAL PAYMENTS & IDENTITIES

Use of digital financial services and identities to enable seamless payments

Digital integration requires reliable and interoperable digital financial services that enable secure digital transactions across platforms. Digital/electronic payments, for instance, are becoming increasingly prevalent, thanks to the rapid surge in the ownership of mobile devices.

In this context, getting payment platforms to "understand" each other and to be "interoperable" is vital to ASEAN's rising digital economies. Achieving regional interoperability requires coordination among multiple stakeholders, a conducive legal and regulatory framework, the support of policymakers and overseers, commercially viable business models, and technological solutions (ideally based on international standards).

Likewise, national identification is important for citizens to prove who they are and allow them to transact and participate in the formal financial sector. Digital IDs further enable populations to take part in the digital economy—whether it is sending/receiving cross-border remittances, making online purchases, or using digital government services.

As such, the indicators in Pillar 3 look at the adoption and use of digital financial services including for banking and other types of financial transactions, as well as whether there are legal frameworks and policies which enable electronic transactions. Pillar 3 also assesses the proportion of national identity adoption as well as the degree to which a digitized identity system is in place.

Figure 5: Pillar 3 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0

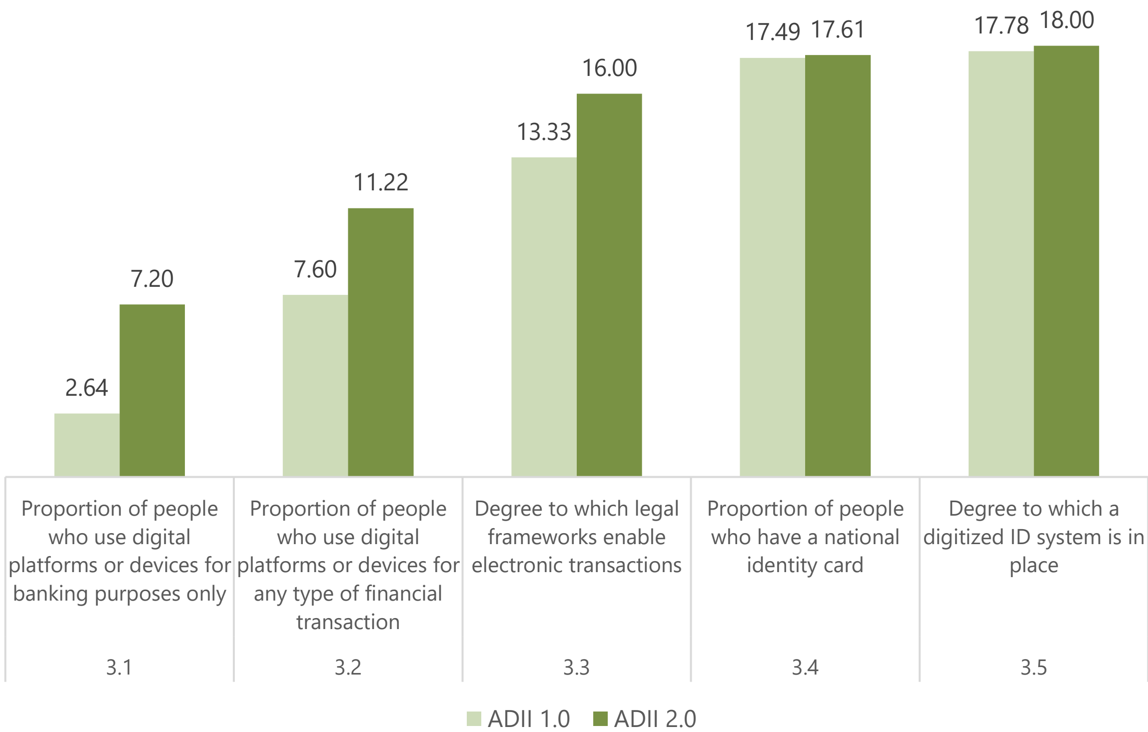


Figure 5 shows a substantial progression in ASEAN's embrace of digital technologies for financial transactions and identity management between ADII 1.0 and ADII 2.0. The most noticeable increase is in the proportion of people who use digital tools for banking and financial purposes (Indicators 3.1 and 3.2), revealing the impact of robust policy efforts aimed at expanding the accessibility of digital finance, improving the public's digital literacy, and creating an enabling environment through supportive legal frameworks. Overall, the improvements seen across all Pillar 3 Indicators has allowed it to take top spot as ASEAN’s strongest-performing Pillar, reflecting the many national and regional efforts that are taking shape and having a tangible impact on the region.

Indicator 3.1: With a score of 7.20, this is ASEAN’s lowest-performing Pillar 3 Indicator. This suggests that there is ample room to grow the region’s digital banking sector, as well as to improve the population’s ability to access and use digital banking tools. The main barrier in the region is the fact that millions of people remain unbanked, with 71% of working adults still receiving their monthly wages in cash[49] and 79% falling under the category of “financially excluded”—with a large proportion of the financially excluded being rural dwellers, micro-enterprises, and women who have no access to credit, insurance, or investment products.[50]

Despite these setbacks, the growth from 2.64 in ADII 1.0 to 7.20 in ADII 2.0 signals a significant leap in the adoption and use of digital banking services across ASEAN. Indeed, while Singapore has typically been the region’s biggest digital banking revenue driver, recent numbers show that digital banks have been successfully growing their user base in the Philippines, Indonesia, Thailand, and Vietnam, thanks to a wide range of innovative policy approaches (regulators granting digital banking licenses, government-driven regulatory sandboxes, partnerships between traditional banks and Fintech companies, etc.).[51]

Indicator 3.2: The score of 11.22 for the use of digital platforms for any financial transaction, as opposed to banking-only transactions, suggests a broader acceptance and utilization of digital platforms for a range of financial services. According to a recent World Economic Forum (WEF) study on digital habits in ASEAN, digital payment applications (e-banking and e-wallets) are the most widely used applications after social media, with three main financial services functions: (1) managing cash flows and expenses; (2) saving for the future; and (3) providing a safety net to get through difficult times.[52] Businesses in the region also recognize the role of financial services in business expansions, expense payments, new customer acquisitions, and customer dispute resolution.

The increase from 7.60 in ADII 1.0 to 11.22 in ADII 2.0 indicates an important upward trend that likely results from regional measures aimed at streamlining and securing the use of digital platforms for financial purposes. In 2022, the central banks of Indonesia, Malaysia, the Philippines, Singapore, and Thailand signed an agreement to enhance cooperation on payment connectivity.[53] The aim is to enable a faster, cheaper, more transparent, and more inclusive cross-border payments system that is expected to expand to include all AMS.[54] Additionally, the *ASEAN Leaders Declaration on Advancing Regional*

[49] Business Times (2023) Removing barriers to the unbanked and accelerating financial inclusion in ASEAN, www.businesstimes.com.sg/international/asean/removing-barriers-unbanked-and-accelerating-financial-inclusion-asean
[50] World Economic Forum (2022) ASEAN Digital Generation Report: Digital Financial Inclusion, www3.weforum.org/docs/WEF_ASEAN_Digital_Generation_Report_2022.pdf
[51] TechWire Asia (2023) Digital banks: What’s driving success in Southeast Asia?, https://techwireasia.com/10/2023/digital-banks-whats-driving-success-in-southeast-asia
[52] World Economic Forum (2022) ASEAN Digital Generation Report: Digital Financial Inclusion, www3.weforum.org/docs/WEF_ASEAN_Digital_Generation_Report_2022.pdf
[53] Monetary Authority of Singapore (2022) Central Banks of Indonesia, Malaysia, Philippines, Singapore and Thailand Seal Cooperation in Regional Payment Connectivity, www.mas.gov.sg/news/media-releases/2022/central-banks-of-indonesia-malaysia-philippines-singapore-and-thailand-seal-cooperation-in-regional-payment-connectivity
[54] eTrade for All (2023) How ASEAN is making instant cross-border payments a reality, https://etradeforall.org/news/how-asean-is-making-instant-cross-border-payments-a-reality

Payment Connectivity and Promoting Local Currency Transaction has demonstrated the region's commitment to setting up a regional cross-border Quick Response (QR) code payment network.[55] The WEF study found that security and safety are foundational requirements that determine the extent to which consumers use digital financial services; to sustain the current upward momentum in coming years, ASEAN will need to focus on policies and frameworks that diversify digital transaction options and ensure the security and ease of online payments.

Indicator 3.3: A score of 16.00 in this area indicates that ASEAN countries are making considerable efforts to develop and implement legal frameworks that support electronic transactions. According to a recent ISEAS study, most AMS perform well in terms of having laws, rules, or regulations that enable e-commerce, e-transactions, e-signatures, digital standards, and/or digital identity systems.[56] Where the region falters, according to the study, is in the efficacy with which these measures are implemented and enforced. In this context, the *ASEAN Agreement on Electronic Commerce*[57] and the *Regional Comprehensive Economic Partnership (RCEP)* provisions on e-commerce[58] should provide guidance to further strengthen regional integration in this area.

The notable improvement from 13.33 in ADII 1.0 to 16.00 in ADII 2.0 reflects the various laws and regulations that AMS have implemented to support and secure electronic transactions. For instance, Viet Nam recently amended its *Law on Electronic Transactions No. 20/2023/QH15 (LoET 2023)* to update and clarify many non-conducive clauses of the previous *Law on E-Transaction (LoET 2005)*.[59] The new law adds new features to the original scope of application, provides more precise definitions, and abolishes unnecessary administrative procedures in the implementation of certain e-transactions. Similar efforts have taken shape in Lao PDR (*Decree on E-Commerce No. 296/GOV 2021*),[60] Myanmar (*E-Commerce Operation Guidelines*),[61] the Philippines (*Internet Transactions Act of 2023*),[62] and Thailand (*Royal Decree on the Operation of Digital Platform Service Businesses that are subject to Prior Notification B.E. 2565, 2022*).[63]

Indicator 3.4: The score of 17.61 (the second highest score in Pillar 3) suggests that a high proportion of ASEAN citizens possess a national identity card. For example, Indonesia's *Kartu Tanda Penduduk (KTP)*,[64] Malaysia's *MyKad*,[65] and Singapore's *National Registration Identity Card (NRIC)*[66] have attained high coverage rates, even in isolated or hard-to-reach areas. Areas of improvement will thus involve bringing national identity systems into the digital age in a useful and impactful manner—ensuring

[55] ASEAN (2023) ASEAN Leaders' Declaration on Advancing Regional Payment Connectivity and Promoting Local Currency Transaction, https://asean.org/asean-leaders-declaration-on-advancing-regional-payment-connectivity-and-promoting-local-currency-transaction
[56] ISEAS (2022) Enabling Domestic Data Flows for E-Commerce in ASEAN Countries, www.iseas.edu.sg/articles-commentaries/iseas-perspective/2022-32-enabling-domestic-data-flows-for-e-commerce-in-asean-countries-by-sithanonxay-suvannaphakdy
[57] ASEAN (2021) ASEAN Agreement on Electronic Commerce officially enters into force, https://asean.org/asean-agreement-on-electronic-commerce-officially-enters-into-force
[58] Enterprise Singapore (2022) Regional Comprehensive Economic Partnership (RCEP), www.enterprisesg.gov.sg/grow-your-business/go-global/international-agreements/free-trade-agreements/find-an-fta/rcep
[59] Lexology (2023) Vietnam's Digital Revolution: How the Amended Law on E-Transactions Will Change the Online Landscape, www.lexology.com/library/detail.aspx?g=b0188642-707a-48ac-83fc-e6bb9918c292
[60] Zico Law (2021) Laos Adopts New E-Commerce Regulation, www.zicolaw.com/resources/alerts/laos-adopts-new-e-commerce-regulation
[61] Tilleke & Gibbins (2023) Myanmar Issues Guidelines for E-commerce Businesses, www.tilleke.com/insights/myanmar-issues-guidelines-for-e-commerce-businesses
[62] Department of Trade and Industry (2023) President Marcos signs Internet Transactions Act (ITA) into law to strengthen e-commerce in PH, www.dti.gov.ph/archives/news-archives/president-marcos-signs-internet-transactions-act-ita-law-strengthen-e-commerce-ph
[63] Nishimura & Asahi (2023) Thailand's Royal Decree on the Supervision of Digital Platform Services, www.nishimura.com/en/knowledge/publications/thailands-royal-decree-on-the-supervision-of-digital-platform-services
[64] WikiWand (2022) Indonesian identity card, www.wikiwand.com/en/Indonesian_identity_card
[65] Employment Hero (2022) MyKad – Malaysia National Registration Identity Card (NRIC), https://employmenthero.com/my/glossary/mykad
[66] Dream Immigration SG (2022) Singapore NRIC Number – National Registration Identity Card, www.dreamimmigrationsg.com/singapore-nric-number-national-registration-identity-card

digital identity systems open the door to a wide range of life-improving services, making these national systems interoperable with one another, and maintaining the security and privacy of citizens' data.[67]

The slight increase from 17.49 in ADII 1.0 to 17.61 in ADII 2.0 may indicate a maturity of the prevalence of national identities, as most of the population already has a national identity card. In this context, further progress may not necessarily come from growing their reach, but their function. This may entail, for example, leveraging the prevalence of identity cards to integrate digital identity solutions and diversify the range of public services they give access to, effectively streamlining government-to-citizen interactions to improve socio-economic inclusivity.[68]

Indicator 3.5: At 18.00, the highest Indicator score in Pillar 3, it is clear that AMS are relatively advanced in implementing digital identity systems. In Singapore, the *Singpass* system serves approximately 97% of Singapore Citizens and Permanent Residents aged 15 and above, allows holders to interact with over 800 government agencies and private-sector organizations across more than 2,700 services, and enables more than 500 million personal and corporate transactions every year.[69] In the Philippines, authorities have accelerated the pace of roll-out of the *Philippine Identification System (PhilSys)* to increase the number of presenceless, paperless, and cashless transactions that can be conducted.[70] For most AMS, policy developments in this area are likely to focus on expanding the use cases for digitized IDs, ensuring interoperability across different services and countries, as well as compatibility with mobile devices.[71]

The very minor increase from 17.78 in ADII 1.0 to 18.00 in ADII 2.0 suggests an ongoing progression towards more integrated and sophisticated digitized ID systems. For example, several countries have started experimenting with ways to leverage AI and Blockchain to improve digital ID verification processes. Indeed, AI's advanced capabilities in facial recognition, behavioral biometrics, and natural language processing can be combined with Blockchain's security and immutability to enhance security and improve user experiences.[72] Likewise, some national agencies are introducing policies aimed at making digital ID systems interoperable across platforms, devices, and jurisdictions.[73]

Much like other ADII Pillars, Pillar 3 should greatly benefit from the finalization of the *ASEAN Digital Economy Framework Agreement (DEFA)*; the fourth of nine priority areas (Digital ID and Authentication) aims to develop a mutually recognizable and interoperable digital identity and electronic authentication framework for ASEAN.[74]

[67] Fulcrum (2022) Promoting Inclusive Growth with Digital Identification in ASEAN, https://fulcrum.sg/promoting-inclusive-growth-with-digital-identification-in-asean
[68] Asian Development Bank (2016) Identify for Development in Asia and the Pacific, www.adb.org/sites/default/files/publication/211556/identity-development-asia-pacific.pdf
[69] Smart Nation (2021) Singpass – Singapore's National Digital Identity, www.smartnation.gov.sg/files/press-releases/2021/media%20factsheet%20on%20singpass%20national%20digital%20identity.pdf
[70] IT News (2023) Philippines to speed up digital national ID rollout, www.itnews.asia/news/philippines-to-speed-up-digital-national-id-rollout-589852
[71] Mobile ID World (2023) Philippines Authorities Aim to Roll Out Mobile IDs By Year's End, https://mobileidworld.com/philippines-authorities-aim-to-roll-out-mobile-id-890914
[72] Daon (2023) The Role of AI in Digital Identity Verification, www.daon.com/resource/the-role-of-ai-in-digital-identity-verification
[73] Access Partnership (2022) Digital Identities and Cross-Border Interoperability, https://accesspartnership.com/digital-identities-and-cross-border-interoperability
[74] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 4 – DIGITAL SKILLS & TALENT

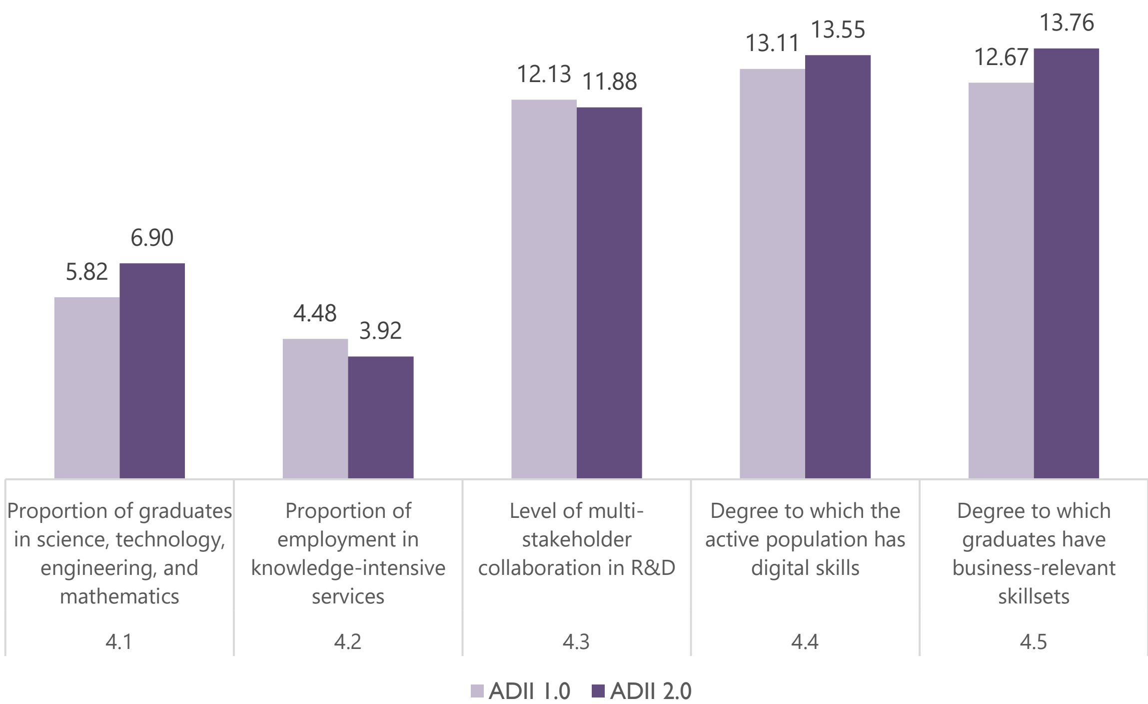
Enhancing digital skills and talent across ASEAN

Digital integration requires a skilled and trained workforce that is capable of driving and sustaining the digitalization of key economic sectors. Indeed, the effectiveness of policies that enable the digital economy hinges on populations having the corresponding knowledge and skills. This includes everything from technical skills and competencies to develop and maintain complex ICT systems and platforms, to digital literacy and capacity to use and apply digital technologies to their full potential.

From coding courses in schools to training and upskilling programs in workplaces, AMS must prepare citizens and businesses to both contribute to and benefit from fast-evolving digital environments. This includes partnering with the private sector to design relevant digital skills roadmaps and accelerating the roll-out of these programs for prioritized sectors.

To this end, Pillar 4 assesses the current level of skills and competencies by looking at the proportion of graduates in science, technology, engineering, and mathematics (STEM) as well as the proportion employed in knowledge-intensive services. The level of multi-stakeholder collaboration involved in R&D is also taken into account to measure the degree to which ideas are shared and collaborated upon. Lastly, there is the measure of digital skills by the population and business-relevant skillsets of graduates.

Figure 6: Pillar 4 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0



As Figure 6 shows, the digital skills and knowledge landscape is highly uneven in ASEAN; combined with the fact that Pillar 4 is the lowest performing pillar in ADII 2.0, this is an area that requires urgent prioritization by ASEAN policymakers, as the lack of a capable digital workforce can impede digital integration, diversification, and transformation in the region. The drop in the proportion of employment in knowledge-intensive services (Indicator 4.2) may point to challenges in job market evolution and a need for more robust policy action to stimulate growth in this sector. Likewise, the slight decline in multi-stakeholder collaboration in R&D (Indicator 4.3) may signal a need for renewed focus on cross-sector partnerships and joint initiatives. Encouragingly, the increase in the active population's digital skills (Indicator 4.4) and graduates' business-relevant skillsets (Indicator 4.5) suggests effective integration of digital competencies into education and workforce development, reflecting ASEAN's commitment to nurturing a future-ready economy. Together, these trends underscore a region in transition, actively engaging with the opportunities and challenges presented by the digital age—with visible gaps pinpointing areas that require further policy refinement.

Indicator 4.1: A moderate score of 6.90 suggests an equally moderate proportion of STEM graduates in ASEAN. Two main challenges emerge for the region: First, the great unevenness in terms of access, quality, cost, outcomes, and prospects for STEM students. According to a recent report on the state of higher education in ASEAN, gaps in digital capacity and unreliable infrastructure constitute a major barrier for STEM students in CLMV economies (Cambodia, Lao PDR, Myanmar, and Viet Nam). This includes patchy and slow internet connection, the high cost of devices, the lack of skills in using digital tools, and a rising mismatch between graduate skills and globally competitive knowledge-intensive labor markets.[75] Second, the wide gender gap that shows no signs of abating when it comes to starting and finishing tertiary studies, including in STEM. According to ASEAN statistics, 19.3% of women compared to 39.8% of men obtain STEM degrees across all ASEAN countries. Similarly, women are the minority across doctoral degree students and often do not go on to work in their doctoral fields.[76] As the ASEAN report on the state of higher education concludes: "Overall, the region faces the challenge of equipping the world's third-largest labor force with the skills needed to support growth and inclusiveness" in a global digital economy.[77]

Encouragingly, the increase from 5.82 in ADII 1.0 to 6.90 in ADII 2.0 shows that progress has undoubtedly been made since the ADII 1.0 report urged ASEAN to "bolster efforts in cultivating digital skills and talent, in particular, the promotion of STEM education and ensure that it leads to a proportionate take-up of employment in digital-related and knowledge-based industries."[78] This increase shows that policy initiatives aimed at further encouraging STEM education do have a palpable impact and should be further pursued. Such policies could include increased funding for STEM programs, scholarships to incentivize students, and partnerships with tech companies to improve employment prospects for graduates. At the regional level, initiatives like the *ASEAN Plan of Action on Science, Technology, and Innovation (APASTI) 2016-2025*[79] should continue promoting STEM education and careers among the youth and facilitating collaborative R&D in these fields.

[75] ASEAN (2022) State of Higher Education in Southeast Asia, https://asean.org/wp-content/uploads/2023/08/The-State-of-Higher-Education-in-Southeast-Asia_11.2022.pdf
[76] ASEAN (2023) Women in STEM in ASEAN, https://uploads.mwp.mprod.getusinfo.com/uploads/sites/62/2023/03/WOMEN-IN-STEM-Infographic-USAID-ASEAN.pdf
[77] ASEAN (2022) State of Higher Education in Southeast Asia, https://asean.org/wp-content/uploads/2023/08/The-State-of-Higher-Education-in-Southeast-Asia_11.2022.pdf
[78] ASEAN (2021) ASEAN Digital Integration Index (ADII) Report 2021, https://asean.org/book/asean-digital-integration-index-report-2021
[79] ASEAN (2016) ASEAN Plan of Action on Science, Technology and Innovation (APASTI) 2016-2025, https://asean.org/book/asean-plan-of-action-on-science-technology-and-innovation-apasti-2016-2025

Indicator 4.2: The relatively low score of 3.92 points to a gap that should be prioritized by ASEAN policymakers, as a low level of employment in knowledge-intensive services risks jeopardizing the region's competitiveness in the global innovation economy. This number also signals that despite many regional initiatives aimed at future-proofing the workforce, there remains a major gap between graduates' knowledge and skills and the applicable know-how that digitalized labor markets value and seek. For example, the *ASEAN Declaration on Human Resources Development for the Changing World of Work and Its Roadmap* highlights the importance of developing human resources through education and lifelong learning for a strong and resilient ASEAN Community.[80] Likewise, the *ASEAN Declaration on Promoting Competitiveness, Resilience, and Agility of Workers for the Future of Work*[81] and the *ASEAN Labour Ministers' Statement on the Future of Work: Embracing Technology for Inclusive and Sustainable Growth*[82] both aim to equip workers with the knowledge and skills to ensure the region's digital economies are not only competitive and dynamic, but also sustainable and inclusive.

The decrease from 4.48 in ADII 1.0 to 3.92 in ADII 2.0 reiterates the urgency with which this discrepancy should be addressed. The drop may point to challenges in job market evolution or to the need for more robust policy action to stimulate growth in the services sector. Whichever the case, a key dynamic should be tackled regionally: the dominance of rural and informal employment in the region. While several AMS are positioning themselves as regional technology hubs, attracting foreign talent and investment through a range of incentives, others are grappling with the challenge posed by unskilled labor, informal and precarious employment, job insecurity, and underemployment.[83]

Indicator 4.3: A score of 11.88 suggests that there is some degree of collaboration between different stakeholders in R&D within ASEAN, but that more can be done in this area. Indeed, ASEAN recognizes that despite varying national contexts, AMS encounter similar challenges; making the sharing of information, knowledge, best practices, and resources an invaluable asset.[84] According to a recent UNCTAD report, ASEAN already has a number of collaborative frameworks and networks in place to engage and empower research communities—the *ASEAN Foresight Alliance*, the *ASEAN Young Scientist Network*, the *ASEAN Network on Metallurgy and Metallic Materials*, the *ASEAN Network on Sustainable and Environmental Materials*, and the *Shared ASEAN High Performance Computing (HPC) Facility*—all it needs is to streamline and systemize the way in which these connections converge towards a common objective.[85]

The decrease from 12.13 in ADII 1.0 to 11.88 in ADII 2.0 suggests a need for revitalized efforts in encouraging partnerships and joint initiatives for R&D collaboration. According to UNCTAD, one way to achieve this is to promote and support talent mobility within ASEAN. Indeed, the *Policy Partnership for Science, Technology, and Innovation (PPSTI)* of the Asia-Pacific Economic Cooperation (APEC) promotes STI cooperation by supporting both digital and people-to-people connectivity through

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[80] ASEAN (2020) ASEAN Declaration on Human Resources Development for the Changing World of Work and Its Roadmap, https://asean.org/wp-content/uploads/2021/08/ASEAN-Declaration-on-Human-Resources-Development-for-the-Changing-World-of-Work-and-Its-Roadmap.pdf

[81] ASEAN (2021) ASEAN Declaration on Promoting Competitiveness, Resilience, and Agility of Workers for the Future of Work, https://asean.org/wp-content/uploads/2021/10/12.-ASEAN-Declaration-on-Promoting-Competitiveness-Resilience.pdf

[82] ASEAN (2021) ASEAN Labour Ministers' Statement on the Future of Work: Embracing Technology for Inclusive and Sustainable Growth, https://asean.org/asean2020/wp-content/uploads/2021/01/ASEAN-Labour-Ministers-Statement-on-the-Future-of-Work-Embracing-Techno....pdf

[83] ASEAN (2023) ASEAN Employment Outlook, https://asean.org/wp-content/uploads/2023/07/ASEAN_employment_outlook_WEB_FIN.pdf

[84] ASEAN (2023) Guidance Document of the ASEAN Declaration on Promoting Competitiveness, Resilience, and Agility of Workers for the Future of Work, https://asean.org/wp-content/uploads/2023/10/Guidance-Document-ASEAN-Declaration-on-Promoting-Competitiveness-Resilience-and-Agility-of-Workers-for-the-Future-of-Work_Final-1.pdf

[85] UNCTAD (2023) Global Cooperation in Science, Technology, and Innovation for Development, https://unctad.org/system/files/information-document/CSTD2023-2024_Issues02_globalcooperation_en.pdf

mobility of researchers and science and technology personnel, thus integrating digitalization and innovation beyond borders and sectors.[86]

Indicator 4.4: The score of 13.55 is promising, indicating that despite the low proportion of employment in knowledge-intensive services flagged earlier, a significant portion of the region's active population does possess digital skills. This is likely due to various initiatives that have been launched to improve access to digital skills in recent years, each with a unique focus and approach. The *ASEAN Digital Innovation Programme*, spearheaded by Microsoft, aims to empower digital innovation and entrepreneurship, providing resources and support to foster a robust digital ecosystem.[87] Likewise, Huawei's *ASEAN Seeds for the Future* program focuses on nurturing ICT talent, offering training and developmental opportunities to students and young professionals in cutting-edge technologies.[88] More internationally, the World Economic Forum's *ASEAN Digital Skills Vision 2020* and UNICEF's *Conference on 21st Century Skills and Youth Participation* aim to close the digital skills gap and prepare the ASEAN workforce for the future, with a focus on inclusive and sustainable access to the digital skills of tomorrow.[89]

The modest increase from 13.11 in ADII 1.0 to 13.55 in ADII 2.0 reflects a positive—but lackluster—trend in digital literacy. This is likely a result of concerted efforts that are hindered by one major obstacle: the pervasive digital divide that still characterizes the region, as major gaps remain between rural/urban, connected/unconnected, and banked/unbanked populations in ASEAN.[90] Indeed, poor ICT infrastructure, limited connectivity, and uneven ICT investment limit the scope and scale of many digital policies in the region. Strong infrastructural and institutional foundations are a fundamental prerequisite for both people and organizations to access and use the transformative potential of digital technologies, a first step towards contributing to the wider digital economy. It is important to note that enhancing digital skills and talent was already a major barrier noted in ADII 1.0, and the fact that Pillar 4 remains the region's weakest pillar suggests that access to tertiary education, skill development, lifelong training, and knowledge transfers remains uneven within and across AMS—despite a number of national and regional initiatives aimed at turning things around.[91]

Indicator 4.5: The high score of 13.76 suggests that graduates in the ASEAN region are perceived to possess skillsets relevant to business needs, making them competitive and sought-after in a highly globalized and rapidly evolving digital economy. Looking at the rapidly evolving field of AI, time, effort, and resources are being devoted to growing countries' ability to provide lifelong skilling, reskilling, and upskilling initiatives to ensure the workforce can keep up with rapidly evolving AI trends. At the same time, schools and universities are introducing a wide range of data analytics and AI-focused courses to equip the upcoming workforce with the necessary skills to embrace AI technologies effectively. For example, the Monetary Authority of Singapore (MAS) has launched a scheme to overcome one of the region's biggest hurdles regarding AI training: the lack of skilled instructors. The *Financial Sector Artificial Intelligence and Data Analytics (AIDA) Talent Development Programme* aims to increase the

[86] UNCTAD (2023) Global Cooperation in Science, Technology, and Innovation for Development, https://unctad.org/system/files/information-document/CSTD2023-2024_Issues02_globalcooperation_en.pdf
[87] ASEAN Foundation (2021) ASEAN Digital Innovation Programme, www.aseanfoundation.org/asean_digital_innovation_programme
[88] Huawei (2022) Fostering Digital Talent to Achieve the Goals of the ASEAN Digital Masterplan 2025, https://e.huawei.com/pl/ict-insights/global/ict_insights/ict33-digital-city/vision/fostering-digital-talent-to-achieve
[89] World Economic Forum (2022) Digital ASEAN, www.weforum.org/projects/digital-asean; and UNICEF (2018) ASEAN-UNICEF Conference on 21st Century Skills and Youth Participation, www.unicef.org/eap/media/3496/file/ASEAN-UNICEF%20Conference%20on%2021st%20Century%20Skills%20and%20Youth%20Participation.pdf
[90] Tech For Good Institute (2022) Driving Digital Literacy in Southeast Asia, https://techforgoodinstitute.org/blog/articles/driving-digital-literacy-in-southeast-asia
[91] East Asia Forum (2022) Addressing the digital divide in ASEAN, www.eastasiaforum.org/2022/07/01/addressing-the-digital-divide-in-asean

number of skilled workers who can develop and innovate AI solutions for the financial industry.[92] It does so by working with financial institutions, established training providers, and educational institutions to co-curate training programs and modules which incorporate the latest developments and trends in AI.

The improvement from 12.67 in ADII 1.0 to 13.76 in ADII 2.0 is notable, and could be attributed to educational reforms and partnerships with industry designed to make curricula more responsive to labor market needs. For example, the *ASEAN Qualifications Reference Framework (AQRF)* and the *ASEAN Work Plan on Education 2016-2020* are initiatives aimed at enhancing the quality and relevance of higher education. This change may also reflect the multiplication of opportunities for private-sector players to provide input on skills that are needed. For example, Amazon Web Services (AWS) has several programs in Indonesia, Singapore, Thailand, and Vietnam.[93]

It is worth noting that Pillar 4 should greatly benefit from the finalization of the *ASEAN Digital Economy Framework Agreement (DEFA)*, as the ninth priority area (Talent Mobility and Cooperation) aims to facilitate digital talent mobility between countries and close collaboration on talent building.[94]

[92] FinExtra (2023) MAS launches AI and data analytics talent programme, www.finextra.com/pressarticle/96920/mas-launches-ai-and-data-analytics-talent-programme
[93] Viet Nam Investment Review (2022) ASEAN countries rush to address digital skills gap, https://vir.com.vn/asean-countries-rush-to-address-digital-skills-gap-97170.html
[94] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 5 – INNOVATION & ENTREPRENEURSHIP

Creating conducive environments and ecosystems to foster innovation and entrepreneurship

Digital integration requires a conducive business environment in which budding digital enterprises can grow into a dynamic, innovative business ecosystem. In a fast-transforming digital world, competitiveness is no longer just about attracting talent and investment, it is also about driving, fostering, and supporting innovation.

From start-ups to government bodies, organizations must be able to think critically and solve problems creatively. Together, these abilities contribute to a country's overall resilience, adaptability, and competitiveness. This includes assisting growing digital enterprises as they navigate the business ecosystem and contribute to digital integration—from the ease of starting a business to complying with digital regulations.

In this context, Pillar 5 focuses on evaluating how well the environment in ASEAN promotes innovation and entrepreneurship—the availability of venture capital, the proportion of GDP used on R&D, how easy or challenging it is for innovative companies to grow, how easy it is to start a business including the procedures required to start up and formally operate an industrial or commercial business, time and costs to complete these procedures, and the paid-in minimum capital requirement, and how intellectual property creation is encouraged and protection is facilitated via a conducive IP ecosystem in the region.

Figure 7: Pillar 5 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0

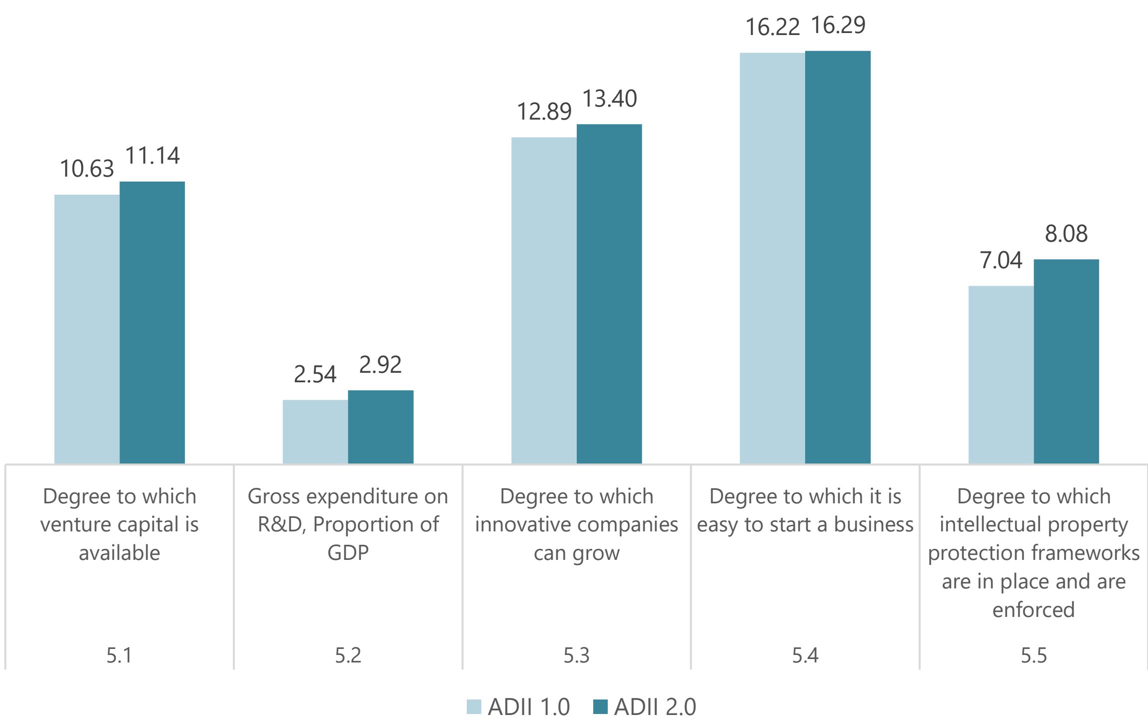

Figure 7 shows that ASEAN has made tangible improvements in creating an enabling environment for digital innovation and entrepreneurship between ADII 1.0 and ADII 2.0. The availability of venture capital (Indicator 5.1), R&D expenditure (Indicator 5.2), the degree to which innovative companies can grow (Indicator 5.3), and the degree to which intellectual property (IP) is protected (Indicator 5.5) have all increased, revealing the efficacy of targeted policies aimed at improving the investment climate. Meanwhile, the ease of starting a business (Indicator 5.4) has barely budged from its top score since ADII 1.0, supporting the idea that AMS are generally highly business-friendly destinations. The improvements made across all five Indicators in Pillar 5 suggest that AMS have made concerted efforts to foster a more innovative and entrepreneur-friendly environment by adopting international standards, encouraging R&D, and supporting the growth of innovative enterprises.

Indicator 5.1: A score of 11.14 indicates a relatively conducive environment for start-ups seeking funding in the ASEAN region. According to recent estimates, the region's growth potential is indeed impressive; Forbes predicts that by 2025, Southeast Asia's technology start-ups could reach a valuation of USD 1 trillion by 2025, trebling from USD 340 billion in 2020.[95] This could be attributed to policy initiatives aimed at strengthening the start-up ecosystem, such as Singapore's enhancement of its *Start-up SG Equity* scheme, which aims to stimulate private-sector investments into innovative, Singapore-based technology start-ups.[96] Such policy measures may have encouraged more venture capital (VC) firms to invest in the region, thereby improving the availability of capital for start-ups.

Looking at the progress made over time, the modest increase from 10.63 in ADII 1.0 to 11.14 in ADII 2.0 suggests an encouraging—but precarious—upward trend. Indeed, this growing confidence among investors may be short-lived; investment in Southeast Asia is experiencing a slowdown, with VC funding dipping by 58.6% in the second quarter of 2023; as a consequence, start-ups in the region raised USD 2.13 billion, much lower than the USD 5.13 billion of 2022.[97] Economists have also expressed concerns over macro-economic factors—namely, potentially lower-than-expected growth in Southeast Asian economies, as well as higher interest rates triggered by escalating inflation.[98] To mitigate these risks, ASEAN policymakers may want to focus on creating more favorable conditions for venture capitalists through tax incentives, co-investment funds, or easing regulations to stimulate an increase in VC activity. There is also an opportunity to attract more international (non-ASEAN) VC firms to engage in the ASEAN market.

Indicator 5.2: The score of 2.92 on R&D expenditure as a proportion of GDP denotes a very low investment in R&D relative to the region's economic output. This is especially salient when compared to the Indicator scores of leading digital economies: from a low of 5.83 in New Zealand to a high of 20.42 in South Korea. This may indicate that AMS prioritize other areas of economic development or face constraints in devoting resources to R&D (or both). To effectively turn innovation-fostering aspirations into action, ASEAN policymakers might consider increasing funding for R&D, providing grants and incentives for research institutions, and encouraging private-sector investment to enhance technological capabilities and innovation.

[95] Business Times (2023) Why South-east Asia still holds much promise for start-ups and venture capital, www.businesstimes.com.sg/international/asean/why-south-east-asia-still-holds-much-promise-startups-and-venture-capital
[96] Startup SG (2023) About Startup SG Equity, www.startupsg.gov.sg/programmes/4895/startup-sg-equity?utm_source=openinnovationnetwork.sg&utm_medium=referral
[97] Business Times (2023) Why South-east Asia still holds much promise for start-ups and venture capital, www.businesstimes.com.sg/international/asean/why-south-east-asia-still-holds-much-promise-startups-and-venture-capital
[98] ASEAN+3 Macroeconomic Research Office (2023) AMRO Maintains ASEAN+3's 2024 Growth Forecast, Warns on Inflation, https://amro-asia.org/amro-maintains-asean3s-2024-growth-forecast-warns-on-inflation

The very moderate rise from 2.54 in ADII 1.0 to 2.92 in ADII 2.0 indicates an encouraging improvement that may reflect an increasing recognition of the value of innovation as a driver of economic growth and productivity.[99] Indeed, policies such as Thailand's *20-Year National Strategy (2017-2036)* and the Philippines' *Science for Change Program* were designed to drive innovation and R&D as a pillar of economic development. But the progress made remains insufficient to allow ASEAN as a whole to stand out as a leading regional/global innovation hub. According to a recent International Monetary FUND (IMF) report, turning things around will require focusing on specific differentiating strengths: "(…) for emerging and developing countries in Asia with widely varying institutional, technological, and firm-level capacities, innovation entails not only the invention of new products and processes but also the diffusion and adoption of existing technologies or practices."[100] This, concludes the report, will allow AMS to close the gap between themselves (emerging digital economies converging towards the global technological frontier) and those on the leading edge (mature digital economies whose purposeful R&D can result in the invention of completely new products and processes).[101]

Indicator 5.3: A notable score of 13.40 in this area suggests that ASEAN provides a supportive ecosystem for the scaling of innovative companies—implying that there are mechanisms and policies in place that encourage the expansion of such businesses through supportive regulations, access to larger markets, and an appetite for technological innovation. The region has indeed seen its fair share of 'Unicorns' (privately held start-up companies with a value of over USD 1 billion)[102] in the last decade, and these continue to be openly pursued by AMS. As researchers point out: "Unicorns contribute towards employment generation, innovation, productivity gains, and cross-border trade for the country that is hosting them. They also draw investments due to their high-growth potential and expected future gains in the event of a buy-out."[103] It is no surprise then, that Indonesia aims to foster 20 Unicorns as big as Gojek and Tokopedia by 2025, that Malaysia aims to have five unicorns in 2025, and that Viet Nam targets to have five tech unicorns by the same year and another five by 2030.[104]

These ambitions may very well become a reality considering that the Indicator score has risen from 12.89 in ADII 1.0 to 13.40 in ADII 20. This increase suggests that ASEAN has effectively become more conducive to the scaling of innovative companies since ADII 1.0, a change that may be linked to regional efforts such as the *ASEAN Economic Community Blueprint 2025*[105] and the *ASEAN Digital Master Plan 2025*,[106] which aim to better integrate economies and facilitate market access—thereby increasing opportunities for sustained growth. Specifically on tech start-ups, the *ASEAN Guidelines on Fostering a Vibrant Ecosystem for Start-Ups Across Southeast Asia*[107] and the *Framework for Promoting the Growth of Digital Start-ups in ASEAN*[108] set out the economic conditions and the policy requirements to develop an enabling ecosystem for digital start-ups in ASEAN.

[99] Journal of the Knowledge Economy (2023) R&D Spending and Economic Policy Uncertainty in Asian Countries, https://link.springer.com/article/10.1007/s13132-023-01285-x
[100] International Monetary Fund (2023) Accelerating Innovation and Digitalization in Asia to Boost Productivity, www.elibrary.imf.org/view/journals/087/2023/001/article-A001-en.xml
[101] International Monetary Fund (2023) Accelerating Innovation and Digitalization in Asia to Boost Productivity, www.elibrary.imf.org/view/journals/087/2023/001/article-A001-en.xml
[102] FinFan (2023) Full list of Southeast Asia's 23 unicorns, https://finfan.vn/News/here-s-the-full-list-of-southeast-asia-s-23-unicorns-2461
[103] Fulcrum (2023) Rise of Unicorns in Southeast Asia, https://fulcrum.sg/aseanfocus/rise-of-unicorns-in-southeast-asia
[104] Fulcrum (2023) Rise of Unicorns in Southeast Asia, https://fulcrum.sg/aseanfocus/rise-of-unicorns-in-southeast-asia
[105] ASEAN (2015) ASEAN Economic Community Blueprint 2025, https://asean.org/book/asean-economic-community-blueprint-2025
[106] ASEAN (2021) ASEAN Digital Master Plan 2025, https://asean.org/book/asean-digital-masterplan-2025
[107] ASEAN (2020) ASEAN Guidelines on Fostering a Vibrant Ecosystem for Start-Ups Across Southeast Asia, https://asean.org/wp-content/uploads/2021/08/ASEAN-Guidelines-on-Fostering-a-Vibrant-Ecosystem-for-Startups-across-Southeast-Asia.pdf
[108] ASEAN (2023) Framework for Promoting the Growth of Digital Start-ups in ASEAN, https://asean.org/wp-content/uploads/2023/03/1.Framework-for-Promoting-the-Growth-of-Digital-Startups-in-ASEAN-1.pdf

Indicator 5.4: At 16.29, this Indicator scores the highest out of the five Indicators under Pillar 5. This high score indicates that ASEAN is a resolutely business-friendly region that makes it relatively quick and easy to start a business (simplified and streamlined business-registration procedures, reduced red tape for various first-time processes, and availability of supportive services and incentivizing measures for entrepreneurs). Research shows that lowering barriers to entry for new businesses fosters a dynamic entrepreneurial ecosystem,[109] which in turn drives the emergence of innovative business models[110]—an objective that AMS is operationalizing through initiatives such as the *ASEAN Work Programme on Starting a Business,* which aims to make business-registration processes more interoperable across the region to ensure smooth and predictable experiences for business owners.[111]

The slight improvement from 16.22 in ADII 1.0 to 16.29 in ADII 2.0 is barely noticeable but indicates sustained efforts in keeping it practical and convenient to start a business and access markets in the region. Two key initiatives that are keeping ASEAN attractive for business owners looking to grow and prosper are: regulatory sandboxes that allow start-ups to fine-tune their potentially risky digital products and services in a controlled environment,[112] and the *Unique Business Identification (UBIN)* system, which allows micro, small, and medium-sized enterprises (MSMEs) to make and receive seamless digital payments within ASEAN.[113] Already launched in Indonesia, Malaysia, the Philippines, Singapore, and Thailand, regulatory sandboxes have successfully helped new entrants in the Fintech sector test their innovative products while giving regulators a chance to develop appropriate rules that manage risk while not stifling innovation.[114] The most recent development in this area was the publication of an OECD paper advocating for AI sandboxes,[115] an initiative that Singapore launched only a few months afterwards.[116] Meanwhile, UBIN is still in the conceptualization stage,[117] but it is poised to streamline cross-border business transactions once implemented.[118]

Indicator 5.5: With an Indicator score of 8.08, it is apparent that while there are measures in place to protect intellectual property (IP) in ASEAN, their enforcement may not be as robust as it could be across all AMS. Research shows that securing the rights of innovators and creators is crucial for fostering an environment where innovation can thrive without the fear of infringement.[119] To this end, ASEAN policymakers may need to strengthen the enforcement of national IP protection measures, develop mechanisms to ensure IP protection is compatible within and across ASEAN, and cooperate internationally to gain access to international best practices and standards. To wit, the *ASEAN Intellectual Property Rights Action Plan 2016-2025* (updated in June 2021) recognizes the difficulties of implementing IP protection measures nationally, trans-nationally, and digitally—and provides a set of deliverables for AMS to further collaborate on: strengthening the financial management of national IP

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[109] Small Business Economics (2020) The influence of ecosystems on the entrepreneurship process: a comparison across developed and developing economies, https://link.springer.com/article/10.1007/s11187-020-00392-2
[110] Journal of Business Venturing Insights (2020) A dynamic analysis of the role of entrepreneurial ecosystems in reducing innovation obstacles for start-ups, www.sciencedirect.com/science/article/abs/pii/S2352673420300482
[111] ASEAN (2020) ASEAN Work Programme on Starting a Business, https://asean.org/wp-content/uploads/2020/12/Adopted-Report-and-Work-Programme-on-Starting-a-Business-in-ASEAN.pdf
[112] The ASEAN Post (2018) Fintech sandboxes in Southeast Asia, https://theaseanpost.com/article/fintech-sandboxes-southeast-asia
[113] Tech For Good Institute (2022) Promoting the Growth of the Digital Start-up Ecosystem in ASEAN, https://techforgoodinstitute.org/events/past-events/promoting-the-growth-of-the-digital-startup-ecosystem-in-asean
[114] APEC (2021) FinTech Regulatory Sandboxes: Capacity Building Summary Report, www.apec.org/docs/default-source/publications/2021/3/FinTech-regulatory-sandboxes-capacity-building-summary-report/221_ec_FinTech-regulatory-sandboxes-capacity-building-summary-report.pdf?sfvrsn=387b04bc_1
[115] OECD (2023) Regulatory sandboxes in artificial intelligence, www.oecd.org/publications/regulatory-sandboxes-in-artificial-intelligence-8f80a0e6-en.htm
[116] IMDA (2023) First of its kind Generative AI Evaluation Sandbox for Trusted AI by AI Verify Foundation and IMDA, www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2023/generative-ai-evaluation-sandbox
[117] Australian Aid (2022) TENDER ANNOUNCEMENT: Benchmarking Guidelines on Unique Business Identification Numbers in ASEAN Member States – Phase I, http://aadcp2.org/tender-announcement-benchmarking-guidelines-on-unique-business-identification-numbers-in-asean-member-states-phase-i
[118] Tech For Good Institute (2022) Promoting the Growth of the Digital Startup Ecosystem in ASEAN, https://techforgoodinstitute.org/events/past-events/promoting-the-growth-of-the-digital-startup-ecosystem-in-asean
[119] Economic Modelling (2021) The link between intellectual property rights, innovation, and growth: A meta-analysis, www.sciencedirect.com/science/article/abs/pii/S0264999321000274

offices, capacity-building for copyright offices, implementing an *ASEAN Copyright Information Network*, and developing mechanisms to detect and fight against IP infringement online.[120]

The small increase observed from 7.04 in ADII 1.0 to 8.08 in ADII 2.0 points to improvements in intellectual property protection in the ASEAN region, which may be attributed to the concerted efforts by ASEAN governments to comply with international IP standards, enhance enforcement mechanisms, and raise public awareness about IP rights. For instance, Viet Nam's amendments to the *Intellectual Property Law* in 2022 were designed to align with the *Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)*, thus strengthening IP protection measures and aligning them with more stringent and systemized approaches.[121] A rapidly evolving issue that may soon become a thorn in digital economies' side is the IP implications of the massive usage of AI-generated content;[122] already a major point of contention in the European Union's seminal *AI Act* negotiations,[123] the topic is currently being tabled in light of the preparatory work for the draft *ASEAN Guide on AI Governance and Ethics*[124] and the draft *ASEAN Guidelines on the Responsible Use of Generative AI*.[125]

Here too, Pillar 5 is expected to benefit from the finalization of the *ASEAN Digital Economy Framework Agreement (DEFA)*. Indeed, the eighth (Cooperation on Emerging Topics) of its nine priority areas covers the establishment of mechanisms for regulatory cooperation for relevant standards and regulations to keep up with technological innovations in emerging topics, including AI.[126]

[120] ASEAN (2021) ASEAN Intellectual Property Rights Action Plan 2016-2025, www.aseanip.org/docs/default-source/content/ASEAN-IPR-Action-Plan-2016-2025-v2.0.pdf
[121] ASEAN Briefing (2022) Vietnam's Intellectual Property Law: New Amendments, www.aseanbriefing.com/news/vietnams-intellectual-property-law-new-amendments
[122] Forbes (2023) Navigating The Generative AI Intellectual Property Landscape, www.forbes.com/sites/forbestechcouncil/2023/10/10/navigating-the-generative-ai-intellectual-property-landscape/?sh=104377e251cf
[123] World Trademark Review (2023) Amended EU AI Act strengthens copyright enforcement against Generative AI but IP issues remain, www.worldtrademarkreview.com/article/amended-eu-ai-act-strengthens-copyright-enforcement-against-generative-ai-ip-issues-remain
[124] Access Partnership (2023) AI Governance in Southeast Asia: Paving a Pragmatic Path, https://accesspartnership.com/ai-governance-in-southeast-asia-paving-a-pragmatic-path
[125] Work-in-progress; unpublished research shared by the AI Asia Pacific Institute (AIAPI) in November 2023.
[126] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

ADII PILLAR 6 – INSTITUTIONAL & INFRASTRUCTURAL READINESS

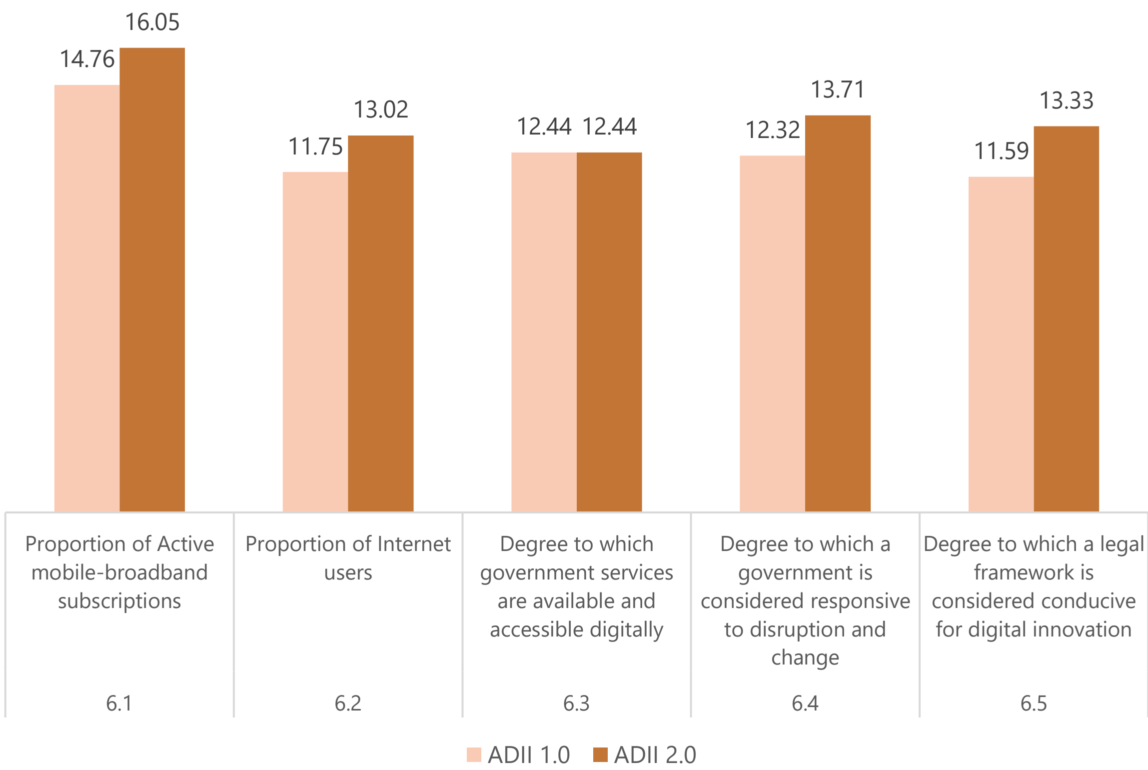
Readying to drive and coordinate actions for digital integration

Digital integration requires AMS to be digitally ready—both in terms of availability of digital infrastructure and in terms of adoption of technology across the public sector (regulators, policymakers, institutions). Indeed, digitally enabled economies are likelier to have the operational and organizational ability to drive and sustain digital transformation at all levels of society.

Together, these elements are indicative of AMS' capacity to effectively enable the coordination of public- and private-sector initiatives, as well as their openness to driving cooperation across regional/global initiatives. Quantifying this ability entails observing the quality of digital infrastructure and the digital readiness of institutions.

As such, Pillar 6 assesses the availability of digital infrastructure and the adoption of technology across the general population and within public sector institutions to drive, coordinate, and sustain digital integration in ASEAN. It looks at the proportion of active mobile-broadband subscribers and Internet users across ASEAN, as well as the availability and accessibility of digital government services. It also evaluates ASEAN governments' responsiveness to disruption and change, and whether legal frameworks effectively enable digital innovation.

Figure 8: Pillar 6 – ASEAN Indicator Scores for ADII 1.0 & ADII 2.0



As Figure 8 illustrates, ASEAN has made notable progress across all Pillar 6 Indicators, save for one that has not moved since ADII 1.0. This encouraging trajectory attests to ASEAN's various commitments to putting digitalization at the top of its economic agenda, though focused policy efforts are still required to fully capitalize on the benefits of a fully digitalized society. Overall, ASEAN has improved the most in terms of digital readiness, responsiveness, and governance (Indicators 6.4 and 6.5), followed by ICT access and connectivity (Indicators 6.1 and 6.2). While these areas have seen significant progress, others, such as the provision and diffusion of digital government services (Indicator 6.3), may require further policy intervention to ensure sustained improvement over time.

Indicator 6.1: With an ADII 2.0 score of 16.05, this is the highest-scoring Indicator under Pillar 6. According to recent research by GSMA, mobile broadband networks are the primary channel to get online for 96% of people in the Asia-Pacific, with operators looking to attain near-complete coverage by spending almost USD 260 billion on their networks by 2030.[127] The picture is a bit more nuanced when zooming into ASEAN: while mobile penetration is high in most ASEAN countries, the region suffers from a significant disparity in terms of connectivity. Indeed, the rural/urban divide persists despite the many efforts that have gone into making broadband connections more accessible; in 2022, only 53% of rural children and adolescents had an internet connection at home, as opposed to 72% of urban kids.[128] From poor or unreliable ICT infrastructure to costly digital devices and broadband subscriptions, there are many reasons for the pervasiveness of the digital divide even amid an encouraging upward trend towards higher Internet usage in ASEAN.[129]

The Increase from 14.76 in ADII 1.0 to 16.05 in ADII 2.0 suggests that there has been a policy-driven effort in the ASEAN region to expand mobile-broadband infrastructure and to make mobile internet more affordable and accessible. Numerous ASEAN-wide initiatives have indeed been launched in this regard: the *Strategic Plan for Information and Media 2016-2025*,[130] the *ASEAN Digital Masterplan 2025*,[131] the *ASEAN Communication Master Plan 2018-2025*,[132] and a wide range of sector-specific initiatives. Most recently, Indonesia[133] and Malaysia[134] have partnered with SpaceX, a satellite-Internet provider, to circumvent the limitations of broadband Internet access and boost Internet connectivity in rural and isolated regions.

Indicator 6.2: At 13.02, the score for Indicator 6.2 follows the same general trend as Indicator 6.1; that of an overall encouraging upward trajectory, but with a number of barriers that hinder sustained and consistent improvements. Apart from accessibility and affordability, a key challenge for the growth of Internet users in ASEAN is Internet quality. Indeed, even when the Internet is readily accessible, it is not necessarily available in a reliable or continuous manner; in many parts of ASEAN, its quality and speed is too poor to allow for Internet-based economic activities to be developed sustainably.[135] According to recent data, Singapore has the highest fixed broadband speeds (globally and regionally), followed by

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[127] GSMA (2023) Digital Societies in Asia Pacific: Harnessing emerging technologies to advance digital nations, www.gsma.com/asia-pacific/resources/apac-digitalsocieties-2023
[128] ASEAN (2021) ASEAN Rapid Assessment: The Impact of COVID-19 on Livelihoods across ASEAN, https://asean.org/book/asean-rapid-assessment-the-impact-of-covid-19-on-livelihoods-across-asean
[129] ASEAN (2022) ASEAN Revs Up: digital transformation, https://asean.org/serial/the-asean-volume-november-2022-number-23
[130] ASEAN (2016) Strategic Plan for Information and Media 2016-2025, https://asean.org/wp-content/uploads/2021/08/14.-May-2016-ASEAN-Strategic-Plan-for-Information-and-Media-2016-20251.pdf
[131] ASEAN (2022) ASEAN Digital Masterplan 2025, https://asean.org/book/asean-digital-masterplan-2025
[132] ASEAN (2021) ASEAN Communication Master Plan 2018-2025, https://asean.org/wp-content/uploads/2021/03/ASEAN-Communication-Master-Plan-2018-2025.pdf
[133] The Business Times (2023) Indonesia, SpaceX launch satellite to boost Internet connectivity, www.businesstimes.com.sg/international/indonesia-spacex-launch-satellite-boost-internet-connectivity
[134] SCMP (2023) Malaysia issues license to Elon Musk's Starlink to bring internet to remote areas, www.scmp.com/news/asia/southeast-asia/article/3228362/malaysia-issues-license-elon-musks-starlink-bring-internet-services-remote-areas
[135] The ASEAN Post (2018) ASEAN's poor mobile internet connectivity, https://theaseanpost.com/article/aseans-poor-mobile-internet-connectivity

Thailand and Malaysia (respectively 6th and 37th globally). At the other end of the spectrum are Myanmar (138th), Cambodia (130th), and Indonesia (124th).[136] When it comes to mobile broadband, Singapore is 19th globally, followed by Brunei Darussalam (23rd) and Malaysia (38th).[137]

The growth observed from 11.75 in ADII 1.0 to 13.02 in ADII 2.0 indicates a successful push towards greater internet penetration within and across AMS. This could reflect the impact of the many connectivity plans and roadmaps cited above, as they have focused on developing ICT infrastructure, Internet accessibility, digital literacy and skills, and overall digital inclusion. On this last point, it is worth noting that Internet adoption and usage has a demonstrated impact on the opportunities that both individuals and organizations can seize in rapidly digitalizing economies—which in turn has a trickle-down effect on innovation and entrepreneurship.[138] Indeed, entrepreneurship in ASEAN faces a set of barriers that make it difficult (or at least uneven) to access the different elements that spur and sustain digital innovation—capital, skills, knowledge, tools, infrastructure, government support. For example, access to business financing is a significant obstacle, particularly for youth, women, and rural populations who tend to struggle to get a business going.[139] AMS have been taking steps to enhance digital inclusion, particularly in terms of alternative credit scoring, digital banking, crowd- and peer-to-peer (P2P) lending, and digital financial literacy platforms.[140] It must be highlighted, however, that none of the solutions are possible if basic, reliable connectivity is not made available.

Indicator 6.3: With a score of 12.44, this Indicator is the lowest-scoring Indicator within Pillar 6, suggesting that there is room for improvement in digitalizing government services, as well as in improving the interface and delivery of digital services. According to the United Nations' *E-Government Survey 2022*, all AMS have made steady improvement since the 2020 edition of the study, demonstrating that the national digital government strategies they have developed has allowed them to provide a wide range of digital government services.[141] In Singapore, for example, the Singapore International Commercial Court (SICC) uses digital technologies such as electronic filing service platforms and advanced litigation technology (evidence and trial management systems) to streamline and optimize the work of international judges.[142] In Malaysia, a number of government-focused digital initiatives are in place: a data exchange hub for ministries to share information and better understand citizens,[143] a platform that centralizes all necessary business registration information and streamlines the attribution of over 1,500 licenses for SMEs, [144] and the provision of services that anticipate citizens' needs in business, education, health, and welfare.[145]

The Indicator score of 12.44 did not change between ADII 1.0 and ADII 2.0, suggesting that while initial efforts to digitize have been implemented, there is a need for continued investment and innovation to further enhance the quality and reach of these services. In this context, there are two main areas that may get thornier for AMS as they gain in digital sophistication: First, the need for greater coordination

[136] Ookla (2023) Median Country Speeds – October 2023, www.speedtest.net/global-index
[137] Ookla (2023) Median Country Speeds – October 2023, www.speedtest.net/global-index
[138] International Monetary Fund (2020) Low Internet Access Driving Inequality, www.imf.org/en/Blogs/Articles/2020/06/29/low-internet-access-driving-inequality
[139] ADB (2022) Realizing the Potential of Over 71 Million MSMEs in Southeast Asia, https://seads.adb.org/solutions/realizing-potential-over-71-million-msmes-southeast-asia
[140] Roland Berger (2021) Bridging the digital divide: Improving digital inclusion in Southeast Asia, www.rolandberger.com/publications/publication_pdf/roland_berger_sea_digital_inclusion.pdf
[141] United Nations (2022) E-Government Survey 2022, https://publicadministration.un.org/egovkb/en-us/Reports/UN-E-Government-Survey-2022
[142] Singapore International Commercial Court (SICC), www.sicc.gov.sg/forms-and-services/use-of-technology-at-the-sicc
[143] GovInsider (2016) Dr Suhazimah Dzazali, Government Chief Information Officer, Malaysia, https://govinsider.asia/intl-en/article/dr-suhazimah-dzazali-government-chief-information-officer-malaysia
[144] GovInsider (2017) Will Malaysia launch a Government Digital Service?, https://govinsider.asia/digital-gov/will-malaysia-launch-a-government-digital-service
[145] GovInsider (2017) Exclusive: Malaysia's Gov Tech plans for 2017, https://govinsider.asia/digital-gov/exclusive-malaysias-gov-tech-plans-for-2017

and collaboration across government bodies and agencies to ensure digital government services are provided in a consistent, continuous, and coherent manner.[146] Second, the need to adopt a more citizen-centric approach to public-service delivery, as well as the need to involve citizens in the policymaking and service-design process; if digital government services do not meet people's on-the-ground needs and priorities, or are not user-friendly and inclusive, then they cannot be considered successful—no matter how advanced or dematerialized they may be.[147]

Indicator 6.4: At 13.71, this Indicator has the second-highest score in Pillar 6, indicating that ASEAN governments are perceived as somewhat agile, adaptable, and responsive to the various disruptions created by the rapid digitalization of society and the economy. Several AMS have indeed risen to the challenge in the past several years, successfully navigating the unprecedented wave of digital advancements spurred by the COVID-19 pandemic,[148] and they are now taking the lead in regard to the AI governance.[149] Indeed, ASEAN has understood the challenges posed by the rise of AI, as well as the urgency of addressing its emerging challenges through effective governance and regulatory frameworks. This is evidenced by the fact that responsible AI deployment and adoption is slowly—but surely—making its way to the top of national and regional agendas in ASEAN, despite AMS having largely different priorities and approaches to AI.[150]

The improvement observed from 12.32 in ADII 1.0 to 13.71 in ADII 2.0 could be the fruit of ASEAN countries having taken concrete steps to make their policymaking processes more agile and responsive. In this sense, the way AMS leveraged digital technologies to navigate the worst of the COVID-19 pandemic was a turning point for the region's burgeoning digital economies. In Singapore, the ubiquity of smartphones made the use of mobile contact-tracing applications an effective means to control the impact of the COVID-19 pandemic,[151] while the *VigilantGantry* contactless system developed by GovTech's Data Science and Artificial Intelligence Division automated temperature-screening processes in high-footfall areas.[152] In Brunei Darussalam, the government integrated AI into its *BruHealth* application to assist with outbreak predicting and planning, enabling the management of policies and resources to control the spread of the virus.[153] In Thailand, the overburdened public healthcare system was able to navigate the worst of the COVID-19 crisis thanks to AI-driven solutions that simplified processes and optimized resource allocation.[154]

Indicator 6.5: The Indicator score of 13.33 suggests that ASEAN has generally established strong legal foundations that drive and support digital innovation; from intellectual property laws to regulatory sandboxes, the region is home to a number of tools and mechanisms that have allowed digital business models to grow and thrive. Initiatives such as the *ASEAN Data Management Framework* are worth highlighting, as they contribute to better supporting digital innovation by creating conducive environments for digital markets, protecting data privacy, and ensuring data can flow between

[146] Institute of Public Administration (2003) E-Government and the Decentralisation of Service Delivery, https://core.ac.uk/download/pdf/71728193.pdf
[147] EY (2023) How can digital government connect citizens without leaving the disconnected behind?, www.ey.com/en_bh/government-public-sector/how-can-digital-government-connect-citizens-without-leaving-the-disconnected-behind
[148] Nikkei Asia (2021) COVID's striking impact on Southeast Asia's digital economy, https://asia.nikkei.com/Opinion/COVID-s-striking-impact-on-Southeast-Asia-s-digital-economy
[149] Access Partnership (2023) Artificial Intelligence: Policies and Priorities for Southeast Asia, https://accesspartnership.com/artificialintelligence-policies-and-priorities-for-southeast-asia
[150] Access Partnership (2023) Responsible AI in Southeast Asia: Tracking Progress and Benchmarking Impact, https://accesspartnership.com/ai-in-sea
[151] BMC Public Health (2023) Use of a digital contact tracing system in Singapore to mitigate COVID-19 spread, https://pubmed.ncbi.nlm.nih.gov/37974135
[152] GovTech Singapore (2020) How GovTech developed fit for purpose temperature scanners – Part 2, www.tech.gov.sg/media/technews/how-gt-developed-fit-for-purpose-temperature-scanners-part-2
[153] Ministry of Health (2020) About BruhHealth, www.moh.gov.bn/SitePages/bruhealth.aspx
[154] YCP Solidiance (2021) How AI Technology is Helping Thailand's Fight Against COVID-19, https://ycpsolidiance.com/article/how-ai-technology-is-helping-thailands-fight-against-covid-19nbsp

countries.[155] On the flow of data across borders, the topic stands out as an enabling area that is particularly strong in the region but that could still benefit from further adjustments. Cross-border data flows are an essential part of the digital economy, considering the central role that data plays in powering all manner of digital products, platforms, services, and devices. From mobile wallets and QR codes to e-government services and automated loan applications, all the emerging digital innovations that have become essential to our daily lives need data to ensure they remain relevant and reliable. In this context, the free and secure movement of data across borders and jurisdictions is central to the growth and differentiation of digital economies. According to a recent report on G20 economies' openness to cross-border data flows, Singapore stands out with a conducive policy and regulatory environment that effectively enables data-driven innovation and improvement—an approach that other digital economies are using as a reference to enable free and secure cross-border data flows.[156] While it does not score as high as other digital economies, the report notes that Indonesia has made some significant progress between 2021 and 2023, likely due to its G20 Presidency in 2022, which saw the government support a more concerted approach to cross-border data flows and data governance, even though the country still has some data localization measures in place.[157]

The increase from 11.59 in ADII 1.0 to 13.33 in ADII 2.0 is indicative of enhancements brought to legal and regulatory frameworks across ASEAN. In this context, the *ASEAN Digital Economy Framework Agreement (DEFA)* is set to consolidate, streamline, and systemize AMS' many ongoing digitalization activities and mechanisms.[158] With its conclusion slated for 2025, it is clear that the *DEFA* aims to be a comprehensive strategy that integrates existing ASEAN digital action plans and provides a clear direction to achieve ASEAN's future digital economy ambitions—all while serving as the basis for AMS to grow their digital economies in a coordinated manner. We understand that the *DEFA* will be designed in a way that recognizes the varied digital economy development challenges facing different AMS and carefully consider the differing circumstances and underlying gaps among them. The *DEFA* will also pay special attention to the CLMV group of economies (Cambodia, Lao PDR, Myanmar, and Viet Nam). Lastly, the *DEFA* will provide evidence-based policy guidance to AMS, drawing from consultations with national authorities and primary research with private sector organizations.

[155] ASEAN (2021) ASEAN Data Management Framework, https://asean.org/wp-content/uploads/2-ASEAN-Data-Management-Framework_Final.pdf
[156] Salesforce (2023) Data Beyond Borders 3.0, www.salesforce.com/content/dam/web/en_au/www/documents/pdf/data_beyond_borders.pdf
[157] Salesforce (2023) Data Beyond Borders 3.0, www.salesforce.com/content/dam/web/en_au/www/documents/pdf/data_beyond_borders.pdf
[158] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

5 | RECOMMENDATIONS

The recommendations provided below identify areas that all AMS can implement, strengthen, or expand to sustain the digitalization momentum they are currently undergoing.

These recommendations are all equally important for ASEAN to maintain its position as a dynamic and forward-looking region that is effectively navigating the ebbs and flows of the digital age. But their intended outcomes will be most meaningful and impactful if AMS conduct their own individual comprehensive self-assessments to ensure they allocate time, effort, funds, manpower, and resources to measures and initiatives that are in line with their own specific needs, priorities, and capabilities

Table 5: ADII 2.0 Recommendations for ASEAN

Focus Areas	Recommendations	Priorities
Pillar 1: Digital Trade & Logistics	1. **Enhance Digital Infrastructure and Connectivity**: ASEAN should focus on upgrading physical infrastructure critical for digital trade. This could include investing in high-speed internet connectivity, modernizing port facilities, and improving road networks. This could be achieved by expanding the *Master Plan on ASEAN Connectivity 2025* so that it includes specific targets for digital infrastructure; such KPIs will provide a more measurable impact on trade efficiency and help bridge the digital divide among AMS. 2. **Strengthen Digital Authentication and Cybersecurity Frameworks**: ASEAN should intensify its efforts in enhancing digital certificates and signatures. This can be achieved by implementing more robust legal and regulatory frameworks that recognize and enforce digital certifications and agreements. For example, a regional digital identity system could be developed to ensure secure and seamless cross-border transactions, boosting trust in intra-regional digital trade. 3. **Adopt Standards and Facilitate Interoperability**: ASEAN should continue to align with international standards and practices to ensure interoperable systems and processes. This would include the implementation of consistent and interoperable trade documentation standards across all AMS. Building upon initiatives like the *ASEAN-Australia Digital Trade Standards Initiative*, ASEAN can further develop policies that ensure seamless integration with global trade systems, thereby reducing trade frictions and enhancing the region's global competitiveness.	• **Indicator 1.2**: Implement a unified digital identity framework across ASEAN countries to enhance secure and trustworthy cross-border digital transactions. • **Indicator 1.3:** Develop a region-wide policy mandating the adoption of standardized, interoperable digital trade documentation formats in line with global best practices. • **Indicator 1.4**: Prioritize the expansion and upgrade of digital connectivity infrastructure across member states, including high-speed internet access and modern logistics facilities.
Pillar 2: Data Protection & Cybersecurity	1. **Make Data Protection and Cybersecurity Laws Interoperable**: Despite the existence of national data protection measures, there is a need for greater regional interoperability to ensure consistency and efficiency in data protection across ASEAN. For example, aligning national laws with the *ASEAN Data Management Framework* and/or the *ASEAN Model Contractual Clauses for Cross-Border Data Flows* can help achieve stronger interoperability of data protection standards—thereby bolstering cross-border business opportunities in the region.	• **Indicator 2.1:** Implement a unified ASEAN-wide data protection regulation, inspired by the *ASEAN Data Management Framework*, to standardize data protection laws across member states. • **Indicator 2.3**: Establish an ASEAN-level program for cybersecurity capacity-building,

2. **Enhance Cybersecurity Cooperation and Integration**: There is an important gap in the convergence of national cybersecurity measures at the regional level. Leveraging the main thrusts of the *ASEAN Cybersecurity Cooperation Strategy 2021-2025*, ASEAN should prioritize the closer integration of national cybersecurity strategies. This could include developing a common framework for cybersecurity standards and practices, encouraging AMS to adopt these standards, and conducting regular regional cybersecurity exercises.

3. **Strengthen Institutional Cybersecurity Capabilities**: There is an urgent need to improve institutional cybersecurity capabilities. ASEAN should prioritize building and enhancing the capacity of governmental and organizational structures dealing with cybersecurity. This could be achieved through initiatives like the *ASEAN CERT Information Exchange Mechanism* and/or the *ASEAN-Singapore Cybersecurity Centre of Excellence*. Emphasis should be on training, staffing, and fostering inter-agency coordination to efficiently respond to the evolving cybersecurity landscape.

focusing on training, resource allocation, and inter-agency collaboration across AMS.

- **Indicator 2.4**: Establish a comprehensive program for intra-regional exchange of technical knowledge and best practices in cybersecurity, fostering collaboration and skill-sharing among AMS to bolster their collective technical cybersecurity capabilities.

Pillar 3: Digital Payments & Identities

1. **Expand Access to Digital Finance and Improve Digital Literacy**: ASEAN should focus on increasing digital finance access, particularly in AMS with high rates of unbanked and underbanked populations. This could involve implementing targeted financial literacy programs and simplifying the process for opening digital bank accounts. This would encourage more people to transition from cash to digital banking, providing them with the necessary tools and knowledge to benefit from the broader digitalization of the economy.

2. **Enhance Security and Interoperability of Digital Payments Systems**: To sustain the rising use of digital platforms for financial transactions, ASEAN should prioritize enhancing the security and interoperability of these systems, as well as developing stronger consumer protection measures to build trust in digital platforms for financial transactions. As such, current efforts on the launch of an interoperable QR code payment network should be sustained and leveraged for other facets and functions of digital finance.

3. **Strengthen Digital Identity Frameworks and Enhance Interoperability**: ASEAN should focus on developing and harmonizing digital identity frameworks across AMS.

- **Indicator 3.1:** Develop a unified digital payment protocol with robust security standards across AMS.
- **Indicator 3.2**: Implement regional initiatives to provide digital banking education and streamline online account opening processes.
- **Indicator 3.5**: Adopt a common framework for digital identity verification, integrating AI and Blockchain technologies, applicable across all ASEAN countries.

This could involve the implementation of interoperable and secure digital ID systems that are compatible with a wide range of services and devices. For instance, leveraging AI and Blockchain technologies for secure and efficient ID verification processes, AMS can enhance user experiences while ensuring data privacy and security.

Pillar 4: Digital Skills & Talent

1. **Improve Access to STEM Education with a Focus on Equity and Inclusivity**: To address the uneven landscape in STEM education and eliminate the gender gap, ASEAN should prioritize laws and policies that ensure equitable access to STEM education, particularly in underrepresented and economically challenged regions like CLMV economies. This could involve investing in digital infrastructure, subsidizing the cost of devices and internet access, and implementing targeted programs to encourage women and minorities to pursue STEM education. An example policy could be the creation of ASEAN-wide scholarships specifically for women and students from disadvantaged backgrounds in STEM fields.

2. **Strengthen Multi-Stakeholder Collaboration in R&D and Knowledge-Intensive Services**: ASEAN needs to foster renewed focus on cross-sector partnerships and joint initiatives in R&D to stimulate growth in the knowledge-intensive services sector. This could be achieved by creating incentives for private sector participation in collaborative research projects and establishing more integrated networks for knowledge-sharing and resource-pooling among AMS. An example initiative could be an ASEAN-wide R&D collaboration fund that incentivizes joint projects between universities, private companies, and government agencies.

3. **Close the Digital Skills Gap through Lifelong Learning and Upskilling Programs**: Given the persistent digital divide and the evolving demands of the digital economy, ASEAN should prioritize policies that promote lifelong learning and upskilling, especially in emerging technologies like AI and data analytics. This could include partnerships with tech companies and educational institutions to develop and deliver training programs that are aligned with current industry needs.

- **Indicator 4.1**: Implement an ASEAN-wide policy to fund and support infrastructure development in underrepresented regions, ensuring equitable access to high-quality STEM education and digital resources.

- **Indicator 4.3**: Establish an ASEAN R&D incentive program that offers tax benefits and funding to companies and institutions participating in cross-border, multi-sector research initiatives.

- **Indicator 4.4**: Launch an *ASEAN Digital Upskilling Platform*, in partnership with leading tech firms, to offer accessible, up-to-date online training in AI, data analytics, and other key digital skills.

An example could be an ASEAN-wide digital skills initiative, co-developed with major tech firms, offering online courses and certifications in AI, cybersecurity, and data science—and accessible to all ASEAN citizens.

Pillar 5: Innovation & Entrepreneurship	1. **Enhance Venture Capital Availability through Incentives**: Despite a conducive environment for start-ups in ASEAN, there is a need to sustain and increase venture capital investments, particularly in light of recent funding slowdowns. ASEAN should introduce more favorable conditions for venture capitalists through tax incentives, co-investment funds, or regulatory easements. Drawing inspiration from Singapore's *Start-up SG Equity* scheme, a similar ASEAN-wide program could be developed to stimulate private-sector investment and to attract more venture capital firms. 2. **Boost R&D Investment and Collaboration**: Given the relatively low R&D expenditure in the ASEAN region, it is crucial to find alternative ways to raise such investment. This can be achieved by providing grants and incentives for research institutions and encouraging private-sector investment in R&D. Following the example of Thailand's *20-Year National Strategy* or the Philippines' *Science for Change Program*, ASEAN can focus on specific areas that draw out AMS' differentiating strengths. 3. **Strengthen Intellectual Property Protection**: With the increasing importance of digital innovation, robust intellectual property (IP) protection plays a significant and vital role in the survival of digital economies. ASEAN should continue enhancing the enforcement of national IP protection measures and enforcing the mechanisms introduced in the *ASEAN IPR Action Plan 2016- 2025 (AIPRAP) v2.0*. Learning from Vietnam's alignment with the *Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP)* when amending its own IP law, similar efforts can be made regionally to follow international IP standards and address emerging challenges, such as IP implications of AI-generated content.	• **Indicator 5.1**: introduce a regional venture capital fund, co-financed by member states, to provide stable and consistent funding to startups across the region. • **Indicator 5.2**: Implement an ASEAN-wide scheme that incentivizes R&D investment by large tech corporations operating in ASEAN, motivating them to actively contribute to the region's innovation landscape. • **Indicator 5.5**: Establish a unified ASEAN IP enforcement body to standardize and enforce IP laws across all AMS, ensuring the measures and mechanisms introduced by the ASEAN Working Group on Intellectual Property Cooperation (AWGIPC) are consistently and effectively implemented. ASEAN should also consider introducing best practices on IP enforcement in the region—notably, the creation of interagency councils on IP Rights to address counterfeit and infringement issues, as well as the implementation of an MoU on eCommerce to protect both consumers and brand owners in the digital space.
Pillar 6: Institutional &	1. **Enhance Rural Connectivity**: ASEAN should prioritize closing the rural-urban digital divide by implementing targeted policies for rural internet connectivity. For instance, ASEAN could establish a rural connectivity fund supported by contributions from telecom operators, which would be used to subsidize the expansion of mobile broadband networks in underserved areas. This approach aligns with the current trend of increasing mobile broadband infrastructure while addressing the region's pervasive digital divide.	• **Indicator 6.1**: Establish a rural connectivity fund to finance the extension of broadband infrastructure in underserved rural and isolated areas. • **Indicator 6.2:** Implement a regional broadband quality benchmarking system to set and enforce

Infrastructural Readiness	2. **Improve Internet Quality and Accessibility**: To address the challenge of inconsistent internet quality across AMS, ASEAN should establish regional standards for broadband speed and reliability. ASEAN could introduce a benchmarking system to measure the internet quality of each AMS, incentivizing improvements. This initiative would support the growth of internet-based economic activities, particularly in countries where internet quality currently lags behind. 3. **Digitalize Government Services with a Citizen-Centric Approach**: ASEAN should focus on enhancing the interface and delivery of digital government services to ensure they are user-friendly and meet the needs of citizens. This could entail AMS establishing a digital service design lab, where citizens can participate in the development and testing of digital government services. This collaborative approach would not only ensure that digital services are more aligned with citizens' needs, but also foster greater public trust and engagement in digital government initiatives.	minimum standards for internet speed and reliability across AMS. • **Indicators 6.3 and 6.4**: Create national digital service design labs to involve citizens in the development and refinement of digital government services.
Next Steps for Future ADII Iterations to Remain Relevant and Impactful for Years to Come	1. **Close alignment of DEFA and ADII processes, priorities, and milestones**: Once completed, the *ASEAN Digital Economy Framework Agreement (DEFA)* will significantly influence initiatives like the ADII by providing a consolidated and systematic approach to digital transformation and integration among AMS. As such, we recommend using the *DEFA* as a guiding framework for ADII's metrics and methodologies—If ADII measures and supports the digital economy goals set by *DEFA*, it is likelier to be able to offer a coherent and coordinated direction for digital growth in the region. It is therefore vital that ADII's development closely aligns with the evolving *DEFA*, adapting its indicators and pillars to reflect *DEFA*'s own processes, priorities, and milestones. 2. **Strengthen and consolidate AMS' and ASEC's data gathering, computing, and publishing capabilities**: This second edition of the ADII report has demonstrated the importance of not relying on third-party organizations and their databases to build data-driven, evidence-based policies for the digital economies of ASEAN. From unavailable or discontinued datasets to evolving methodologies, third-party databases can be used as a starting point, but should not be the only source of quantitative resources. As such, we recommend sustaining the consultative and participative process developed by the data completion mechanism to gradually replace all third-	• Conduct a preliminary mapping exercise to see which ADII Pillars and Indicators already align with *DEFA* provisions. • Determine whether existing Indicators or Pillars should be kept, updated, or replaced to better align with *DEFA* actions and provisions. • Where possible/necessary, develop new sets of Indicators that effectively capture the priority areas outlined by the *DEFA* (e.g., interoperability of various systems, use of digital standards, AI operationalization, etc.). • Leverage the Data Completion Committees (DCCs) that were created for the ADII 2.0 data completion exercise to develop/complete any

party databases with datasets that are completely produced, selected, and computed by ASEAN, for ASEAN.

quantitative/qualitative assessments that do not currently exist.

- Leverage the cross-sectoral discussions and consultations that were created during the ADII 2.0 data completion exercise to regularly revisit and adjust future iterations of the ADII.

6 | LOOKING AHEAD: SUSTAINING AND EXPANDING THE ADII

The ADII was developed as a living document that can evolve to best reflect and address ASEAN's changing priorities and needs. As such, this section examines the various ways in which the ADII methodology can be sustained and expanded so that it continues to deliver reliable and actionable evidence for years to come.

In terms of continuity and sustainability, this entails AMS having the necessary resources, methodologies, processes, and know-how to drive the end-to-end collection, computation, and publication of ADII datasets—thereby overcoming the reliance on third-party data sources that are not always available or viable.

In terms of expansion and extension, AMS need the ability and confidence to revisit ADII indicators and pillars periodically, readjusting and complementing the datasets based on the region's evolving priority areas. This may entail adding new indicators and pillars to complement existing ones, as well as removing or replacing some of them.

ADII CONTINUITY & SUSTAINABILITY

The question of sustaining the ADII over time came to light during the preparation of ADII 2.0, when several third-party datasets that had been used for ADII 1.0 were no longer updated or available.

As a response, the ADII data completion exercise was initiated all through 2022-2023, bringing together AMS, the ASEAN Secretariat (ASEC), and industry representatives to adjust the way the ADII data was produced, collected, and computed to reduce dependency on third-party entities and their databases.

Most notably, the data completion exercise saw the creation of Data Completion Committees (DCCs) to support AMS in their quest to provide accurate quantitative and qualitative assessments of their digital capabilities.

Comprised of several government and non-government experts recognized for their experience and expertise, the DCCs coordinated with a wide range of stakeholders to find alternative data sources. In some cases, this entailed extracting and computing data points from existing government databases. In other cases, this entailed developing a new survey with a methodology that is as close as possible to the original third-party survey.

Beyond the data completion exercise for ADII 2.0, the process and purpose of the DCCs may serve as a blueprint for two long-term objectives of the ADII framework:

1. **A framework for AMS to keep future iterations of the ADII updated (continuity)**. The processes developed for this data completion exercise will be useful for AMS to overcome any methodological obstacles related to data availability and reliability. By being able to reproduce third-party methodologies, AMS will be better equipped to keep updating the ADII as ASEAN's digital integration efforts are tracked, measured, and assessed in future iterations of the ADII.

2. **A model for AMS to build/strengthen their data-gathering capabilities (sustainability)**. Looking beyond third-party databases (and even the ADII itself), the ability to gather, structure, normalize, and publish datasets related to the evolution of the digital economy is an invaluable asset for all AMS. Ensuring that key aspects of the digitalization of the region's economies are systematically tracked and formally captured will allow AMS to design strong, future-proof, and evidence-based policies.

ADII EXPANSION & EXTENSION

The importance of adjusting the ADII came to light during the preparation of ADII 2.0, when AMS and ASEAN grew the number of national, regional, and multilateral initiatives aimed at supporting the rise of digital economies in the region.

Specifically, the prominence of emerging initiatives such as the *ASEAN Digital Economy Framework Agreement (DEFA)* made it clear that the ADII must evolve to effectively align with new, high-priority areas of focus.[159]

Touted as the first major regionwide digital economy agreement in the world, the *DEFA* seeks to offer a comprehensive roadmap to empower businesses and stakeholders across ASEAN, through accelerating trade growth, enhancing interoperability, creating a safe online environment, and increasing participation of MSMEs. As such, key topics such as digital trade, cross-border e-commerce, cybersecurity, digital identities, digital payments, and AI are being considered to ensure a future-proof *DEFA*.

Beyond the *DEFA*'s specific areas of focus, there are two main ways for the ADII methodology to adjust its approach to cover rapidly emerging and evolving digitalization issues:

[159] ASEAN (2023) Framework for Negotiating ASEAN Digital Economy Framework Agreement, https://asean.org/wp-content/uploads/2023/09/Framework-for-Negotiating-DEFA_ENDORSED_23rd-AECC-for-uploading.pdf

1. **Modifying current ADII Indicators and Pillars to integrate priority DEFA issues and dimensions (expansion)**. Some of the *DEFA* provisions are already present in existing ADII Pillars, they just need to be completed with additional Indicators. We will examine existing third-party databases to see which elements may be measured and tracked by AMS DCCs and added to existing ADII Pillars.

2. **Creating new ADII Indicators and Pillars to align with priority DEFA issues and dimensions (extension)**. There are several instances of none of the *DEFA* provisions being covered or represented by existing ADII Indicators and Pillars. In such cases, we will examine existing third-party databases to see which elements may be measured and tracked by AMS DCCs and added as new ADII Pillars.

No matter which *DEFA* provisions end up being negotiated and finalized in 2025, it is important that the ADII is able to evolve to fit new ASEAN-wide actions and priorities.

In addition, to ensure the ADII remains accurately representative of AMS' evolving needs and priorities, the ACCED meetings will continue to provide an important platform to regularly highlight key issues or trends that may impact digital integration and discuss how they should be reflected in future iterations of the ADII.

ANNEX A:
ADII 2.0 COUNTRY FINDINGS

This section explores individual AMS findings, looking at their strengths and weaknesses based on the progress shown by their ADII scores. It highlights some of the areas AMS should be prioritizing based on these scores, and presents examples of prominent initiatives that should be launched or sustained.

This section also indicates the main data gaps that were identified during the development of the ADII and recognizes the efforts taken by AMS to bridge these gaps.

BRUNEI DARUSSALAM

Brunei Darussalam has made significant strides in the creation of a robust and inclusive digital economy. It launched the *Digital Economy Masterplan 2025 (DEM 2025)* in 2020,[160] with the aim to drive socioeconomic growth through three flagship projects that will serve as the backbone to this new ecosystem: the *Digital ID (DID)*, the *National Information Hub (NIH)*, and the *Digital Payments Hub (DPH)*. To date, the *NIH* has been launched, while the *DID* and the *DPH* are expected to be ready by Q3 2024 and Q1 2024 (respectively).

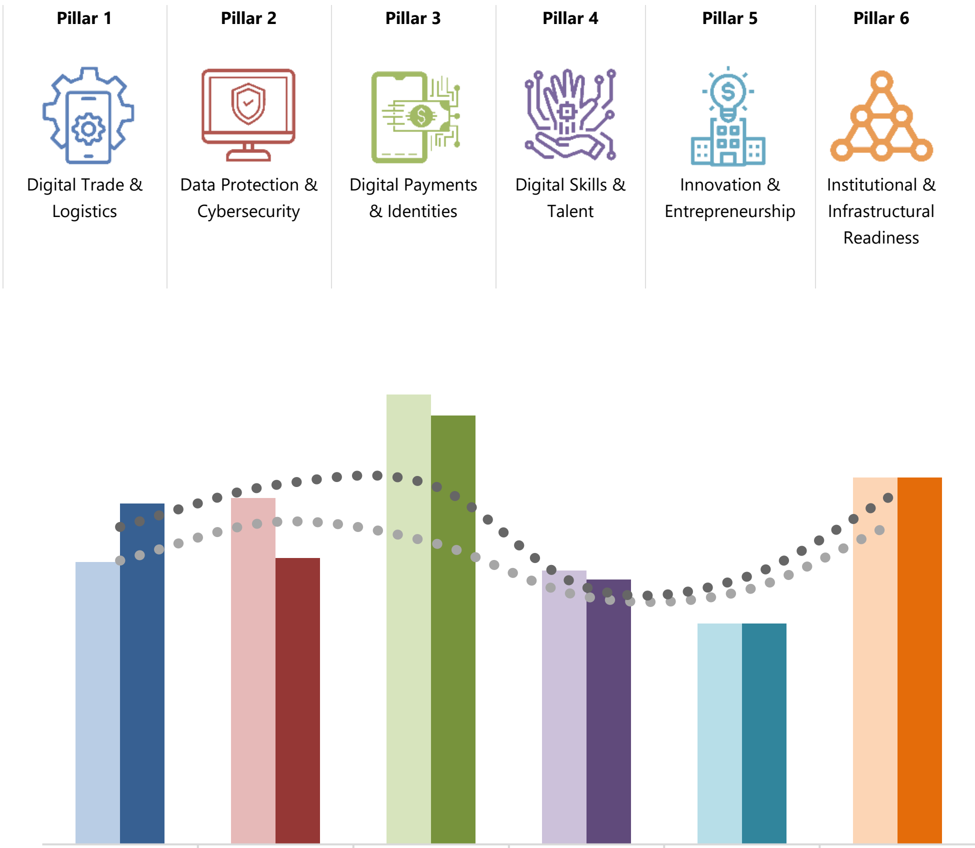
Other notable initiatives include the government's digital transformation of the Public Health Information Management System, the *Brunei Healthcare Information Management System (Bru-HIMS)*. Under the concept of "One Patient One Record", the government's *Bru-HIMS* initiative aims to have unify patient data, including their medical history, prescription records, diagnostic test results, appointments, etc. from public healthcare facilities into a single Electronic Patient Record. To date, 99% of the country's population has a record with *Bru-HIMS*, and 85% of all public hospitals work processing has been digitalized. Looking ahead, there are plans to integrate *Bru-HIMS* with the *BruHealth* mobile application to provide a more holistic view of patients' health history.[161] Originally developed for contact-tracing during the COVID-19 pandemic and for anything COVID-19 related in Brunei Darussalam, the *BruHealth* application has evolved into a health promotion and management tool, allowing users to manage medical appointments and health records.[162]

Despite these achievements, there are still some concerns over the protracted introduction of national data privacy and cybersecurity regulations. The Authority for Info-Communications Technology Industry of Brunei Darussalam (AITI), the statutory board responsible for overseeing the development of the country's ICT industry and protection of consumer's interests, has been in the process of developing a new *Personal Data Protection (PDP) law* since 2021. The upcoming *PDP* law will seek to govern the collection, use, and disclosure of personal

[160] Ministry of Transport and Infocommunications (2020) Digital Economy Masterplan 2025, www.mtic.gov.bn/DE2025/documents/Digital%20Economy%20Masterplan%202025.pdf
[161] Ministry of Health (n.d.), Bru-HIMS, www.moh.gov.bn/SitePages/Bru-HIMS.aspx
[162] Ministry of Health (n.d.), BruHealth, www.moh.gov.bn/SitePages/bruhealth.aspx

data by private organizations, in a way that will recognize the right of individuals to protect their personal data and related obligations by private sector organizations.

Figure 9: Brunei Darussalam and ASEAN Scores – ADII 1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Brunei Darussalam (ADII 1.0)	54.97	67.46	87.56	53.31	42.99	71.42
Brunei Darussalam (ADII 2.0)	66.39	55.70	83.49	51.52	42.99	71.42
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 9 shows that the best performing ADII 2.0 pillar for Brunei Darussalam is Pillar 3, with a score of 83.49 that surpasses the ASEAN average of 69.95. The score is more reflective of the country's efforts to advance digital finance adoption than the advancements made in the implementation of a digitized national ID system.[163] There are several notable products that may have contributed to the increase in adoption of digital financial services in Brunei Darussalam: various QR Code Payments launched by local banking institutions, Pocket Brunei (a popular digital payments service provider), Takaful Brunei Mobile App and TAIB Islamic Insurance App (two online insurance providers), and MoneyMatch (a currency exchange service provider). Other milestones that may have contributed to the country's relatively high score include the introduction of a *Digital Payments Roadmap* by the Central Bank of Brunei Darussalam, and the country's entry into the *Global Financial Innovation Network (GFIN)*.

[163] Brunei Darussalam has opted to retain the same score as ADII 1.0 for indicator 3.5 on degree to which a digitized ID systems in place.

Brunei Darussalam's second-best performing pillar is Pillar 6, which at 71.42 remains above the ASEAN average of 68.20. This score is reflective of efforts to improve the country's Internet connectivity, resulting in a notable surge in the adoption of fixed broadband services. For example, broadband subscriptions for households and businesses grew from 51% in 2019 to 88% in 2023. During the same period, coverage for 4G mobile broadband improved from 91% to 99.3%,[164] and more recently, in June 2023, the country achieved a new milestone in connectivity with the launch of its 5G network.[165] The Pillar 6 score is also the result of the implementation of strategic enablers and flagship digital projects outlined in the *ASEAN Digital Economy Masterplan 2025*, as well as the longer-term vision set out in *Wawasan 2035*, which emphasizes a well-connected population for long-term sustainable development.[166]

The two lowest-performing pillars for Brunei Darussalam are Pillars 2 and 5, with a score of 55.7 for Pillar 2 (below the ASEAN average of 69.99) and a score of 42.99 for Pillar 5 (below the ASEAN average of 51.27). The relatively low scores for both pillars are indicative of the challenge of fostering a conducive regulatory environment for digital activities. On Pillar 2, the recent launch of *Cybersecurity Order 2023* does provide measures to prevent, manage, and address cybersecurity threats and incidents,[167] but it may not be sufficient to foster trust across key economic sectors. On Pillar 5, the continued absence of a *PDP* law may be hindering the development of innovative business models that involve the collection, use, disclosure, and processing of personal data.[168]

Looking at ADII changes over time, the biggest improvement for Brunei Darussalam was for Pillar I, with a score that went from 54.97 to 66.39. A notable initiative that may have contributed to this achievement is the digitalization of Muara Port, the country's largest deep-water port and main international trading gateway.[169] Digital features introduced since 2022 include the ability to initiate a service online via an application; the use of AI to allocate pilot/ pilot boats, and tugboats; frequent updates on vessel locations; and timely issuance of invoices upon completion of pilots' jobs.[170] Other government-led initiatives that may have contributed to this progress are those focused on promoting e-commerce adoption. This includes the establishment of *eKadaiBrunei*, an online directory for local e-commerce delivery and logistics companies, and the annual cybershop fair by AITI to showcase products and services by local e-commerce businesses.[171]

Conversely, Brunei Darussalam saw its ADII score drop for Pillars 2, 3, and 4, even as the average ASEAN score improved during that same period. The score for Pillar 2 is 55.7 in ADII 2.0, below the ASEAN average of 69.99 and a sharp drop from 67.46 in ADII 1.0. This may be linked to the absence of a comprehensive data protection regime, even as other AMS have introduced important regulations. For example, Indonesia passed its *PDP* bill in September 2022, and Viet Nam issued *Decree No. 13/2023/ND-CP on Personal Data Protection* in April 2023. The score may also not be capturing the full impact of the recently introduced *Cybersecurity Order* (published in May 2023).[172]

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[164] AITI (2023) Statistics, www.aiti.gov.bn/events-and-publications/statistics/.
[165] AITI (2023) Press Release – 5th Generation (5G) Mobile Services Now Available Nationwide www.aiti.gov.bn/news/2023/5th-generation-5g-mobile-services-now-available-nationwide
[166] MTIC (2020) Launching of Digital Economy Masterplan 2025, www.mtic.gov.bn/Lists/News/NewItemDisplay.aspx?ID=714
[167] Cybersecurity Brunei, Cybersecurity Order 2023, www.csb.gov.bn/cyber-security-order
[168] AITI (n.d.) Personal Data Protection, www.aiti.gov.bn/regulatory/pdp/
[169] ASEAN Ports Association (2022) 46th ASEAN Ports Association (APA) Meeting, Country Paper: Maritime and port authority of Brunei Darussalam, https://apaport.org/api/download/resource/Brunei%20Darussalam%20National%20Single%20Window%20System%20(BDNSW)%20And%20Digitalization%20In%20Muara%20Port_79ED7742-4CDB-4ECC-A97E-2E5D3E389472.pdf
[170] Borneo Bulletin (2022) Brunei News – New digital services aim to help with port traffic https://borneobulletin.com.bn/new-digital-services-aim-to-help-with-port-traffic
[171] AITI (2020) Online Marketplaces For Micro, Small And Medium Enterprises (MSMEs) https://www.aiti.gov.bn/events-and-publications/press-releases/online-marketplaces-for-micro-small-and-medium-enterprises-msmes
[172] Cybersecurity Brunei (2023) Cybersecurity Order 2023, www.csb.gov.bn/cyber-security-order

For Pillar 3, the score dropped from 87.56 in ADII 1.0 to 83.49 in ADII 2.0 but remained above the ASEAN average of 69.95. This may be explained by gaps in the provision of measures that enable or support electronic transactions, which are outweighed by the advancements made in fostering widespread adoption of digital financial services—all while other AMS made slower progress during the same period. For Pillar 4, the main setback that led to a slight drop of the Pillar score was the relatively poor performance in Indicators 4.1 (proportion of STEM graduates), 4.2 (proportion of knowledge-intensive employment), and 4.3 (multi-stakeholder collaboration in R&D)—all essential markers of a digitally skilled workforce as well as drivers of a future-ready labor market.

Lastly, it is worth highlighting that the scores for Pillars 5 and 6 have not progressed much since ADII 1.0, mainly because in the absence of updated datasets to complete the missing/outdated data entries, Brunei Darussalam's Data Completion Committee (DCC) preferred to retain the scores from the previous ADII iteration (namely, the data for Indicators 1.4, 1.5, 3.5, 4.3, 4.4, 4.5, 5.1, 5.3, 5.4, 6.4, and 6.5 remained unchanged).[173]

[173] For more details on missing/outdated datasets and the methods that were used to address them, see the section titled "ADII 2.0 Data Completion Exercise".

CAMBODIA

Cambodia has made significant progress in its digitalization journey, with notable increases in Internet users and mobile subscribers. According to the Ministry of Post and Telecommunications (MPTC), as of 2020, the country boasts 16.3 million Internet users, a substantial increase from the 5 million reported in 2004, and 20.8 million mobile subscribers, up from 2.5 million in 2004.[174] Social media platforms are widely used in the country, with approximately 12 million users, constituting about 70% of the population as of 2020. The COVID-19 pandemic accelerated digital adoption in Cambodia, prompting the government to accelerate digital transformation effort.[175] Nevertheless, substantial gaps remain in digital infrastructure, digital governance, and digital skills and literacy.

Cambodia's digital transformation strategy is centered around the *Digital Economy and Society Policy Framework 2021-2035*, which rests on two foundations (digital infrastructure and reliability and confidence in digital systems) and on three pillars (digital citizens, digital government, and digital businesses).[176] Of the three pillars, the government has prioritized the establishment of a digital government to promote widespread adoption and use of digital technology, thereby catalyzing the digital transformation of the economy and society. In line with this, the government published the *Cambodia Digital Government Policy 2022-2035*, a strategic document reflecting its long-term vision for achieving complete digital government transformation. The policy outlines 10 key strategies and establishes a digital government committee, overseen by the MPT, to steer the digital transformation process.[177]

To tie these efforts together, the government has established the National Council for Digital Economy and Society, which comprises three committees: 1. The Digital Economy and Business Committee (led by the Ministry of Economy and Finance); 2. The Digital Government Committee (led by the Ministry of Post and Telecommunications); and 3. The Digital Security Committee (to be established).[178]

In terms of digital governance, the government has been actively preparing policies, laws, and regulations to launch digital services for citizens and stakeholders. For example, the government enacted the *Law on Electronic Commerce (E-Commerce Law)* and the *Law on Consumer Protection (Consumer Protection Law)* in 2019. The *E-Commerce Law* regulates domestic and cross-border e-commerce activities for Cambodia, establishes legal certainty for electronic transactions, and enacts several important protections for online consumers. Meanwhile, the *Consumer Protection Law* establishes rules to guarantee the rights of consumers and to ensure that businesses conduct commercial competition in a fair manner.[179] The law applies to the sale of goods and services, including those conducted online. Beyond these, the government has also been preparing a draft *Law on Information Technology Crimes*, a draft *Law on Cybersecurity*, and a draft *Law on Access to Information*.[180]

Regarding digital skills and literacy, the government is prioritizing reforms to human resources in the civil service by providing digital technology skill training for existing officials and recruiting digital talent. Notable initiatives include capability-building efforts by the MPTC through the Cambodia Academy of Digital Technology to train

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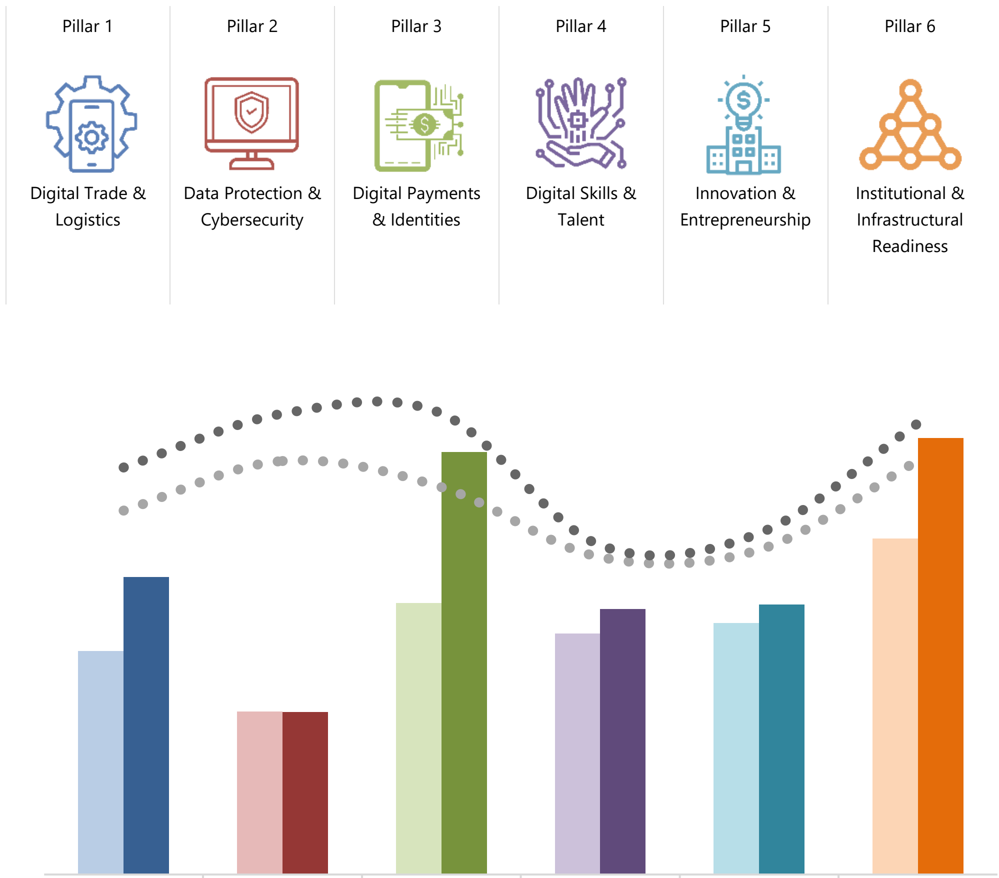
[174] Ministry of Post and Telecommunications (2022) Cambodia Digital Government Policy 2022-2035, https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf
[175] Cheng Savuth, Oum Sothea (2023) Digital transformaiton in Cambodia, www.jstor.org/stable/27211228
[176] Ministry of Post and Telecommunications (2021) Cambodia Digital Economy and Society Policy Framework 2021-2035, https://mptc.gov.kh/en/documents/policies/31605/
[177] Ministry of Post and Telecommunications (2022) Cambodia Digital Government Policy 2022-2035, https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf
[178] National Council for Digital Economy and Society (2023) About DEBC, https://digitaleconomy.gov.kh/about/debc?lang=en
[179] Tilleke & Gibbins (2019) Cambodia enacts a new E-commerce law and a consumer protection law, www.tilleke.com/insights/cambodia-enacts-new-e-commerce-law-and-consumer-protection-law/#:~:text=The%20Consumer%20Protection%20Law%20applies,real%20rights%20over%20immovable%20property.
[180] Ministry of Post and Telecommunications (2022) Cambodia Digital Government Policy 2022-2035, https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf

civil servants on digital skills and literacy. As of 2021, 6,700 civil servants have been provided digital literacy training by the Academy.[181]

Additionally, MPTC, with support from the Ministry of Education, Youth, and Sports, is developing the *Digital Skills Development Roadmap* to address the country's digital skills shortage.[182] Set to be launched in March 2024, the Roadmap will introduce the *Cambodia Digital Skill Competencies Framework*, a multi-stakeholder initiative that outlines four main strategies: 1) To attract students to STEM and digital disciplines; 2) To align curricula with digital specializations; 3) To provide internationally recognized certification programs; and 4) To prepare students with high-demand skills and knowledge.

Figure 10: Cambodia and ASEAN Scores – ADII1.0 & ADII 2.0



Cambodia (ADII 1.0)	33.91	24.76	41.20	36.56	38.19	50.97
Cambodia (ADII 2.0)	45.14	24.65	64.12	40.31	41.02	66.26
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 10 shows that the best performing pillar for Cambodia is Pillar 6. The score of 66.26, just slightly below the ASEAN average of 68.20, may reflect the country's introduction of the *Digital Economy and Society Policy*

[181] Ministry of Post and Telecommunications (2022) Cambodia Digital Government Policy 2022-2035, https://asset.cambodia.gov.kh/mptc/media/Cambodia_Digital_Government_Policy_2022_2035_English.pdf

[182] Kiripost (2023) Cambodia digital skills development roadmap to address skills shortage, https://kiripost.com/stories/cambodia-digital-skill-development-roadmap-to-address-skills-shortage

Framework 2021-2035. This framework outlines the government's vision for a vibrant digital economy and society and aims to promote digital adoption and transformation across various stakeholders, including the state, citizens, and businesses, to spur new economic growth and enhance social welfare in a "new normal". The framework outlines five key goals: developing infrastructure, fostering digital citizens, establishing a digital government, building trust and reliability in the digital system, and enabling digital businesses.[183]

At the same time, the *Digital Government Policy Framework 2022-2035* accelerated the country's digital transformation, with the creation of the Digital Government Committee under MPTC and the Digital Economy Department under MEF to enhance the government's digital infrastructure and information systems.

The country's second-best performing Pillar across the six Pillars is Pillar 3; Cambodia scored 64.12, reflective of the rise of digital payments in Cambodia as more people transition from cash to digital payments. In 2022, the National Bank of Cambodia reported that there were 19.5 million registered e-wallet accounts, and that the total number of online transactions reached one billion for the first time, with the total value of transactions amounting to USD 272.8 billion.[184] The rise in the use of digital payments is also likely related to the growing popularity of the *Bakong* system, Cambodia's peer-to-peer fund transfer service that was officially launched in 2020. *Bakong* uses a standardized QR-code system to initiate digital payments, which has enabled the country to initiate digital transactions with neighboring countries such as Malaysia, Viet Nam, and Thailand.[185]

At the other end of the spectrum is Pillar 2, Cambodia's lowest performing ADII 2.0 Pillar. With a score of 24.65, Cambodia is well below the ASEAN average of 69.99 despite the *E-Commerce Law* and the *Consumer Protection Law* being enacted to safeguard consumers when transacting online. This discrepancy between legislative efforts and below-average outcomes may be explained by the fact that neither of these laws specifically target data protection, while the laws concerning personal data protection, cybersecurity, and information technology security remain in draft form.

Looking at changes over time, it is clear that Cambodia has made progress across five of the six ADII 2.0 Pillars, though in very different proportions.

The biggest improvement has been in Pillars 3 and 6. For Pillar 3, Cambodia's score improved from 41.2 in ADII 1.0 to 64.12 in ADII 2.0, likely thanks to the sharp rise in digital payments adoption, as well as the growing popularity of the *Bakong* system, which has made domestic and cross-border peer-to-peer funds transfer safe, convenient, and secure. For Pillar 6, the country's score rose from 50.97 in ADII 1.0 to 66.25 in ADII 2.0, an increment that is likely reflective of the introduction of the *Digital Government Policy* in 2022 as well as the creation of a new Digital Government Committee under MPTC.

Cambodia also advanced in Pillar 1, with a score that went from 33.91 in ADII 1.0 to 45.14 in ADII 2.0. This growth is likely a result of recent and ongoing initiatives undertaken by both the government and the private sector to enhance the country's trade and logistics productivity. Notable recent efforts include an agreement between the Ministry of Commerce and the Chinese conglomerate, Alibaba, that was signed in October 2023 to bolster

[183] National Council for Digital Economy and Society (2021) Building a digital economy and society, https://digitaleconomy.gov.kh/intro-detail?lang=en#:~:text=Cambodia's%20Digital%20Economy%20and%20Society,economic%20growth%20and%20promote%20social
[184] The Star (2023) Cambodia records sharp rise in digital payments in 2022 says national bank report, www.thestar.com.my/aseanplus/aseanplus-news/2023/05/27/cambodia-records-sharp-rise-in-digital-payment-in-2022-says-national-bank-report
[185] The Asset (2023) Bakong, Cambodia's CBDC, goes global, www.theasset.com/article/49794/bakong-cambodia-s-cbdc-goes-global

collaboration in e-commerce, cloud computing, digital talent, and tourism.[186] Other initiatives that also likely contributed to this positive trend include: the launch of the *Cambodia National Single Window* in 2019 to expedite and streamline trade and customs processes through digital technologies; the launch of the *Online Business Registration* portal in 2020 to simplify business registration processes;[187] and a partnership agreement between the Logistics Business Association of Cambodia and the Cambodia Digital Tech Association to boost the adoption of digital technologies in the transportation and logistics sector.[188]

The smallest change was observed in Pillar 2; this is very likely due to the absence of a specific law on data protection, while a specific law on cybersecurity is still in the making. As such, addressing the gaps observed in data protection and cybersecurity should be prioritized for this Pillar to improve in the next iteration of the ADII.

[186] Yahoo (2023) Alibaba inks deal with Cambodia via Jack Ma's trade initiative as e-commerce giant dobules down on Southeast Asia, https://finance.yahoo.com/news/alibaba-inks-deal-cambodia-via-093000144.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAALmo4NXLnV9fw48jO2xQeFlhMsBKu-zYAiBIBKAs5pr6jT9IiZwvFUHr1dXYp8G1ReJawvu4WJ8HG8vBDnR4PzpIVGlLHkBikhKE5Taaxb2uAwtjTRZw7wFaUX2mLoOwMWca7KcCcIRaLBa_CiZdynuidDUvhpHaaikDE6o4mxnq
[187] Ministry of Economy and Finance (n.d.) Disruptive digital technology and Cambodia's international trade: empirical analysis and policy implications, www.nbc.gov.kh/download_files/macro_conference/english/S4_Disruptive_Digital_Teachnology_and_Cambodia_International_Trade.pdf
[188] Khmer Times (2021) Digital technology boosted in transport, www.khmertimeskh.com/50931081/digital-technology-boosted-in-transport

INDONESIA

Indonesia's digital economy is rapidly growing and maturing, positioning the country as a major digital economy player in ASEAN. In 2022, the country's digital economy was recorded at USD 77 billion, representing some 40% of the digital economy market share in ASEAN.[189]

This growth is greatly influenced by polices and legal frameworks aimed at driving the country's digital capabilities. For example, the *Digital Indonesia Roadmap 2021-2024*[190] sets the direction to advance the development of digital infrastructure, digital government, digital economy, and digital society. Most recently, the National Development Planning Ministry (Bappenas) unveiled the *Indonesian Digital Development Master Plan for 2023-2045*, focusing on strategies for digital industry development and fostering digital talents and R&D, among others. [191]

To embrace the coming wave of AI, the Indonesian government further launch the *Indonesia National AI Strategy 2020-2045 (Stranas KA)*, which focuses on providing a national direction to the way AI can be leveraged. The strategy covers four main areas: Ethics and Policy, Talent Development, Infrastructure and Data, and Industrial Research and Innovation. These developments are expected to support priority sectors, such as health services, bureaucratic reform, education, food security, mobility, and smart cities.[192]

In line with this, the Ministry of Communication and Informatics issued *Circular Letter Number 9 of 2023, on December 19, 2023, regarding the Ethics of Artificial Intelligence*. This circular letter serves as a guide for the ethical use and utilization of AI for both public and private Electronic System Operators (PSE).[193]

[189] Antara News (2023) Indonesia's digital economy has grown rapidly: Finance Ministry, https://en.antaranews.com/news/291867/indonesias-digital-economy-has-grown-rapidly-finance-ministry
[190] KOMINFO (2021) Minister of Communication and Information Presents Indonesia's Digital Roadmap in ATxSG, www.kominfo.go.id/content/detail/35713/siaran-pers-no240hmkominfo072021-tentang-menkominfo-paparkan-roadmap-digital-indonesia-dalam-atxsg/0/siaran_pers
[191] Antara News (2023) Bappenas unveils document to support digital transformation, https://en.antaranews.com/news/261485/bappenas-unveils-document-to-support-digital-transformation
[192] Safene (2022) Priorities and Challenges of Indonesia's Artificial Intelligence National Strategy (Stranas KA), https://safenet.or.id/2022/05/priorities-and-challenges-of-indonesias-artificial-intelligence-national-strategy-stranas-ka
[193] Mondaq (2023) Draft Circular Letter On Ethical Guidelines For Use Of Artificial Intelligence, www.mondaq.com/new-technology/1401050/draft-circular-letter-on-ethical-guidelines-for-use-of-artificial-intelligence

Figure 11: Indonesia and ASEAN Scores – ADII1.0 & ADII 2.0

	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Indonesia (ADII 1.0)	49.67	78.43	59.73	45.64	48.81	62.44
Indonesia (ADII 2.0)	58.91	93.12	62.47	40.67	50.16	69.59
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 11 shows that almost all of Indonesia's ADII 2.0 scores are above the ASEAN average, save for Pillar 4. Indonesia's strongest performance is in Pillar 2, followed by Pillar 6. However, more efforts are required to improve the scores of Pillars 4 and 5.

The high Pillar 2 score is likely due to the many data protection laws and regulations that Indonesia has recently enacted. The *Personal Data Protection Law (PDPL)*, which took effect in October 2022, covers data ownership rights, data access, and obligations related to the collection, storage, processing and transfer of personal data of Indonesian users.[194] The law also introduces requirements for notifications to the regulator on cross-border data transfers, as well as setting penalties for personal data breaches.[195] Other than the *PDPL*, data protection is covered by the *Electronic Information and Transactions Law (EIT Law)* and *Government Regulation No. 71 of 2019 on the Implementation of Electronic Systems and Transactions (GR 71)*.[196]

[194] Norton Rose Fulbright (2022) Highlights of Indonesia's Personal Data Protection Law, www.nortonrosefulbright.com/en/knowledge/publications/31bce8f0/highlights-of-indonesias-personal-data-protection-law

[195] Thales (2023) Personal Data Protection (PDP) Law of Indonesia, https://cpl.thalesgroup.com/compliance/apac/indonesia-personal-data-protection-law

[196] Baker and McKenzie (2023) Indonesia: Key Data Privacy and Security Laws, https://resourcehub.bakermckenzie.com/en/resources/data-privacy-security/asia-pacific/indonesia/topics/key-data-privacy-and-security-laws

On the cybersecurity front, Indonesia does not currently have a standalone cybersecurity law, but key aspects of cybersecurity are covered by the *EIT Law* and *GR 71*. In addition, there are sectoral cybersecurity legislations, such as *Nomor 29/SEOJK.03/2022 (SEOJK 29)*, which covers cybersecurity requirements for the financial sector.[197]

Indonesia's second-best performing pillar is Pillar 6.[198] The high score is driven by high scores in the number of active mobile-broadband subscriptions (Indicator 6.1) and the degree to which government services are available and accessible digitally (Indicator 6.3). On the first Indicator, Indonesia has long recognized that around 23% of its population still has limited or no access to the internet;[199] this has prompted several initiatives to enhance internet access and connectivity throughout the country. For example, The Ministry of Communication and Informatics is in the process of building fiber optic cables and Base Transceiver Stations (BTS).[200] The government has also partnered with SpaceX to launch *SATRA-1*, a communications satellite, to support upstream digital infrastructure growth in rural and remote parts of the country.[201] On the second Indicator, the Coordinating Ministry for Economic Affairs is improving the implementation of the *Electronic-Based Government System (SPBE)* to modernize government administration, supported by the construction of Indonesia's *National Data Centre (PDN)*. Functions such as cloud computing, big data analytics, AI, blockchain and the metaverse will be included as part of the PDN ecosystem.[202]

Conversely, Indonesia's weakest performance was in Pillar 4. While the Indonesian government has initiated several programs to enhance digital skills and nurture talents, these efforts will take time to bear visible results. To this end, the government has been exploring additional plans and partnerships to expand the scope and reach of digital education and training. One such initiative is the *Skills for Jobs Program* launched by the Coordinating Ministry for Economic Affairs and Microsoft, which aims to provide free digital literacy skills and job training to over 1 million Indonesian by 2024.[203] The Ministry of Communication and Informatics is also continuing its efforts in fostering digital skills through three distinct programs: *Siberkreasi National Movement for Digital Literacy*,[204] *Digital Talent Scholarship*, and *Digital Leadership Academy*.[205]

Looking at changes over time, Indonesia saw the greatest improvement in Pillar 2, with the score increasing from 78.42 in ADII 1.0 to 93.12 in ADII 2.0. As discussed above, this upward trajectory is a testament to Indonesia's efforts to introduce a comprehensive legislative framework around data protection. It stands to reason that a bigger increase could be expected to take shape in a future iteration of the ADII study if Indonesia were to launch a standalone Cybersecurity legislation.

Another Pillar which has remarkably progressed over time is Pillar 1, which saw its score increase from 49.67 in ADII 1.0 to 58.91 in ADII 2.0. This is likely reflective of the fact that the Indonesia National Single Window Agency has been dedicating more resources to the *National Logistics Ecosystem (NLE)*. For example, roads, bridges, airports, and harbors will be provided with internet access points to the *Digital Broadcasting System (DBS)*.[206]

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197 ASEAN Briefing (2023) New Cybersecurity Rules for Financial Institutions in Indonesia, www.aseanbriefing.com/news/new-cybersecurity-rules-for-financial-institutions-in-indonesia

198 It is worth noting that Indonesia had no available data for two of the five Pillar 6 Indicators (Indicators 6.4 and 6.5), a gap that may undermine the full extent of the country's performance in this area.

199 Telecom Review (2023) SATRIA-1: How Indonesia's First Satellite Aims to Accelerate Internet Connectivity, www.telecomreviewasia.com/news/featured-articles/3487-satria-1-how-indonesia-s-first-satellite-aims-to-accelerate-internet-connectivity

200 OpenGov Asia (2023) Indonesia's Three Digital Development Focus Areas in 2023, https://opengovasia.com/indonesias-three-digital-development-focus-areas-in-2023

201 Reuters (2023) Indonesia and Space X launch satellite to boost connectivity, www.reuters.com/technology/space/indonesia-spacex-launch-satellite-boost-internet-connectivity-2023-06-19

202 OpenGov Asia (2023) Indonesia's Three Digital Development Focus Areas in 2023, https://opengovasia.com/indonesias-three-digital-development-focus-areas-in-2023

203 Microsoft (2023) Supporting Indonesia's Inclusive Digital Economy through Skills, https://news.microsoft.com/id-id/2023/06/17/supporting-indonesias-inclusive-digital-economy-through-skills

204 Tech for Good Institute (2021) National-level priorities to grow the digital economy: Spotlight on Indonesia, https://techforgoodinstitute.org/blog/articles/advancing-digital-economy-through-national-level-priorities-spotlight-on-indonesia

205 HRM Asia (2022) Indonesia strives to nurture digital talents, https://hrmasia.com/indonesia-strives-to-nurture-digital-talents

206 The Jakarta Post (2023) National Logistics Ecosystem optimizes 2024 state budget, www.thejakartapost.com/business/2023/10/24/national-logistics-ecosystem-optimizes-2024-state-budget.html

The only marginal dip in performance is in Pillar 4. As covered previously, while the government has implemented many initiatives to equip the population with digital skills and knowledge, results from such programs take time to bear fruit. In addition, more support is needed from the Indonesian government to overcome certain barriers that make such initiatives less impactful than they could be: inequitable access to quality digital infrastructure, prohibitively costly digital devices, unreliable distance-learning systems, and overly expensive educational content, among others.[207] Another obstacle that Indonesia must tackle is the question of whether the digital skills taught are in line with labor market needs; if what is being taught does not translate into knowledge- and specialist-driven jobs, then the various initiatives launched will only create a generation of under-employed workers.

[207] Technode (2022) The Race for Digital Learning Leaves Many Students Behind- But not anymore, with Indonesian Schools Going Digital without the Internet, https://technode.global/2022/08/22/the-race-for-digital-learning-leaves-many-students-behind-but-not-anymore-with-indonesian-schools-going-digital-without-the-internet/

LAO PDR

While Lao PDR has been described as a digitally nascent country,[208] it is a rapidly growing economy that is making steady progress in the digitalization of both society and economy. For example, Internet penetration rates have increased from 20% in 2015 to 62% in 2023, and the number of mobile phone users has risen from 3.5 million to 6.4 million during the same period.[209] This has facilitated the expansion of e-commerce and digital services, leading to increased and accelerated overall trade, as digital enablers have enhanced trade in both physical goods and digital services.

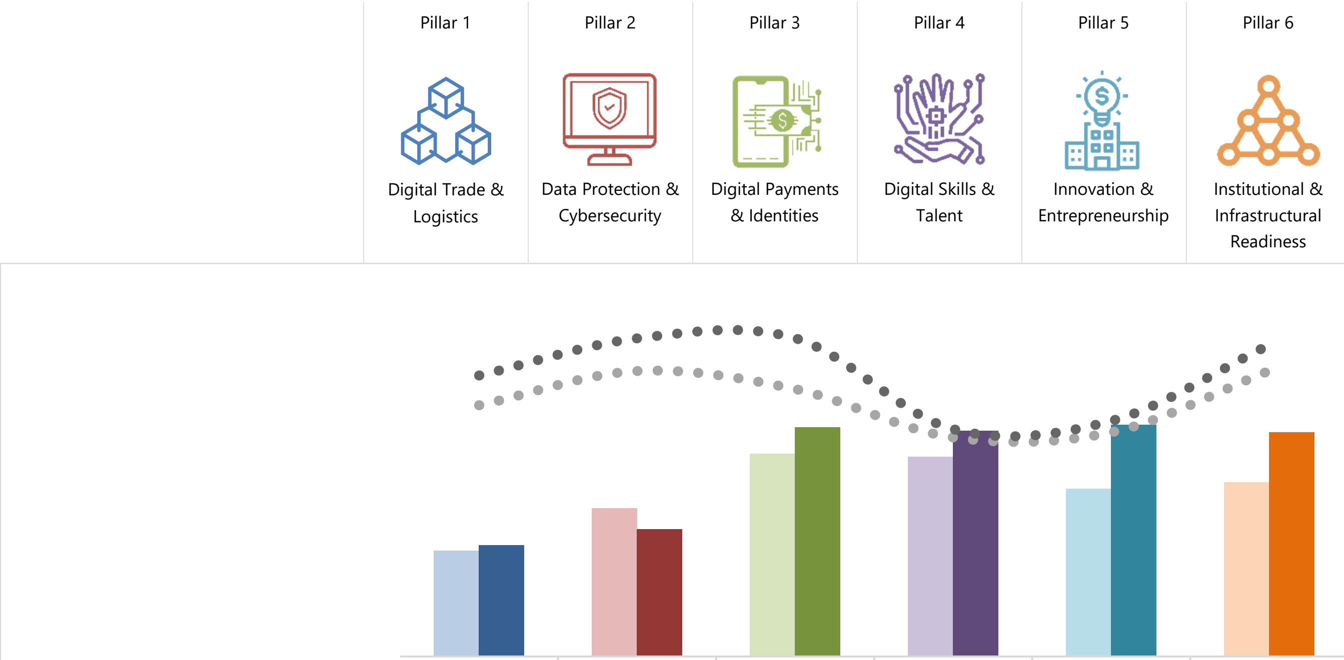
On the policy level, the government has made substantial progress and implemented various national strategies and frameworks, such as the *National Digital Economic Development Vision 2021-2040*, the *National Digital Economy Strategy 2021-2030*, and the *National Digital Economy Plan 2021-2025*.[210] Notably, the *National Digital Economy Strategy 2021-2030* establishes eight priorities, including: develop digital technology infrastructure; develop platforms for digital government, e-payments, data exchanges; develop digital-ready human resources, and ensure cyber-safety and that information and privacy are protected. Furthermore, Lao PDR the 2017 *Law on Electronic Data Protection* establishes a comprehensive regulatory framework for data privacy and governance.[211]

Despite these commendable efforts, challenges persist in terms of the affordability and quality of internet services in rural areas, as well as the digital skills of the workforce. For example, training and change management opportunities to enhance the digital skillsets of ministry and provincial government employees are limited.[212] To address relatively low digital literacy rates, the government launched the first-ever *Digital Literacy Camp* to introduce students to e-learning and important topics like online safety and cybersecurity.[213]

Lao PDR remains committed to bridging these gaps and further enhancing its digital landscape to support economic development and create a digital economy, digital society, and digital government. The Ministry of Technology and Communications is preparing to expand 5G trials and develop Lao PDR as a connecting hub and digital gateway within the region.[214] The Ministry of Technology and Communications also launched the Gov-X application in August 2023, a new mobile application that aims to improve the delivery and accessibility of government services to citizens, businesses, and government officials.[215]

[208] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf
[209] Laotian Times (2023) Digital 2023 Report on Laos Released, https://laotiantimes.com/2023/03/13/digital-2023-report-on-laos-released-internet-mobile-and-social-media
[210] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf
[211] Lao Services Portal (2017) Law on Electronic Data Protection 25/NA, http://lsp.moic.gov.la/?r=site/displaylegal&id=289#:~:text=The%20law%20on%20Electronic%20Data%20Protection%20defines%20the%20principles%2C%20regulations,data%20are%20safe%20and%20correct.
[212] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf
[213] UNICEF (2023) In Lao PDR, a digital transformation of education has begun, www.unicef.org/laos/stories/lao-pdr-digital-transformation-education-has-begun
[214] The Star (2023) Lao govt pushes for urgent digital transformation, www.thestar.com.my/aseanplus/aseanplus-news/2023/08/04/lao-govt-pushes-for-urgent-digital-transformation
[215] Lao News Agency (2023) Laos launches Gov-X- a one stop mobile application for public e-services, https://kpl.gov.la/En/detail.aspx?id=76191

Figure 12: Lao PDR and ASEAN Scores – ADII 1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Lao PDR (ADII 1.0)	23.22	32.58	44.53	43.89	36.91	38.27
Lao PDR (ADII 2.0)	24.51	27.96	50.39	49.63	50.99	49.28
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 12 shows that Lao PDR's ADII 2.0 scores are all under ASEAN averages, suggesting there is much room for improvement across all Pillars. However, it must be noted that Lao PDR's scores for Pillars 4 and 5 are very close to the ASEAN average, with very small differences of less than a point. This suggests that Lao PDR is largely keeping pace with the region in the areas of Digital Skills & Talent and Innovation & Entrepreneurship.

The highest scoring Pillar for Lao PDR is Pillar 5 (ADII 2.0 score of 50.99), closely followed by Pillar 3 (ADII 2.0 score of 50.39). Pillar 5 attests to Lao PDR's many efforts to create a conducive policy and business environment for innovation and entrepreneurship. Organizations including the Lao National Chamber of Commerce and Industry (LNCCI) and the Lao ICT Commerce Association (LICA) have undertaken some initiatives to build the country's start-up ecosystem. For instance, the Ministry of Technology and Communications (MTC) signed an MoU with LICA in September 2021 to promote Lao start-up ecosystems and innovative entrepreneurship,[216] building on other programs such as the *Lao Digital Forum*—which are held every quarter on topics such as e-commerce and EdTech.[217] Lao PDR has significant potential to grow further in this pillar as it is reported that only approximately 30% of ministries/provinces have undertaken any initiatives to promote an entrepreneurship ecosystem among young entrepreneurs and university and technical and vocational education training (TVET) students.[218]

On Pillar 3, Lao PDR has also multiplied the number of enabling and supportive initiatives. In early 2023, the central bank, Bank of the Lao PDR, signed an MoU with Japanese blockchain company Soramitsu to begin a

[216] Vietntianne Times (2021) Govt, LICA to power up digital economy in Laos with MOU signing, www.vientianetimes.org.la/freeContent/FreeConten_Govt_LICA_198.php
[217] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf
[218] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf

proof-of-concept for a Central Bank Digital Currency (CBDC),[219] which was followed by a successful trial of the "*Digital Lao Kip*" in August 2023.[220] The Central Bank has also been working on laying the foundations for digital payments; in November 2022, an MoU was signed between Bank of the Lao PDR and the National Bank of Cambodia (NBC) to facilitate cross-border transactions between the two jurisdictions. These developments follow the launch of the *Lao Payment and Settlement System (LaPASS)* in June 2020, which allows real-time settlement of payments and cheque clearing,[221] and the introduction of a Lao QR Code in early 2020. [222] In the area of digital IDs, while Lao PDR does not currently have a digital ID system, the Ministry of Technology and Communications has signed a MoU with a private company to conduct a feasibility study on the use of blockchain technology for digital IDs, among other use cases.[223] Most recently in February 2023, the government initiated the *Electronic Registration and Vital Statistics System (eCRVS)* project, which aims to have at least 70% of the country's birth certificates digitalized by 2024.[224]

The two lowest performing ADII 2.0 Pillars for Lao PDR are Pillars 1 and 2. For Pillar 1, there is significant potential to promote digitally enabled trade and logistics. Several paperless trade facilitation mechanisms are still nascent in Lao PDR, a gap that makes it more difficult to capture digital trade opportunities without automated customs and electronic submission of customs declarations.[225] Nevertheless, Lao PDR has made notable progress in digital trade. For instance, the *Lao National Single Window (LNSW)* has been fully operational since February 2021, helping to simplify and streamline the import, export, and transit of all goods and cargo at all international borders and airports. The *Lao Trade Portal*, which was set up in 2012, has also helped to reduce the burden of cross-border trade in bringing clarity to trade-related law and continues to be useful in 2022, with more than 100,000 visits in the first four months of 2022.[226]

For Pillar 2, there remain notable developments that demonstrate efforts and ambitions for Lao PDR to increase its scores in Data Protection & Cybersecurity. In enhancing protection against cyber-threats, the government has upgraded its *Computer Emergency Response Team (LaoCERT)* to the Department of Cyber Security and is drafting a national cybersecurity strategy.[227] Notably, one of the *National Digital Economy Strategy 2021-2030*'s eight priorities includes ensuring cyber-safety and protecting information and privacy.

Lao PDR's score in Pillar 4 on Digital Skills and Talent is not far from the ASEAN average. Part of the *National Digital Economy Development Plan (2021-2025)* targets the increase of the percentage of workers with digital skills and expertise from 0.3% to 1% of the current workforce. Currently, several universities and colleges provide ICT degrees, but they are a small proportion. In its efforts to grow the talent pool, the Lao ICT Commerce Association is working with universities and vocational centers to expand the ICT/digital workforce by introducing computer classes in the educational curriculum.[228]

[219] Nikkei Asia (2023) Laos begins digital currency trial with Japan blockchain company, https://asia.nikkei.com/Business/Markets/Currencies/Laos-begins-digital-currency-trial-with-Japan-blockchain-company
[220] Laotian Times (2023) Laos Conducts Trial of Digital Kip And Launches QR Payment with Cambodia Bank, https://laotiantimes.com/2023/08/21/laos-conducts-trial-of-digital-kip-and-launches-qr-payment-with-cambodian-bank/
[221] J&C Group (2020) BOL Launches Real Time Financial System, https://jclao.com/bol-launches-real-time-financial-system/
[222] FinTech News Singapore (2023) Laos' Fintech Revolution is Still Bank-Led, For Now, https://fintechnews.sg/68354/laos/laos-fintech-revolution-is-still-bank-led-for-now/
[223] World Bank (2022) Positioning the Lao PDR for a Digital Future, https://thedocs.worldbank.org/en/doc/c01714a0bc2ca257bdfe8f3f75a64adc-0070062022/original/Positioning-The-Lao-PDR-for-a-Digital-Future-11-10-22.pdf
[224] The Laotian Times (2023) Govt. Unveils Data Management System for Digital Birth Registration, https://laotiantimes.com/2023/03/07/govt-unveils-data-management-system-for-digital-birth-registration
[225] UNESCAP (2022) Readiness Assessment for Cross-Border Paperless Trade: Lao People's Democratic Republic, https://repository.unescap.org/bitstream/handle/20.500.12870/4369/ESCAP-2022-PB-Readiness-assessment-cross-border-paperless-trade-LaoPDR.pdf?sequence=1&isAllowed=y
[226] World Bank (2022) Ten years old and growing strong – Happy Birthday to the Lao Trade Information Portal, https://blogs.worldbank.org/trade/ten-years-old-and-growing-strong-happy-birthday-lao-trade-information-portal
[227] The Star (2022) Laos backs Asean efforts to address cyber threats, www.thestar.com.my/aseanplus/aseanplus-news/2022/11/02/laos-backs-asean-efforts-to-address-cyber-threats
[228] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf

Lao PDR is also undertaking several initiatives to boost its score for Pillar 6. For instance, the Lao President appointed a *National Digital Transformation Committee* in June 2023, with the mandate to set guidelines, strategies, policies, and mechanisms to guide and implement the transformation of the digital economy at central and local government levels.[229] Key digital government developments have also been implemented such as creating the *Lao National Internet Center* to centralize telecommunication and internet services, the *National Internet System*, the *National Data Center*, and the data center in the Dong Markhai area.[230]

Looking at changes between ADII 1.0 and ADII 2.0, Lao PDR has made positive progress for all pillars except Pillar 2.

With an increase from 36.91 in ADII 1.0 to 50.99 in ADII 2.0, Pillar 5 has not only undergone the biggest improvement out of the six Pillars, but it has also brought Lao PDR much closer to the ASEAN average. This is testament to Lao PDR's strong progress in promoting innovation and entrepreneurship over the years. Notably, Lao PDR's intellectual property (IP) protections were substantially enhanced. In February 2022, Lao PDR further improved its framework for enforcing IP rights through border measures against infringing goods, publishing new customs instructions that added industrial designs to the list of safeguarded IP rights for the Lao customs department. This means that an IP owner can now request the Lao customs department to take action on products infringing a protected industrial design under the customs border measures.[231]

On the other hand, Lao PDR's Pillar 2 score dipped between ADII 1.0 and ADII 2.0, a change that is likely driven by the level of legislative and regulatory capabilities. This downward trend may be reflective of Lao PDR's relatively less comprehensive regulatory framework for data protection and privacy, while other AMS like Viet Nam and Thailand have introduced some very thorough mechanisms in this area. As mentioned, Lao PDR is in the middle of drafting a cybersecurity strategy; once completed and implemented, the Pillar 2 score should experience a slight improvement in a future iteration of the ADII study.

[229] Laotian Times (2023) Lao President Appoints New Committee for Country's Digital Transformation, https://laotiantimes.com/2023/06/08/lao-president-appoints-new-committee-for-countrys-digital-transformation/
[230] UNDP (2022) Digital Maturity Assessment – Lao PDR Supporting Digital Government Transformation, www.undp.org/sites/g/files/zskgke326/files/2022-08/UNDP_LaoPDR_DMA_2022.pdf
[231] Managing IP (2022) Lao customs expands border measures to protect IP rights, www.managingip.com/article/2av9jyh6ust2rl2cqdgqo/sponsored-content/lao-customs-expands-border-measures-to-protect-ip-rights

MALAYSIA

Malaysia's digital economy has significantly matured in recent years. In 2023 alone, Malaysia saw an influx of digital economy investments, totaling some USD 6 billion,[232] inching closer towards its goal of expanding the contribution of the digital economy sector to 25.5% of the national GDP by 2025.[233]

The development of Malaysia's digital economy is supported by various national frameworks, including the *National Fourth Industrial Revolution Policy* (2021), the *Malaysia Digital Economy Blueprint* (2021), and the *Digital Investments Future5 (DIF5) Strategy 2021-2025*.[234] In 2022, a national strategic initiative, *Malaysia Digital*, was launched with the objective of attracting investments, talents, and businesses. Under the *Malaysia Digital Catalytic Programmes (Pemangkin)*, nine focus areas have been identified as sectors with high growth potential, which include: trade, agriculture, services, cities, health, finance, content, tourism, and Islamic digital economy.[235]

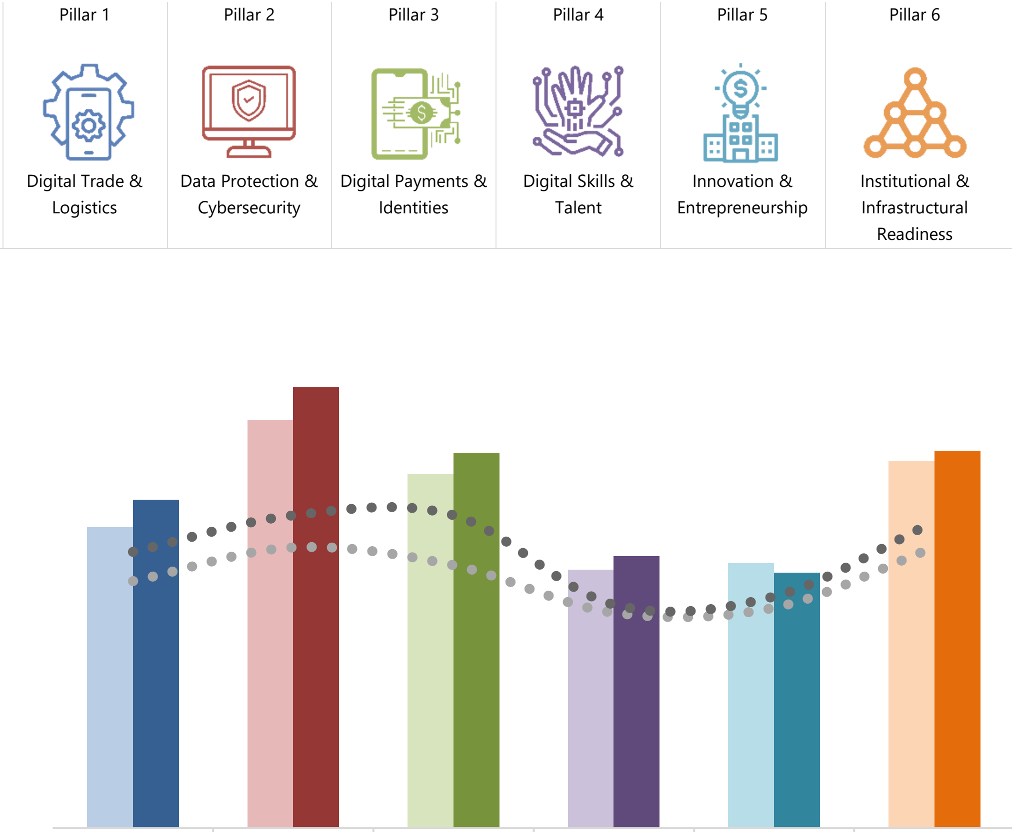
To support the development of these sectors, technology enablers such as blockchain, automation, AI have been identified as key investment areas.[236] Through the Malaysia Digital Economy Cooperation (MDEC), the government is expected to allocate more than USD 50.8 million to support the implementation of initiatives and programs under *Pemangkin*.[237]

Apart from setting the direction for digital economy, the Malaysian government has also been actively working on easing payment policies. For example, the Bank Negara Malaysia (BNM) is a part of a collaborative project, *Project Mandala*, which explores automating compliance procedures to promote an efficient regulatory environment for cross-border payments with other central banks and financial institutions.[238] Recently, BNM launched a real-time payment systems linkage with the Monetary Authority of Singapore (MAS) to enable cross-border QR payments to merchants on Singapore's *PayNow* and Malaysia's *DuitNow*. The collaboration is expected to lay a foundation for a scalable cross-border payment network across ASEAN.[239]

Other initiatives have also been introduced to support digital infrastructure, including the *National Fiberisation and Connectivity Plan (2019-2023)* and *JENDELA (2020-2025)*. The successful implementation of these policies initiatives is set to deliver improvements in mobile broadband speed, rise in 4G population coverage, and fiber connectivity. Looking ahead, the government is planning to increase the coverage of 5G networks, as well as ensure that Internet connectivity can be accessed by all by 2050. [240]

[232] Forbes (2023) MDX 2023: Celebrating the growth of Malaysía Digital Health, www.forbes.com/sites/malaysia-digital-economy-corporation/2023/10/26/mdx2023-celebrating-the-growth-of-malaysias-digital-economy/?sh=25606c2228c2
[233] New Straits Times (2023) Digital economy expected to contribute 25.5 per cent to GDP by 2025, www.nst.com.my/news/nation/2023/05/910000/digital-economy-expected-contribute-255-cent-gdp-2025
[234] ISEAS Yusof Shak Institute (2022) Economic Working Paper: Strategic Policies for Digital Economic Transformation: The Case of Malaysia, www.iseas.edu.sg/wp-content/uploads/2022/10/ISEAS_EWP_2022-6_Lee.pdf
[235] Ministry of Communications and Digital (2022) Catalysing Malaysia's Digital Economy, www.kkd.gov.my/en/public/news/22878-catalysing-malaysia-s-digital-economy
[236] Technode (2023) Malaysia's MDEC Eyes $230M Digital Investments by 2025, https://technode.global/2023/04/18/malaysias-mdec-eyes-230m-digital-investments-by-2025
[237] The New Straits Times (2023) Govt identifies RM1 billion potential investments in nine sectors under Pemangkin, www.nst.com.my/news/nation/2023/04/900499/govt-identifies-rm1-billion-potential-investments-nine-sectors-under
[238] Monetary Authority Singapore (2023) BIS and central bank partners to explore protocols for embedding policy and regulatory compliance in cross-border transactions, www.mas.gov.sg/news/media-releases/2023/bis-and-central-bank-partners-to-explore-protocols-for-embedding-policy-and-regulatory-compliance
[239] Monetary Authority Singapore (2023) Launch of Cross-border Real-time Payment Systems Connectivity between Singapore and Malaysia, www.mas.gov.sg/news/media-releases/2023/launch-of-cross-border-real-time-payment-systems-connectivity-between-singapore-and-malaysia
[240] Malaysian Communications and Multimedia Commission (2023) Digital Connectivity: Ensuring Everyone Gets Connected: A Glance at Malaysia's Digital Connectivity Journey, www.mcmc.gov.my/skmmgovmy/media/General/pdf2/Insight-Digital-Connectivity.pdf

Figure 13: Malaysia and ASEAN Scores – ADII1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Malaysia (ADII 1.0)	67.35	91.27	79.20	57.85	59.22	82.18
Malaysia (ADII 2.0)	73.49	98.71	83.98	60.85	57.16	84.41
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 13 shows that Malaysia's digital integration scores continue to be above the regional ASEAN average, with Pillar 2 standing out as significantly above the ASEAN average. Though Malaysia's scores in Pillars 4 and 5 are higher than the ASEAN average, there is notable room for improvement in both areas.

The highest scoring pillar for Malaysia is Pillar 2; at 98.71, this score is significantly above the ASEAN average and reflects the government's prioritization of data privacy and cybersecurity. To strengthen the current data protection and privacy framework, the Ministry of Communications and Digital reviewed and introduced various new amendments to the *Personal Data Protection Act (PDPA)* in addition to those proposed in 2022. The new amendments include increasing penalties for data misuse and PDPA breach, as well as enhancing the enforcement power of the Personal Data Protection Department (JPDP) as a statutory commission. The amendments are tabled to be discussed in early 2024.[241] Additionally, the Personal Data Commissioner issued the *General Code of Practice of Personal Data Protection (General CoP)* at the end of 2022 to govern data users that are covered by the PDPA (such as those in healthcare, banking and financial, and aviation).[242]

[241] Global Compliance News (2023) Malaysia: 90 Days under Malaysia Madani – Personal Data Protection redux, www.globalcompliancenews.com/2023/03/21/https-insightplus-bakermckenzie-com-bm-technology-media-telecommunications_1-malaysia-90-days-under-malaysia-madani-personal-data-protection-redux_03032023/
[242] Baker MacKenzie (2023) Malaysia: New legal requirements under the General Code of Practice of Personal Data Protection, https://insightplus.bakermckenzie.com/bm/technology-media-telecommunications_1/malaysia-new-legal-requirements-under-the-general-code-of-practice-of-personal-data-protection

Apart from data protection, the Malaysian government is also committed to cybersecurity, though results have been much more moderate in this area. In 2020, the *Malaysia Cyber Security Strategy* had set the objective of introducing a new cybersecurity law, but there is currently no standalone legislation that covers cybersecurity matters. The government has, however, announced plans to introduce a draft *Cybersecurity Bill* in 2024 to address this major legislative gap. It must be highlighted that various legislations and guidelines do exist at the sector level. These include the *Risk Management in Technology (RMiT) Policy Document* issued by the Central Bank of Malaysia,[243] and the *Guidelines on Management of Cyber Risk* released by the Securities Commission Malaysia.[244]

Malaysia's weakest pillar is Pillar 5, despite having made significant efforts to create an environment that promotes entrepreneurship and innovation. Indeed, various overarching frameworks such as the *National Entrepreneurship Policy 2030*, the *Malaysia Digital Economy Blueprint*, and the *12th Malaysia Plan* directly or indirectly aim to foster a conducive environment for innovative digital business models. MDEC has also made strategic investments through *Pemangkin* to encourage and attract domestic tech talents and investments in nine high growth sectors. Likewise, the *Digital Hub Programme*, which aims to "grow start-ups, global tech companies, accelerators, talent builders and investors",[245] is another ongoing effort.

The Ministry of Entrepreneurship Development and Cooperative (Kuskop) has also introduced efforts to support SMEs, most notably through the *Professional Training and Education for Growing Entrepreneurs program (PROTEGE)*. The program includes the *New Generation Entrepreneurs Online Bootcamp (N-GENE)* and the *Student-Preneur Biz Apprenticeship Program (SPACE 2.0)*.[246] *N-GENE* is an initiative that aims to assist unemployed graduates and underprivileged entrepreneurs to cultivate business skills through preparatory training and business guidance.[247] Similarly, *SPACE 2.0* is designed to enable final-year students to become entrepreneurs by providing training, business guidance, and a business start-up grant for qualified students.[248] Overall, the *PROTEGE* program is expected to benefit more than 12,450 students in 2023.[249]

Looking at changes over time, Malaysia has shown the biggest increase in Pillar 2. This can be attributed to the multiple efforts aimed at improving existing legislative frameworks and developing new cybersecurity and data protection mechanisms. On the data protection front, apart from the *Personal Data Protection Act (PDPA)* introduced in 2013, the Ministry of Communications and Digital has issued other subsidiary legislations to complement the *PDPA*. For instance, the *Personal Data Protection (Fees) Regulations 2013* was announced to regulate the maximum fees that may be imposed by data users on data subjects for access and correction of data.[250] On the cybersecurity front, the government plans to introduce a standalone legislation that will update and complement cybersecurity laws that may no longer adequately cover all current cyber-threats, including the *Communications and Multimedia Act 1998*, the *Computer Crimes Act 1997*, and the *Digital Signatures Act 1997*.[251]

243 Bank Negara Malaysia (2023) Risk Management in Technology (RMiT) Policy Document, www.bnm.gov.my/-/risk-management-in-technology-rmit-policy-document
244 Securities Commission Malaysia (2016) Guidelines on Management of Cyber Risk, www.sc.com.my/api/documentms/download.ashx?id=9aaddb2e-aa13-409a-a47f-8d0124afd229
245 GovInsider (2017) Malaysia wants to attract tech entrepreneurs, https://govinsider.asia/intl-en/article/malaysia-mdec-digital-hub-tech-entrepreneur-programme
246 Malay Mail (2023) Entrepreneur Development Ministry allocated RM5m to implement entrepreneurship programmes for graduates, www.malaymail.com/news/malaysia/2023/03/13/entrepreneur-development-ministry-allocates-rm5m-to-implement-entrepreneurship-programmes-for-graduates/59366
247 Centre For Entrepreneur Development And Research (CEDAR) Sdn Bhd (2021) N-GENE (New Generation Entrepreneurs Online Bootcamp), www.cedar.my/n-gene-new-gen-entrepreneur-online-bootcamp.html
248 Benefit Portal by the Ministry of Finance (2023) Apprentice Entrepreneur Program (SPACE 2.0), https://manfaat.mof.gov.my/perniagaan/bizme-space-2.0
249 Malay Mail (2023) Entrepreneur Development Ministry allocated RM5m to implement entrepreneurship programmes for graduates, www.malaymail.com/news/malaysia/2023/03/13/entrepreneur-development-ministry-allocates-rm5m-to-implement-entrepreneurship-programmes-for-graduates/59366
250 Linklaters (2022) Data Protected- Malaysia, www.linklaters.com/en/insights/data-protected/data-protected---malaysia
251 LPP Law (2019) Cybersecurity Law and Framework in Malaysia, https://lpplaw.my/insights/e-articles/cybersecurity-law-and-framework-in-malaysia/

The only score that dipped for Malaysia was Pillar 5, which went from 59.22 in ADIII 1.0 to 57.16 in ADIII 2.0. While not a major drop, the downward trajectory should be a key area of focus for Malaysia. Indeed, despite making myriad efforts in encouraging entrepreneurship and innovation, the country still faces several challenges. For example, the lack of venture capital and private equity funding is a pressing issue that prevents entrepreneurs from accessing sustainable financing to maintain and expand their businesses.[252] Moreover, limited budgets allocated to R&D remain an impediment. According to the Ministry of Science, Technology and Innovation, the Malaysian government only allocated 1% of national GDP to R&D, which is markedly lower than other digital economies, especially non-ASEAN ones such as Japan and South Korea.[253]

[252] The Star (2023) Long way to go for country's VC scene, www.thestar.com.my/business/business-news/2023/04/08/long-way-to-go-for-countrys-vc-scene
[253] The Edge Malaysia (2023) Lack of R&D funding stopping Malaysia from achieving high-tech nation status by 2030, says Chang, https://theedgemalaysia.com/node/685945

MYANMAR

In the past decade, Myanmar has implemented a series of policies that support its digital economy and ecosystem and demonstrate its commitment to leveraging digital technologies to enhance its socioeconomic development, catch up with global standards, and widen the range of benefits and opportunities for as many people as possible.

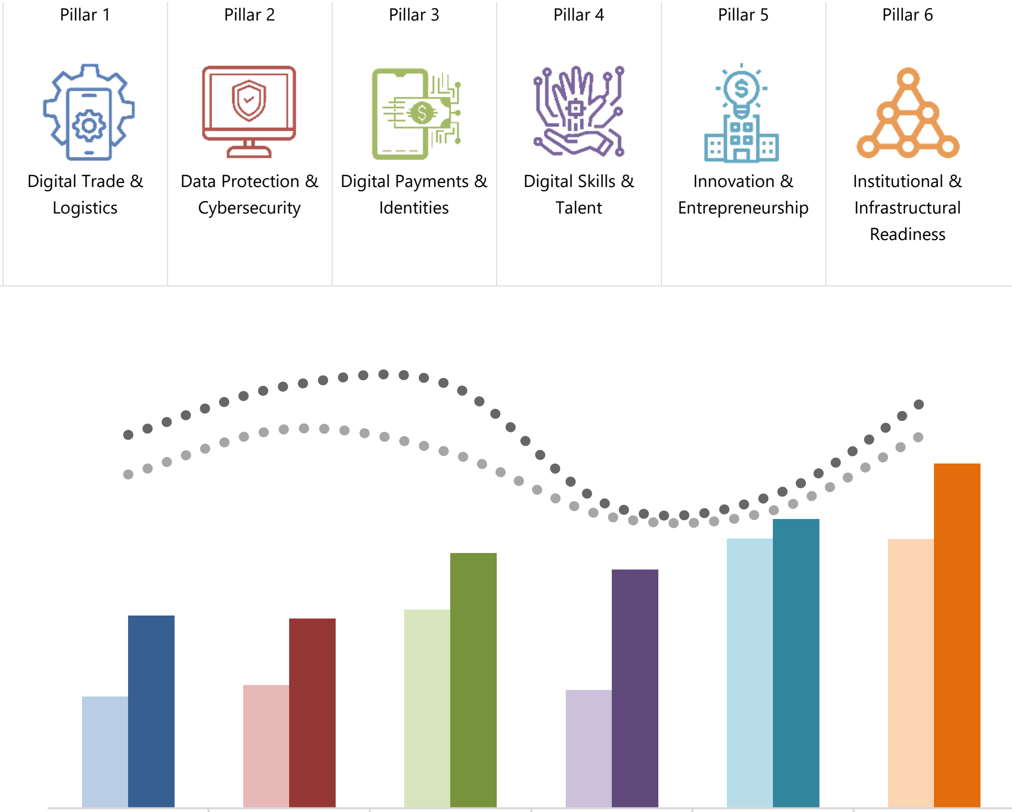
In 2013, the government of Myanmar reformed its telecommunications sector via the *Telecommunications Law* to liberalize the mobile market.[254] The introduction of competition within the sector resulted in visible improvements in internet penetration and smartphone adoption; in 2023, the country counts 22 million internet users, a penetration rate of 44%, and 64.6 million mobile connections.[255] To sustain this encouraging digitalization momentum, the government has formulated several other policies and plans to support digital infrastructure development. For example, in 2018, the *Universal Service Strategy (2018-2020)* was announced to guarantee universal access to telecommunications services in Myanmar[256]—though it remains unclear when this draft will be formalized.[257]

Apart from improving digital infrastructure, the government has set a clear direction for its digital economy. The *Myanmar Digital Economy Roadmap (2018-2025)* was released in 2019 to guide "digital transformation, digital government, digital trade and innovation to develop a digital economy across all sectors". Specifically, the Roadmap sets ambitious targets for the country's digital transformation. Indeed, it aims to: raise digital transformation across business sectors to 10% by 2020 and up to 30% by 2025; increase the utilization of digital technologies by SMEs to 20% in 2020 and 50% in 2025; and enhance digital financial service transactions to 15% in 2020 and 30% in 2025. Additionally, the Roadmap targets substantial increases in foreign direct investment in the digital economy, aiming for USD 8 billion by 2020 and USD 12 billion by 2025.[258] In addition, the government has developed initiatives to digitalize public services, including the *Myanmar e-Governance Master Plan (2016-2020)*, the *Myanmar National Web Portal*, and the *National Data Centre*.[259]

While there are many policies and initiatives in place, the impact of these plans is undercut by the lack of concrete steps to turn them into action. According to the Myanmar Centre for Responsible Business, most of these plans tend to run into issues when it comes to their operationalization and implementation. In other cases, it is more a matter of the plans quickly becoming outdated or simply not being renewed.[260] Moreover, significant gaps in ICT infrastructure,[261] as well as restrictions on internet access, have weighed down the country's digitalization ambitions.[262]

[254] World Bank Blogs (2015) Myanmar's telecom sector takes off, https://blogs.worldbank.org/ppps/myanmars-telecom-sector-takes
[255] Freedom House (2023) Freedom on the Net 2023: Myanmar, https://freedomhouse.org/country/myanmar/freedom-net/2023
[256] Ministry of Transport and Communications Post and Telecom Department (2023) Unversal Service Strategy for Myanmar (2018-2022), www.motc.gov.mm/sites/default/files/Universal%20Service%20Strategy%20%28Draft%29_0.pdf
[257] Myanmar Centre for Responsible Business (2019) Policy Brief: The Legal and Policy Framework for Information Communication Technology (ICT) in Myanmar: Implications for Human Rights, www.myanmar-responsiblebusiness.org/pdf/2019-Policy-Brief-Myanmar-ICT-Legal-Framework_en.pdf
[258] Digital Economy Development Committee (2019) Myanmar Digital Economy Roadmap, https://myanmar.gov.mm/documents/20143/9096339/2019-02-07+DEDC+RoadMap+for+Websites.pdf/
[259] Myanmar Centre for Responsible Business (2019) Policy Brief: The Legal and Policy Framework for Information Communication Technology (ICT) in Myanmar: Implications for Human Rights, www.myanmar-responsiblebusiness.org/pdf/2019-Policy-Brief-Myanmar-ICT-Legal-Framework_en.pdf
[260] Myanmar Centre for Responsible Business (2019) Policy Brief: The Legal and Policy Framework for Information Communication Technology (ICT) in Myanmar: Implications for Human Rights, www.myanmar-responsiblebusiness.org/pdf/2019-Policy-Brief-Myanmar-ICT-Legal-Framework_en.pdf
[261] Oxford Business Group (2020) Myanmar's new digital strategy improves ICT development and network readiness, https://oxfordbusinessgroup.com/reports/myanmar/2020-report/economy/connection-roadmap-new-digital-strategy-for-government-trade-and-investment-to-improve-sector-development-and-network-readiness
[262] The World Bank (2022) Digital & Telecom: Myanmar Infrastructure Monitoring, https://openknowledge.worldbank.org/server/api/core/bitstreams/a04b035a-1194-5886-9be8-47272d62622e/content

Figure 14: Myanmar and ASEAN Scores – ADII 1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Myanmar (ADII 1.0)	18.51	20.41	32.93	19.58	44.65	44.60
Myanmar (ADII 2.0)	31.94	31.44	42.29	39.51	47.93	57.05
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 14 shows that Myanmar's ADII 2.0 Pillar scores are consistently below the ASEAN average. Of the six pillars, Myanmar's strongest performance is in Pillar 6, followed by Pillar 5, and its weakest performance is in Pillar 2, followed by Pillar 1.

Regarding Pillar 6, the government of Myanmar has introduced a range of plans and initiatives to improve the digitalization of the government. In September 2023, the Ministry of Transport and Communications released a draft *e-Governance Master Plan 2030* to establish a national strategic vision for digital government.[263] Indeed, the United Nation's *E-Government Development Index*, which assesses e-government development globally, recognizes that Myanmar has made some progress in digitalizing public services, specifically in areas concerning the provision of online services, telecommunications infrastructure, and human capital development.[264]

In relation to Pillar 2, the relatively high score reflects the fact that various aspects of data protection and privacy are covered by several existing legislations (such as the *Financial Institutions Law*, the *Telecommunications Law*, the *Electronic Transactions Law*).[265] Nevertheless, having a standalone data protection and privacy law in place

[263] MITV (2023) Consultation Workshop: Myanmar E-Governance Master Plan 2030 (Draft), www.myanmaritv.com/news/consultative-workshop-myanmar-e-governance-master-plan-2030-draft
[264] United Nations (2023) UN E-government Knowledgebase, E-Government Development Index (EGDI), https://publicadministration.un.org/egovkb/en-us/About/Overview/-E-Government-Development-Index
[265] DLA Piper (2022) Data Protection Laws of the World: Myanmar, www.dlapiperdataprotection.com/index.html?t=law&c=MM

would undoubtedly streamline, clarify, and complement any existing measures, especially those that were written before the advent of cross-border, trans-jurisdictional data-driven business models and algorithms in day-to-day life.[266]

Similarly, there is no general legislation in place to govern cybersecurity matters in Myanmar. In 2022, the Ministry of Transport and Communications released the revised draft *Cyber Security Law* to set out measures to "regulate activities in the cyberspace, protect critical infrastructure, and establish licensing requirements for cybersecurity and digital platform service providers".[267] The draft was circulated for consultation with relevant stakeholders, but no clear timelines have yet been defined.[268]

Looking at changes over time, Myanmar has made some encouraging progress on all fronts, with the biggest score increase observed for Pillar 4.

Myanmar has indeed multiplied the number of plans and initiatives focusing on digital skills development. In 2020, under Myanmar's *Sustainable Development Plan (2018-2020)*, the government set objectives to develop a comprehensive Technical and Vocational Education Training (TVET) curricular and enable greater access to TVET to under-represented groups.[269] In 2022, the Ministry of Science and Technology released a draft of the *Science, Technology, and Innovation Strategic Plan (2022-2027)* to accelerate the development of digital skills and nurture talents in the field of science and technology.[270] To keep pace with the regional average, more efforts are required to remove existing impediments, such as lack of digital infrastructure, ICT tools, and insufficient trainers and teachers to deliver digital skills training.[271]

In addition to Pillar 4, Myanmar has improved the ADII 2.0 score of Pillar 1, with the score increasing from 18.51 in ADII 1.0 to 31.94 in ADII 2.0. This improvement is likely due to changes made in existing digital trade and logistics systems. For example, the Trade Department of Myanmar added export licenses processing to Myanmar's *Tradenet 2.0* portal, an online system for domestic and international traders to process trade license applications.[272] As a result of these changes, most companies are now able to apply for export and import licenses online. *Tradenet 2.0* also offers 24-hour services for other licensing procedures, such as export and import registrations.[273]

[266] Myanmar Centre for Responsible Business (2019) Policy Brief: A Data Protection Law that Protects Privacy: Issues for Myanmar, www.myanmar-responsiblebusiness.org/pdf/2019-Policy-Brief-Data-Protection_en.pdf
[267] Allen & Gledhill (2022) Myanmar Cyber Security Bill seeks to regulate online activity and access to information, www.allenandgledhill.com/mm/publication/articles/21705/cyber-security-bill-seeks-to-regulate-online-activity-and-access-to-information
[268] Allen & Gledhill (2022) Myanmar Cyber Security Bill seeks to regulate online activity and access to information, www.allenandgledhill.com/mm/publication/articles/21705/cyber-security-bill-seeks-to-regulate-online-activity-and-access-to-information
[269] The Government of the Republic of the Union of Myanmar(2018) Ministry of Planning and Finance. Myanmar Sustainable Development Plan (2018-2030), https://themimu.info/sites/themimu.info/files/documents/Core_Doc_Myanmar_Sustainable_Development_Plan_2018_-_2030_Aug2018.pdf
[270] MITV (2022) Book Launching Ceremony: Science, Technology & Innovation Strategic Plan (2022-2027), www.myanmaritv.com/news/book-launching-ceremony-science-technology-innovation-strategic-plan-2022-2027
[271] International Conference: Digitalising and Greening TVET For Sustainable Development (2022) Issues and Innovative Practices in Digitalising TVET in Myanmar, https://sea-vet.net/images/International_Conference_2022/Panel_1/3._DR.SAI_PPT.pdf
[272] MITV (2023) Trade Net 2.9: Online Import and Expert Licensing Procedures, www.myanmaritv.com/news/trade-net-20-online-import-and-export-licensing-procedures
[273] MITV (2023) Trade Net 2.9: Online Import and Expert Licensing Procedures, www.myanmaritv.com/news/trade-net-20-online-import-and-export-licensing-procedures

THE PHILIPPINES

The Philippines' digital economy has experienced massive growth in recent years, with a 47% growth in Gross Merchandise Value (GMV) from USD 8 billion in 2019 to USD 20 billion in 2022.[274] This rapid growth is set to continue its upward trend, with an expected 20% growth in GMV from USD 20 billion in 2022 to USD 150 billion by 2033.[275] The rapid growth underscores the pivotal role that the Philippines' digital economy plays in the country's socio-economic development.

On the policy front, the need for digital transformation to accelerate post-pandemic economic recovery remains a key policy priority, and the digitalization of key sectors including finance, agriculture, healthcare, and public service delivery remains a critical part of the national economic strategy.[276] To drive the growth of its industries, the Philippines has been actively pursuing foreign investment to grow and strengthen its digital infrastructure.[277]

In tandem with fostering digital innovation, the Philippine government has made significant strides in establishing regulatory safeguards to manage the use of data and technology. Notably, the Philippines passed its *Data Privacy Act* in 2012,[278] significantly ahead of many other Asian economies. Since then, its National Privacy Commission (NPC) has issued a circular governing the registration of data processing system and data protection officer, notification regarding Automated Decision-Making (ADM) or profiling, and the NPC seal of registration.[279]

The Philippines has also responded to perceived gaps in its existing framework by developing new legislation; notably, the *Internet Transactions Act of 2023* (*Republic Act No. 11967*), which governs B2B and B2C internet transactions and applies to various entities, including digital platforms, e-marketplaces, e-retailers, online merchants, and consumers. The legislation delineates specific obligations for involved parties and establishes a comprehensive redress mechanism and liability regime.

These obligations encompass ensuring transaction visibility, safeguarding data privacy, and adhering to relevant laws.[280] There is also the *Philippine Creative Industries Development Act (PCIDA),* known as *Republic Act No. 11904*, which is set to support a collaborative environment for government and local creatives. This policy promotes the development of the Philippine creative industries by protecting and strengthening the rights and capacities of creative firms, artists, artisans, creators, workers, indigenous cultural communities, content providers, and stakeholders. Additionally, the *National Cybersecurity Plan 2023-2028,* developed to safeguard vulnerable targets against cyberattacks,[281] is expected to be released.[282]

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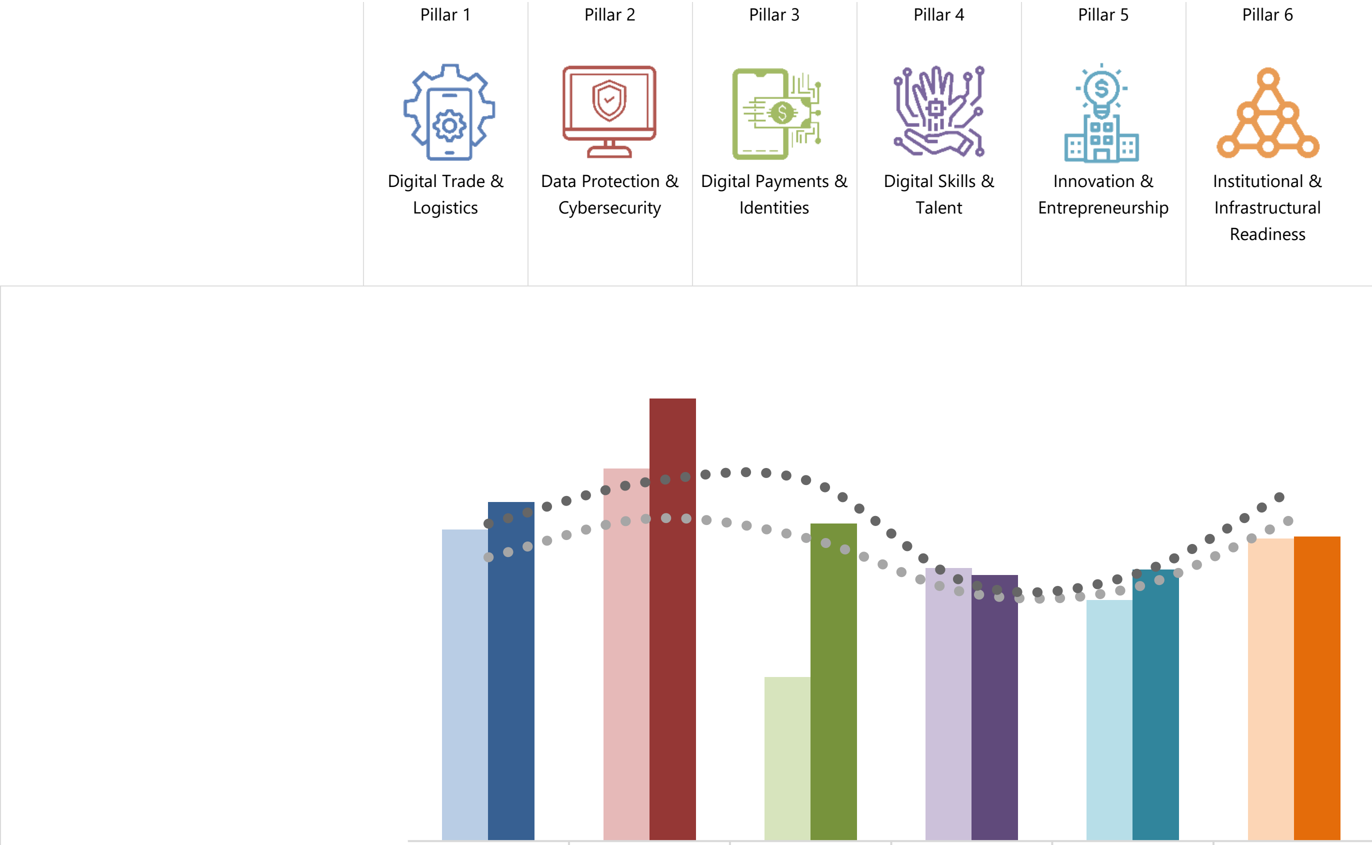
274 Google, Temasek, and Bain & Company (2022) e-Conomy SEA 2022, https://services.google.com/fh/files/misc/philippines_e_conomy_sea_2022_report.pdf
275 Digital Transformation ng Pilipinas (2023) Gov't online shift may boost Philippine digital economy, www.bworldonline.com/technology/2023/01/05/496484/govt-online-shift-may-boost-philippine-digital-economy
276 The Philippines National Economic and Development Authority (2023) Philippine Development Plan 2023-2028, https://pdp.neda.gov.ph/wp-content/uploads/2023/01/PDP-2023-2028.pdf
277 The Straits Times (2023) Marcos wraps up US trip with a deeper alliance – and reassurances for China, www.straitstimes.com/world/united-states/marcos-wraps-up-us-trip-with-deepened-alliance-and-reassurances-for-china
278 The Philippines National Privacy Commission (2023) Data Privacy Act of 2012, https://privacy.gov.ph/data-privacy-act
279 The Philippines National Privacy Commission (2022) Registration of Personal Data Procession System, Notification regarding automated decision-making or profiling, designation of data protection officer, and the national privacy commission seal of registration, https://privacy.gov.ph/wp-content/uploads/2023/05/Circular-2022-04-1.pdf
280 Official Gazette of the Republic of the Philippines (2023) Internet Transactions Act of 2023, www.officialgazette.gov.ph/downloads/2023/12dec/20231205-RA-11967-FRM.pdf
281 Inquirer.Net (2023) PH draws up 5-year cybersecurity plan, https://newsinfo.inquirer.net/1809372/ph-draws-up-5-year-cybersecurity-plan
282 OpenGov Asia (2023) The Philippines National Cybersecurity Plan Shaping a Secure Cyber Landscape, https://opengovasia.com/the-philippines-national-cybersecurity-plan-shaping-a-secure-cyber-landscape

Figure 15: The Philippines and ASEAN Scores – ADII1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Philippines (ADII 1.0)	60.61	72.49	31.89	53.13	46.93	58.89
Philippines (ADII 2.0)	66.00	86.15	61.83	51.77	52.89	59.26
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 15 shows that the Philippines fares comparably well in its digital integration with four of its ADII 2.0 Pillars performing above the regional average (Pillars 1, 2, 4, and 5). Meanwhile, Pillars 3 and 6 have improved since ADII 1.0, but these two pillars perform below the average Pillar scores for ASEAN.

The highest scoring Pillar for the Philippines is Pillar 2. With a score of 86.15, the Philippines surpasses the ASEAN average of 69.99. The Philippines has indeed emerged as a regional frontrunner in data privacy; not only because it was among the first in ASEAN to enact a *Data Privacy Act* in 2012,[283] but also because it is one of two AMS to adopt international standards on data protection. Indeed, the Philippines adheres to the *Asia-Pacific Economic Cooperation (APEC) Cross-Border Privacy Rules (CBPR) System*[284] and participates in the *Global CBPR Forum*.[285] The Philippines has also drawn-up a 5-year national cybersecurity plan, the *National Cyber Security Plan 2023-2028*, which emphasizes a risk-based approach and focuses on identifying individual Critical Information Infrastructure based on its significance and potential economic impact if breached.[286]

[283] National Privacy Commission (2022) PH leads ASEAN's move to protect privacy, https://privacy.gov.ph/ph-leads-asean-move-to-protect-privacy
[284] National Privacy Commission (n.d.) The APEC CBPR System: An Overview, https://privacy.gov.ph/the-apec-cbpr-system-an-overview
[285] Allen & Overy (2023) Global Cross Border Privacy Rules and the new initiatives to support data free flow with trust, www.allenovery.com/en-gb/global/blogs/data-hub/global-cross-border-privacy-rules-and-the-new-initiatives-to-support-data-free-flow-with-trust#:~:text=Just%20over%20a%20year%20ago,CBPR%20Forum%20(the%20Forum)
[286] CICC (n.d.) Stakeholders vow to work for a cyber-resilient ICT nation, https://cicc.gov.ph/news/stakeholders-vow-to-work-for-a-cyber-resilient-ict-nation/

The second-best performing Pillar for the Philippines is Pillar 1; the Philippines scored 66.00, well above the ASEAN average score of 61.82. This is reflective of the country's efforts to foster digital trade. Notably, the Bureau of Customs (BOC) has actively engaged in regional initiatives, implementing paperless trade by participating in the *Framework Agreement on Facilitation of Cross-border Paperless Trade in Asia and the Pacific (CPTA)*.[287] In April 2023, the BOC and the Bureau of Plant Industry (BPI) also took the lead in facilitating the expansion of the *ASEAN Single Window Agreement* through the *Electronic Phytosanitary (e-Phyto) Certificate* system with Indonesia and Thailand.[288]

The two lowest-performing Pillars for the Philippines are Pillars 3 (61.83) and 6 (59.26), both of which are below the corresponding average Pillar scores for ASEAN.

While Pillar 3 is the lowest performing Pillar, it must be noted that the Philippine government has made significant efforts in driving digital ID implementation, as well as in operationalizing digital payments and digital banking. The Philippine Statistics Authority (PSA) continues to amplify its efforts in delivering the benefits of being registered to the *Philippine Identification System (PhilSys)* and is helping realize the goal of digital transformation through the continuous rollout of the *Electronic Philippine IDs (ePhilIDs)*. Over 80 million Filipinos, representing about 87.5% of the population aged five years and above, have been registered to the *PhilSys*. To this end, the Philippine Postal Corporation (PHLPost) has distributed 38.51 million Philippine IDs while the PSA has already issued 40.61 million ePhilIDs.[289] As of December 2023, a total of 83,032,994 Filipinos have been registered, representing 90.3% of the target set for 2023 (92 million).

The Philippines government has also been expanding the ID system across government agencies and social benefits programs to accelerate the digital transformation of government services and the distribution of social benefits. Notably, in July 2023, the PSA signed an MoU with the Social Security System to adopt *PhilSys* in social security services and transactions.[290] The *PhilSys* registry has been integrated into the *National Household Targeting System for Poverty Reduction (Listahanan)* to identify potential social protection beneficiaries.[291] Several government agencies, including the Department of Social Welfare and Development (DSWD), the Department of Foreign Affairs, and the central bank of the Philippines, have pledged their support for ePhilIDs. Notably, the DSWD has established a central database of individuals living on the streets with the help of biometrics.[292]

The pilot authentication services also commenced in April 2023 for applicants requesting Civil Registry Documents (CRD). Out of 15,156 applicants that underwent biometric authentication, a total of 14,976 were successfully authenticated. Moreover, the Department of Information and Communications Technology (DICT) reported a total of 972,821 eGovPh App subscribers. A total of 669,871 successfully verified their accounts, and 437,739 subscribers successfully generated their digital ID as of December 2023. In relation to the authentication of the eGovPH App through PhilSys, the System Integrator (SI) reported a total of 7,972,307 successful authentications as of December 2023.

[287] Philippine News Agency (2023) PH ranks 2nd in Southeast Asia for digital trade facilitation, www.pna.gov.ph/articles/1207342
[288] BOC (2023) BOC, BPI lead regional exchange of e-Phyto Certificate in ASEAN, https://customs.gov.ph/boc-bpi-lead-regional-exchange-of-e-phyto-certificate-in-asean
[289] Philippine News Agency (2023) Over 80M Filipinos now registered to PhilSys, www.pna.gov.ph/articles/1210125
[290] Biometric Update (2023) Philippines ID system expands across government agencies and social benefit programs, www.biometricupdate.com/202308/philippines-id-system-expands-across-government-agencies-and-social-benefit-programs
[291] Biometric Update (2023) Philippines ID system expands across government agencies and social benefit programs, www.biometricupdate.com/202308/philippines-id-system-expands-across-government-agencies-and-social-benefit-programs
[292] Biometric Update (2023) Philippines ID system expands across government agencies and social benefit programs, www.biometricupdate.com/202308/philippines-id-system-expands-across-government-agencies-and-social-benefit-programs

On digital payments, the Bangko Sentral ng Pilipinas (BSP) has launched multiple key initiatives to encourage broader adoption of digital payments, increasing the proportion of people who use digital payments from 31.3% in 2021to 42.1% in 2022.[293] Key initiatives include the introduction of multiple-batch settlements on BSP's electronic funds transfer system, *PESONet*. This initiative facilitates faster funds transfers, leading to an increase in transactions from 5.5 million to 7 million from 2020 to 2021.[294] In addition, the *Bills Pay PH* policy, launched in December 2022, streamlines transactions from various payment service providers, improving interoperability and eliminating the need for physical bill payments.[295]

On innovation and entrepreneurship (Pillar 5), the Philippines also shows great progress thanks to various guiding laws and regulations, including the *Philippine Innovation Act* and the *Innovative Startup Act*.[296] At the same time, the *Philippine Development Plan 2023-2028* defined two key priorities to drive innovation: (a) to nurture a supportive environment for R&D through the building of Knowledge, Innovation, Science and Technology (KIST) parks, university science and technology parks, technology business incubators, laboratories, science and technology research facilities, and other infrastructure; and (b) to establish and promote innovation hubs and other similar collaborative platforms that will ignite and accelerate digital innovation and entrepreneurship. Together, such measures have allowed venture capital to reach innovative businesses, thus facilitating their growth and diversification.

Turning to Pillar 6, challenges related to limited infrastructure investment and provision are being addressed. Notably, government-funded R&D is set to be changed through the review of the *Government Procurement Reform Act (GPRA)*, which aims to encourage innovators to participate in the public procurement process by providing a conducive policy environment and by positioning government organizations as early adopters of innovative, fit-for-purpose digital solutions. Additionally, the Philippine government has rolled out the *Build, Build, Build (BBB)* program aimed at enhancing the country's mobility, connectivity, and economic growth.[297] The BBB Program operationalizes the *Philippine Development Plan 2017-2022*'s target of accelerating public infrastructure expenditure from 5.1% of GDP in 2016 to 7.4% of GDP by the end of 2022.[298] Notable achievements in the road sector are the completed road sections of the *Central Luzon Link Expressway (CLLEX)*, the *Cavite Laguna Expressway*, and the *Manila Cavite Toll Expressway*.[299]

Looking at the changes between ADII 1.0 and ADII 2.0, the Philippines has made progress across five out of six pillars between ADII 1.0 and ADII 2.0, with the exception of Pillar 4. The most notable improvement has been in Pillar 3, where the country's score almost doubled from 31.89 in ADII 1.0 to 61.83 in ADII 2.0. As noted above, this substantial improvement is likely attributed to the Philippine government's key initiatives to foster a wider adoption of digital payments, digital banking, and digital IDs.

However, there was a marginal dip in the Philippines' Pillar 4 score, as it went from 53.13 in ADII 1.0 to 51.77 in ADII 2.0. Despite this decline, the country's score remains higher than the ASEAN average of 50.01. This is likely due to the challenge of bridging the skills gap in response to rapid technological advancements, especially when it comes to ensuring the skills acquired are in adequacy with the needs and priorities of the labor market.

[293] Asian Banking & Finance (2023) Philippines on track to meet digital payments goal: central bank, https://asianbankingandfinance.net/banking-technology/exclusive/philippines-track-meet-digital-payments-goal-central-bank
[294] GovInsider (2023) From coins to clicks: The Philippines' journey to digi-payments, https://govinsider.asia/intl-en/article/from-coins-to-clicks-the-philippines-journey-to-digi-payments
[295] GovInsider (2023) From coins to clicks: The Philippines' journey to digi-payments, https://govinsider.asia/intl-en/article/from-coins-to-clicks-the-philippines-journey-to-digi-payments
[296] Department of Trade and Industry (2019) Innovative Startup Act. https://ecommerce.dti.gov.ph/wp-content/uploads/2020/11/IRR-of-RA-11337-Innovative-Startup-Act.pdf
[297] Philippine News Agency (2022) 'Build, Build, Build' continues: Building more for better lives, www.pna.gov.ph/articles/1179572
[298] https://pdp.neda.gov.ph/wp-content/uploads/2021/02/Pre-publication-copy-Updated-PDP-2017-2022.pdf
[299] Philippine News Agency (2022) 'Build, Build, Build' continues: Building more for better lives, www.pna.gov.ph/articles/1179572

Notably, the newly established Inter-Agency Council (IAC) for Development and Competitiveness of the Philippine Digital Workforce aims to serve as the primary body responsible for enhancing the skills and competitiveness of the Filipino digital workforce. The IAC is also mandated to conduct a nation-wide of digital skills mapping to serve as basis in the formulation of an upcoming *Digital Technology and Digital Skills Roadmap*. This Roadmap will contain strategies to ensure that Filipino workers are equipped with digital skills that are on par with global standards, as well as outline plans for the upskilling, reskilling, and training of the Filipino workforce.[300]

In addition, the Philippine government has been actively implementing policy initiatives aimed at bolstering digital skills and talent. Notably, the Philippine government has taken a significant step towards enhancing its digital workforce's competitiveness with the enactment of *Republic Act (RA) No. 11927* on 30 July 2022, also known as *The Philippine Digital Workforce Competitiveness Act*, and the publication of its *Implementing Rules and Regulations (IRR)* on 27 October 2023.

Various programs and initiatives are enrolled under the law with different national government agencies and other relevant stakeholders on board to lead the activities. One of which is the conduct of the nationwide Digital Technology and Digital Skills Mapping initiative which aims to identify the available digital skills and competencies of the current workforce, including those that are gender related. It also targets to analyze the skills gaps by matching the supply of digital skills based on needs assessment exercises with the industries, and available labor market intelligence data vis-a-vis the current labor market demand, and evolving and emerging jobs and skills. The initiative also intends to determine the training needs of the current workforce for reskilling, upskilling, retooling, and other training programs, analyze the demographics of the Philippine workforce in digital technology and sectors, and identify the available and needed digital platforms, and ICT infrastructure, among others, to ensure accessibility of the digital workforce.

Lastly, in a bid to enhance cybersecurity expertise, the Department of Information and Communications Technology (DICT) has partnered with Google Philippines to empower civil servants with globally competitive skills for the growing cybersecurity field in the country.[301]

[300] Congress of the Philippines (2021) Republic Act No. 11927 - Act to Enhance the Philippine Digital Workforce Competitiveness, Establishing for the Purpose an Inter-Agency Council for Development and Competitiveness of Philippine Digital Workforce and for Other Purposes, https://legacy.senate.gov.ph/republic_acts/ra%2011927.pdf

[301] Philippine Information Agency (2023) DICT ensures no Filipino left behind upskilling, digital literacy push, https://pia.gov.ph/news/2023/07/24/dict-ensures-no-filipino-left-behind-upskilling-digital-literacy-push

SINGAPORE

Singapore stands out as a leading digital economy in ASEAN. In 2022, the digital economy, measured by value added for information and communications sector and the digitalization of the rest of the economy, contributed USD 79 billion, representing approximately 17% to the country's gross domestic product (GDP), an increase from USD 43.2 billion or 13% of GDP in 2017. Key areas of growth within the information and communications sector are e-commerce, online services, and gaming, while most of the value added from digitalization of key economic sectors stems from finance and insurance, wholesale trade, and manufacturing.[302]

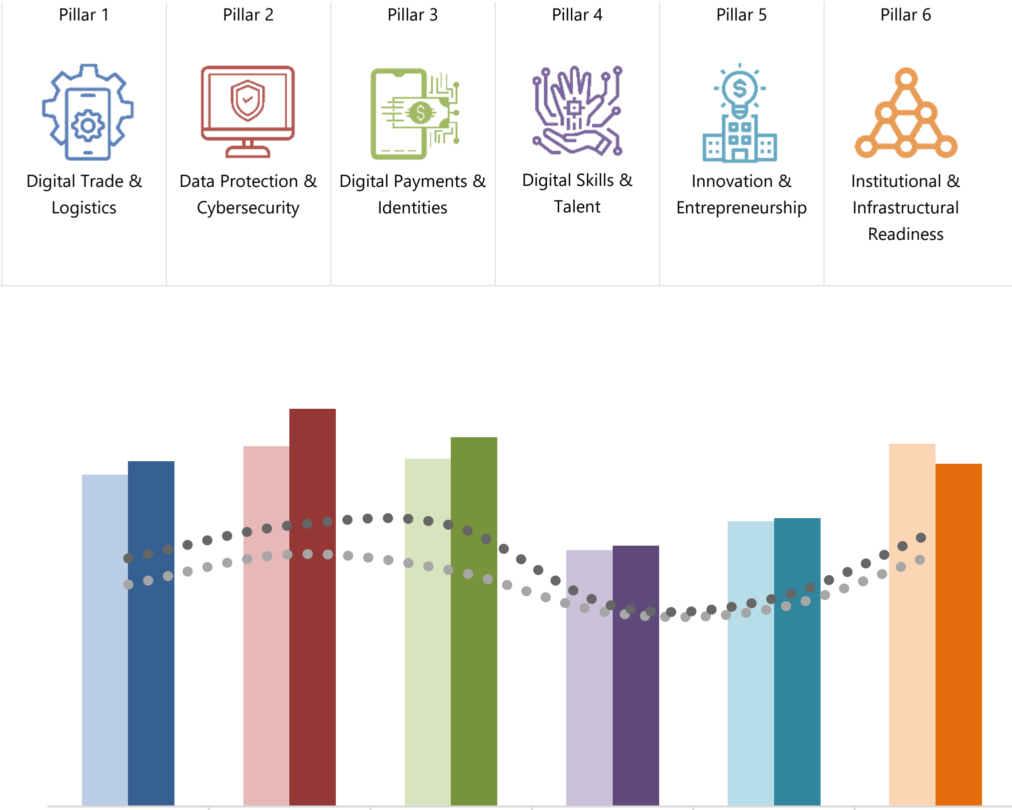
The Singapore government has played a pivotal role in steering the course of the country's digital transformation. Since the launch of the *Smart Nation Vision* in 2014, numerous initiatives have been introduced spanning AI, business and finance, digital government services, health, safe digital spaces, transportation, and smart urban living. Each initiative is designed to leverage technology for digital transformation across the three pillars of Singapore's *Smart Nation Vision*: digital government, digital economy, and digital society.[303] A stand-out initiative is the introduction of Singapore's national digital ID, *Singpass*, globally recognized as an exemplar of a robust foundational digital ID.[304]

Beyond this, the government has introduced policies aimed at fostering innovation and entrepreneurship in the digital economy. Notably, recognizing the significance of SMEs' contribution to the GDP and workforce employment, the government launched the *SME Go Digital* initiative in 2017, with the aim to help SMEs to adapt to changing times and capitalize on advancements in digital solutions that can enhance their operations and productivity.[305] Central to the *SME Go Digital* initiative are sector-specific industry digital plans (IDP) that offer SMEs step-by-step guides to identify suitable digital solutions. These plans include training programs designed to equip employers and their employees with the necessary skillset along their digitalization journey. Success from the IDPs, and more broadly, from the *SME Go Digital initiative*, is perhaps evident from the increased digitalization of SMEs over the past five years, which grew from 74% in 2018 to 94% in 2022.[306]

Elsewhere, Singapore has been investing significant resources in strengthening the public sector's engineering expertise and building up the government's capabilities in emerging technologies. In 2019, the government established five capabilities centers to support the development of infocomm technologies and smart systems, with each center focused on a specific area: (1) application design, development, and deployment; (2) cybersecurity; (3) data science and AI; (4) government ICT infrastructure; and (5) smart city technology. The five areas are integral to Singapore's digital government blueprint and *Smart Nation* agenda, and aim to deliver innovative, citizen-centric products and services for the next phase of the country's digital transformation.[307]

[302] IMDA (2023) Singapore Digital Economy Report 2023, www.imda.gov.sg/-/media/imda/files/infocomm-media-landscape/research-and-statistics/sgde-report/singapore-digital-economy-report-2023.pdf
[303] Smart Nation Singapore (n.d.) Initiatives, www.smartnation.gov.sg/initiatives/strategic-national-projects
[304] World Bank Group, GovTech (2022) National digital identity and government data sharing in Singapore – A case study of Singpass and APEX, www.developer.tech.gov.sg/assets/files/GovTech%20World%20Bank%20NDI%20APEX%20report.pdf
[305] IMDA (n.d.) SMEs Go Digital, www.imda.gov.sg/how-we-can-help/smes-go-digital
[306] IMDA (2023) Singapore Digital Economy Report 2023, www.imda.gov.sg/-/media/imda/files/infocomm-media-landscape/research-and-statistics/sgde-report/singapore-digital-economy-report-2023.pdf
[307] GovTech (2023) Singapore digital government journey, www.developer.tech.gov.sg/our-digital-journey/singapore-digital-government-journey/overview.html

Figure 16: Singapore and ASEAN Scores – ADII1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Singapore (ADII 1.0)	82.64	89.70	86.60	63.79	71.08	90.36
Singapore (ADII 2.0)	86.00	99.09	91.98	64.95	71.81	85.41
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 16 shows that Singapore scores above the ASEAN average across all six ADII pillars, with Pillar 2 its strongest overall ADII 2.0 Pillar and Pillar 4 its weakest.

The high score of Pillar 2 likely reflects the country's robust regulatory framework for data protection and cybersecurity, overseen respectively by the Personal Data Protection Commission (PDPC) and the Cyber Security Agency of Singapore (CSA). The PDPC regularly issues guidelines and guides to assist organizations in complying with the country's *Personal Data Protection Act*, including for new or emerging technologies. For example, in July 2022, the PDPC issued a guide to help organizations apply basic anonymization techniques on datasets,[308] a guide to help organizations use security cameras and biometric recognition systems responsibly and safeguard individuals' biometric data,[309] and a guide addressing personal data protection considerations for blockchain design to help organizations understand how to comply with the PDPA when deploying blockchain applications.[310]

[308] PDPC (2022) Guide to Basic Anonymisation, www.pdpc.gov.sg/News-and-Events/Announcements/2022/03/Guide-to-Basic-Anonymisation-Now-Available
[309] PDPC (2022) Guide on the Responsible Use of Biometric Data in Security Applications, www.pdpc.gov.sg/News-and-Events/Announcements/2022/05/Guide-on-the-Responsible-Use-of-Biometric-Data-in-Security-Applications-Now-Available
[310] PDPC (2022) Guide on personal data protection considerations for blockchain design, www.pdpc.gov.sg/news-and-events/announcements/2022/07/guide-on-personal-data-protection-considerations-for-blockchain-design

The PDPC was also responsible for publishing the first and second editions of the *Model AI Governance Framework*, which sets out implementable guides for organizations to address key ethical and governance issues when deploying AI solutions.[311] Another initiative under PDPC's purview is the *Data Protection Trustmark (DPTM)*, a voluntary enterprise-wide certification enabling organizations to showcase their commitment to accountable data protection practices. This multi-faceted approach underscores Singapore's commitment to fostering a secure and ethical digital landscape.[312]

On cybersecurity, the CSA has been tasked to review and update Singapore's cybersecurity posture to supporting a growing digital economy, which may involve expanding Singapore's *Cyber Security Act* to improve threat awareness levels, protect virtual assets such as systems hosted on the cloud, and secure important digital infrastructure, beyond critical information infrastructure. In tandem with this, Singapore has introduced its *Safer Cyberspace Masterplan 2020* and the *Cybersecurity Strategy 2021*. The Masterplan outlines 11 initiatives to better mitigate and address cyber-threats, several of which have been implemented, such as CSA's internet hygiene portal and the cybersecurity labelling scheme.[313] The Strategy, meanwhile, complements the Masterplan and outlines Singapore's updated goals and approach to cyber-space, including building a vibrant cybersecurity ecosystem through research and innovation, and growing a robust cyber-talent pipeline.[314] The implementation of these initiatives, among others, is likely to have contributed to the high scored achieved in this pillar.

With a score of 64.95, Pillar 4 is Singapore's lowest-performing ADII 2.0 Pillar, yet it remains above the ASEAN average of 49.87. This is likely reflective of the government's concerted efforts to work closely with industry and local universities to cultivate a skilled ICT workforce tailored for Singapore's digital economy. For example, Workforce Singapore, a statutory board under the Ministry of Manpower, has published job transformation maps that offer insights into the technologies shaping the future of jobs across various sectors.[315] These maps guide workers and employers on successfully navigating changes through job redesign and reskilling of professionals.[316]

In conjunction with the job transformation maps is the national program *TechSkills Accelerator (TeSA)*, designed to equip individuals with digital skills to stay competitive. *TeSA* also provides guidance for employers in redesigning tech jobs for at-risk workers. Led by the Infocomm Media Development Authority (IMDA) in collaboration with other government agencies, the local trade union, and industry, TeSA offers reskilling courses for in-demand digital skills.[317] In September 2023, IMDA appointed 5 Training Partners to scale upskill/reskill effort of 18,000 individuals over the next three years in AI and related tech skills, such as software engineering, cloud, and mobility, which are complimentary to AI/GenAI. IMDA will collaborate with the appointed training partners, including local universities, to create a suite of reskilling courses incorporating critical skills required by AI professionals, such as AI coding for programmers and natural language processing for data analytics. The training initiative includes partnerships with technology suppliers, such as multinational tech companies, and tech product start-ups. Over 20 leading companies, across diverse industries such as financial services, retail, hospitality, and education, have expressed their intention to collaborate with IMDA and the training partners to reskill their employees.[318]

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[311] PDPC (2023) Singapore's Approach to AI Governance, www.pdpc.gov.sg/help-and-resources/2020/01/model-ai-governance-framework
[312] IMDA (2023) Data protection trustmark certification, www.imda.gov.sg/how-we-can-help/data-protection-trustmark-certification
[313] CSA (2020) Safer Cyberspace Masterplan, www.csa.gov.sg/Tips-Resource/publications/2020/safer-cyberspace-masterplan
[314] CSA (2021) National Cybersecurity Strategy 2021, www.csa.gov.sg/Tips-Resource/publications/2021/singapore-cybersecurity-strategy-2021
[315] IMDA (2023) Jobs transformation map, www.imda.gov.sg/how-we-can-help/techskills-accelerator-tesa/jobs-transformation-map
[316] Workforce Singapore (2023) Jobs transformation map, www.wsg.gov.sg/home/employers-industry-partners/jobs-transformation-maps
[317] Skills Future (2023) TechSkills Accelerator, www.skillsfuture.gov.sg/initiatives/early-career/tesa
[318] IMDA (2023) IMDA leads nationwide AI skill to build AI talent pool, www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2023/imda-leads-ai-skilling-to-build-ai-talent-pool

Looking at trends over time, Singapore has demonstrated positive progress across all ADII Pillars, except for Pillar 6. The most substantial improvement is observed in Pillar 2, which increased from 89.7 in ADII 1.0 to 99.09 in ADII 2.0. This positive trajectory underscores the government's commitment to fortifying the country's digital defenses and creating a secure and trusted digital space. For example, in addition to the PDPC guidelines and CSA initiatives detailed above, the score is also reflective of the positive impact from amendments to the PDPA introduced in 2021; these amendments consider technological advancements, new business models, and global development in data protection legislation. They also enhance PDPC's enforcement authority and raise the financial penalty cap for breaches of the PDPA.[319]

Apart from Pillar 2, Singapore has also made notable improvements in its score for Pillars 1 and 3. On Pillar 1, Singapore's score improved from 82.64 in ADII 1.0 to 86.00 in ADII 2.0, an upward trend that likely reflects the positive impact of national initiatives such as *InvoiceNow*,[320] the *Networked Trade Platform*,[321] *TradeNet*,[322] and the much broader *Logistics Industry Digital Plan*.[323] Each of these initiatives has aimed to enhance the productivity and resilience of the local logistics industry through greater adoption and innovative use of digital technologies. For Pillar 3, Singapore's score has improved from 86.6 in ADII 1.0 to 91.98 in ADII 2.0, reflecting Singapore's increasingly mature digital payments sector, where 90% of consumer payments occur via digital channels, coupled with the widespread use of its national digital ID, Singpass.

Singapore saw a minor decline in performance for Pillar 6, dropping from a score of 90.36 in ADII 1.0 to 85.41 in ADII 2.0. The score is likely the result of a combination of factors, including the decision to maintain the scores from ADII 1.0 for indicator 6.5, and updated data from a national survey with similar methodologies for indicator 6.4. Nevertheless, the country still scores above the ASEAN average of 68.2, reflecting Singapore's generally strong performance in this Pillar. Progress for Pillars 4 and 5 is minimal and reflects Singapore's decision to maintain the score from ADII 1.0 for three of the five indicators measured for both pillars.

[319] Morgan Lewis (2022) Singapore personal data protection act changes have implications for health sector, www.morganlewis.com/pubs/2022/08/singapore-personal-data-protection-act-changes-have-implications-for-healthcare-sector
PDPC (2022) Amendments to enforcement under personal data protection act (PDPA) in updated advisory guidelines and guide, www.pdpc.gov.sg/news-and-events/announcements/2022/09/amendments-to-enforcement-under-the-personal-data-protection-act-in-updated-advisory-guidelines-and-guide

[320] Launched in 2019, InvoiceNow is a nationwide e-invoicing network that neables companies of all sizes to process invoices in more efficient manner, reducing business costs and shortening payment cycles. InvoiceNow operates on the open standard Peppol framework, which directly transmits e-invoices in a standard digital format across different finance systems. IMDA (2023) InvoiceNow, www.imda.gov.sg/how-we-can-help/nationwide-e-invoicing-framework/invoicenow

[321] Officially launched in 2018, the Networked Trade Platform (NTP) is a one-stop trade and logistics ecosystem developed by Singapore Customs and GovTech. The platform brings players across the trade value chain into a single platform to enable end-to-end digital trade. The NTP is a key aspect of the government's efforts to drive industry wide digital transformation. GovTech Singapore (2023) Networked Trade Platform, www.tech.gov.sg/products-and-services/networked-trade-platform/

[322] TradeNet is Singapore's National Single Window for trade declaration. It provides a single platform for Singapore's trade and logistics community to fulfill all import, export, and transhipment related regulatory requirements. TradeNet (2023) Welcome to TradeNet, www.tradenet.gov.sg/tradenet/login.jsp

[323] The Logistics industry digital plan is part of the SME Go Digital program by IMDA to help SMEs go digital. The IDP provides a step-by-step guide on digital solutions SMEs can adopt at each stage of growth. IMDA (2023) Logistics Industry Digital Plan, www.imda.gov.sg/how-we-can-help/smes-go-digital/industry-digital-plans/logistics-idp

THAILAND

Thailand is one of the countries in ASEAN that has witnessed the fastest growth in digital infrastructure and accessibility, a trend that has been further accelerated by the COVID-19 pandemic.[324] The Thailand Board of Investment (BOI) estimates that the country's digital economy is poised for greater growth and is expected to contribute about 25% of Thailand's gross domestic product (GDP) by 2027, up from 17% in 2018.[325]

Digitalization has transformed numerous service sectors in Thailand, with most notable advancements observed in financial services. The central bank plans to issue new digital banking licenses in 2024 to provide better customer experiences and increase the reach of financial services across Thailand.[326]

The Thai government's commitment towards technological development is evident through its many digital economy plans and smart city initiatives. It is focused on advancing the adoption of new technologies under the *Thailand 4.0* vision, which includes AI, 5G, Smart Cities, with various initiatives developed by the Ministry of Digital Economy and Society (MDES). A major push by Thailand towards digitalization is reflected in the MDES' *Thailand Digital Economy and Society Development Plan (2018-2037)*, a 20-year national digital blueprint for digital transformation of government operations and businesses. Numerous digital priorities were recently unveiled in September 2023, including: the *Go Cloud First* policy, digital identities (*ThaID*), the *Big Data Institute*, and the *City Data Platform*, all focused on enhancing digital infrastructure.[327]

These initiatives provide strong impetus to Thailand's digital economy, which ranks 38th in the Institute for Management Development (IMD)'s Digital *World Competitiveness Ranking* in 2021, the highest place it has achieved.[328]

[324] ISEAS (2023) Governing the Digital Economy in Thailand: Domestic Regulations and International Agreements, www.iseas.edu.sg/wp-content/uploads/2023/06/ISEAS_Perspective_2023_58.pdf
[325] Thailand Board of Investment (2017) Data Center and Cloud Service in Thailand, www.boi.go.th/index.php?page=business_opportunities_detail&topic_id=128004&language=en
[326] Google, Temasek, and Bain & Company (2023) Country spotlight: Thailand, https://services.google.com/fh/files/misc/thailand_e_conomy_sea_2023_report.pdf
[327] Ministry of Digital Economy and Society (2023) "Prasert" Digital Minister explains government policy to prepare to use technology to develop the country; emphasis on transparency and fairness, www.mdes.go.th/news/detail/7327--%E0%B8%9B%E0%B8%A3%E0%B8%B0%E0%B9%80%E0%B8%AA%E0%B8%A3%E0%B8%B4%E0%B8%90--%E0%B8%A3%E0%B8%A1%E0%B8%A7-%E0%B8%94%E0%B8%B4%E0%B8%88%E0%B8%B4%E0%B8%97%E0%B8%B1%E0%B8%A5-%E0%B8%8A%E0%B8%B5%E0%B9%89%E0%B9%81%E0%B8%88%E0%B8%87%E0%B8%99%E0%B9%82%E0%B8%A2%E0%B8%9A%E0%B8%B2%E0%B8%A2%E0%B8%A3%E0%B8%B1%E0%B8%90%E0%B8%9A%E0%B8%B2%E0%B8%A5-%E0%B9%80%E0%B8%95%E0%B8%A3%E0%B8%B5%E0%B8%A2%E0%B8%A1%E0%B8%99%E0%B8%B3%E0%B9%80%E0%B8%97%E0%B8%84%E0%B9%82%E0%B8%99%E0%B9%82%E0%B8%A5%E0%B8%A2%E0%B8%B5%E0%B8%9E%E0%B8%B1%E0%B8%92%E0%B8%99%E0%B8%B2%E0%B8%9B%E0%B8%A3%E0%B8%B0%E0%B9%80%E0%B8%97%E0%B8%A8-%E0%B9%80%E0%B8%99%E0%B9%89%E0%B8%99%E0%B8%84%E0%B8%A7%E0%B8%B2%E0%B8%A1%E0%B9%82%E0%B8%9B%E0%B8%A3%E0%B9%88%E0%B8%87%E0%B9%83%E0%B8%AA%E0%B9%80%E0%B8%9B%E0%B9%87%E0%B8%99%E0%B8%98%E0%B8%A3%E0%B8%A3%E0%B8%A1
[328] Bangkok Post (2021) Thailand achieves highest digital competitiveness rank to date, www.boi.go.th/index.php?page=business_opportunities_detail&topic_www.bangkokpost.com/business/2190243/thailand-achieves-highest-digital-competitiveness-rank-to-date#:~:text=This%20year%2C%20the%20company%20has,has%20achieved%20since%20ranking%20startedid=128004&language=en

Figure 17: Thailand and ASEAN Scores – ADII1.0 & ADII 2.0

	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Thailand (ADII 1.0)	83.34	87.91	68.73	43.76	56.09	62.61
Thailand (ADII 2.0)	87.37	93.84	91.38	45.68	57.35	74.21
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 17 shows that Thailand performs strongly across all but one ADII 2.0 Pillars, with scores performing above the ASEAN average except for Pillar 4. It is notable that Thailand's highest performing pillars—Pillars 2, 3, and 1)—are significantly above the ASEAN average, signifying that Thailand is a leading digital economy in these areas. Much like most other AMS, Thailand's weakest Pillar is Pillar 4, which suggests that there is room for improvement.

Thailand's high performance in Pillar 2 can be attributed to its comprehensive legal and regulatory environment for data protection and cybersecurity. Indeed, the *Personal Data Protection Act (PDPA)* became fully enforceable in June 2022 and constitutes a significant milestone in data protection given that it is considered the first Thai law designed to govern data protection in the digital age.[329] Meanwhile, on the cybersecurity front, 2022 saw the National Cyber Security Agency (NCSA) roll out 34 of the 40 proposed subordinate regulations of the Cybersecurity Act (2019).[330] The latest measures include *Standards for Defining the Attributes of Cybersecurity for Data or Information Systems* and *Standards and Guidelines for Promoting the Development of Cybersecurity Service Systems*. In addition, the Cabinet approved the *Cybersecurity Policy and Operation Plan 2022-2027*, a guiding masterplan aimed at monitoring and strengthening national cybersecurity.[331] Looking at sector-specific

[329] International Trade Administration (2022) Thailand Personal Data Protection Act, www.trade.gov/market-intelligence/thailand-personal-data-protection-act#:~:text=The%20PDPA%20is%20considered%20the,storage%2C%20and%20data%20consent%20protocols.
[330] OpenGov Asia (2022) Thailand Proposes Additional Regulations to Boost Cybersecurity, https://opengovasia.com/thailand-proposes-additional-regulations-to-boost-cybersecurity
[331] Bangkok Post (2022) Agency assesses Thailand's organisational preparedness, www.bangkokpost.com/business/2461752/agency-assesses-thailands-organisational-preparedness

measures, the Thai Parliament updated its *Royal Decree on Measures to Prevent and Suppress Technology Crime B.E.* in 2023, granting more power to relevant agencies to suppress financial crimes and requiring banks to suspend any suspicious transactions immediately.[332]

Regarding Pillar 3, Thailand scored strongly as well, signifying successful efforts in advancing digital payments and identities in the country. The Bank of Thailand, with the aim of promoting financial inclusion and competition within the Thai financial market, is set to issue three virtual bank licenses in 2024.[333] Digital payments also continue to grow, with an estimated 95% of total transactions in Thailand among consumers to be done via online channels in 2023.[334] In promoting digital identities, Thailand is pursuing the first phase of its *Digital ID framework (2022-2024)* and is targeting 10 million registrants by the end of 2024. In May 2023, DOPA published the *ThaID-D.DOPA* mobile application (the national ID scheme operated by the Department of Provincial Administration, DOPA), which gives users access to a digital version of their national ID card.[335]

As for Pillar 1, Thailand's high score may be influenced by the piloting of its blockchain-based *National Digital Trade Platform (NDTP)* in 2022, which aims to digitize the trade process to speed up imports and exports and improve access to trade finance for SMEs.[336] Thailand is also notable for its implementation of digital trade legislation, with the current regulatory framework on digital trade encompassing various dimensions, including electronic transactions, consumer protection, privacy, data protection, cybercrime, and the adoption of the United Nations Commission on International Trade Law (UNCITRAL)'s model law on e-commerce.[337]

Pillar 4, Thailand's lowest performing ADII 2.0 Pillar, also scores below the ASEAN Pillar average score, suggesting that despite many initiatives in this domain, there is some catching up to do. Efforts are indeed being made to enhance Thailand's performance on that front. To address talent gaps, the Office of National Higher Education, Science, Research, and Innovation Policy Council (NXPO) is driving public policies to support digital talent development, including the *Science, Research and Innovation Fund* and the soon-to-be-launched *Higher Education Fund*. Meanwhile, the *Equitable Education Fund (EEF)* provides 2,500 scholarships annually to support students from low-income families to enter vocational colleges.[338] These scholarships, as well as funding from the private sector, can be tapped to foster digital talent development. At the same time, Thailand's National Digital Economy and Society Commission (NDESC) has created 11 skill standards, which have been adopted by ASEAN, to help businesses assess employees' skillsets against industry norms and prepare for AI-driven growth.[339]

In promoting innovation and entrepreneurship under Pillar 5, Thailand is focused on encouraging innovation and SMEs to adopt technology in their businesses.[340] The Digital Economy Promotion Agency (DEPA) has been key in educating, providing funding and assisting entrepreneurs to get started on digitalization. For instance, it has committed to continue supporting the private sector to develop digital content and services that are

[332] Pattaya Mail (2023) Parliament updates cybercrime law to target online scams and financial crimes, www.pattayamail.com/thailandnews/parliament-updates-cybercrime-law-to-target-online-scams-and-financial-crimes-437662
[333] The Nation (2023) Thai central bank to issue 3 virtual bank licences next year: governor, www.nationthailand.com/thailand/economy/40029536
[334] Bangkok Post (2023) Rise of online payments accelerates march to cashless society, www.bangkokpost.com/business/general/2480872/rise-of-online-payments-accelerates-march-to-cashless-society
[335] Bangkok Post (2023) 'ThaID' officially launches to spearhead Thailand 4.0, www.bangkokpost.com/life/tech/2602885/thaid-officially-launches-to-spearhead-thailand-4-0
[336] Ledger Insights (2022) Thailand launches national trade blockchain, integrates Japan's TradeWaltz, Singapore's NTP, www.ledgerinsights.com/blockchain-thailand-trade-tradewaltz-singapore-ntp/
[337] ADB (2022) Aid for Trade in Asia and the Pacific: Leveraging Trade and Digital Agreements for Sustainable Development, www.adb.org/publications/aid-trade-asia-pacific-trade-digital-agreements
[338] Office of National Higher Education Science Research and Innovation Policy Council (2022) MHESI and MDES join forces to strengthen an ecosystem for digital talent development, www.nxpo.or.th/th/en/14288/
[339] OpenGovAsia (2023) Driving AI Competence: Thailand's National Standards for ICT Professionals, https://opengovasia.com/driving-ai-competence-thailands-national-standards-for-ict-professionals/
[340] The FinLab (2022) How Ecosystem Cooperation Keeps The Digitalisation Momentum Going, https://thefinlab.com/success-stories/how-ecosystem-cooperation-keeps-the-digitalisation-momentum-going/

competitive and helpful to tourists, partnering with a platform that uses AI and big data to analyze tourists' preferences and travel demands, to effectively support SMEs.[341] In 2022, DEPA also jointly established the first 5G innovation center and laboratory with telecom operator AIS in Thailand Digital Valley in the Eastern Economic Corridor (EEC) with the aim of strengthening the country's 5G ecosystem and improving workforce skills. Looking ahead, Thailand is set to introduce more initiatives. For instance, in November 2023, the digital minister announced a 50% cut in corporate tax for companies that invest in a smart-city province to incentivize smart city development.[342]

Turning to Pillar 6, numerous digital priorities have been announced to enhance digital infrastructure: the *Go Cloud First* policy, the *Big Data Institute*, and the *City Data Platform*, to name but the biggest ones.[343] Institutional arrangements have also been established to support digital transformation; in May 2020, a national 5G committee was formed with the aim to develop a roadmap for 5G adoption and enhance cooperation among relevant agencies for 5G development.[344] In February 2021, the government created the *Thailand National AI Strategy Working Committee*[345] to drive the AI development and application by accelerating infrastructure development, technology and innovation development and skill development through collaboration between the public and private sectors.[346]

Looking at changes over time, Thailand has made positive progress for all pillars from ADII 1.0 to ADII 2.0. The biggest score increase was for Pillar 3, where it went from 68.73 in ADII 1.0 to 91.38 in ADII 2.0—raising it significantly above the average Pillar score for ASEAN. This is a testament to Thailand's strong progress in promoting digital payments and identities, as described earlier in this section. Implementing significant Fintech initiatives such as virtual bank licenses, promoting cross-border payment links,[347] and its digital ID program has helped positioned Thailand as a forerunner in the region.

It is also notable that Thailand's improvements in Pillar 6 have brought it above the ASEAN average, suggesting that the increase was contributed by increases in Thailand's proportion of Internet users and the responsiveness of the Thai government to disruption and change. Indeed, the internet penetration rate for Thai households stands at 91% at the end of 2023, up from 89.1% in early 2022.[348]

[341] The Nation (2022) State agency-backed digital platform helps keep SMEs competitive in tourism sector, https://www.nationthailand.com/thailand/policies/40021574

[342] The Nation (2023) Thailand offers 50% tax cut to boost smart cities, www.nationthailand.com/thailand/policies/40033035

[343] Ministry of Digital Economy and Society (2023) "Prasert" Digital Minister explains government policy to prepare to use technology to develop the country; emphasis on transparency and fairness, www.mdes.go.th/news/detail/7327--%E0%B8%9B%E0%B8%A3%E0%B8%B0%E0%B9%80%E0%B8%AA%E0%B8%A3%E0%B8%B4%E0%B8%90--%E0%B8%A3%E0%B8%A1%E0%B8%A7-%E0%B8%94%E0%B8%B4%E0%B8%88%E0%B8%B4%E0%B8%97%E0%B8%B1%E0%B8%A5-%E0%B8%8A%E0%B8%B5%E0%B9%89%E0%B9%81%E0%B8%88%E0%B8%87%E0%B8%99%E0%B9%82%E0%B8%A2%E0%B8%9A%E0%B8%B2%E0%B8%A2%E0%B8%A3%E0%B8%B1%E0%B8%90%E0%B8%9A%E0%B8%B2%E0%B8%A5-%E0%B9%80%E0%B8%95%E0%B8%A3%E0%B8%B5%E0%B8%A2%E0%B8%A1%E0%B8%99%E0%B8%B3%E0%B9%80%E0%B8%97%E0%B8%84%E0%B9%82%E0%B8%99%E0%B9%82%E0%B8%A5%E0%B8%A2%E0%B8%B5%E0%B8%9E%E0%B8%B1%E0%B8%92%E0%B8%99%E0%B8%B2%E0%B8%9B%E0%B8%A3%E0%B8%B0%E0%B9%80%E0%B8%97%E0%B8%A8-%E0%B9%80%E0%B8%99%E0%B9%89%E0%B8%99%E0%B8%84%E0%B8%A7%E0%B8%B2%E0%B8%A1%E0%B9%82%E0%B8%9B%E0%B8%A3%E0%B9%88%E0%B8%87%E0%B9%83%E0%B8%AA%E0%B9%80%E0%B8%9B%E0%B9%87%E0%B8%99%E0%B8%98%E0%B8%A3%E0%B8%A3%E0%B8%A1

[344] Telecom.com (2020) Thailand forms committee for 5G development led by PM Prayut Chan-o-cha, https://telecom.economictimes.indiatimes.com/news/thailand-forms-committee-for-5g-development-led-by-pm-prayut-chan-o-cha/75937948

[345] OECD AI Policy Observatory (2022) Thailand's AI strategy to boost economic and social wellbeing, https://oecd.ai/en/wonk/thailand-ai-strategies

[346] National Science and Technology Development Agency (2022) Prime Minister chairs the first meeting of National AI Committee, www.nstda.or.th/en/news/news-years-2022/prime-minister-chairs-the-first-meeting-of-national-ai-committee.html

[347] Bank of Thailand (2023) Cross-border Payment Linkages, www.bot.or.th/en/financial-innovation/digital-finance/digital-payment/cross-border-payment.html

[348] National Statistical Office of Thailand (2023) ICT Use in Thailand (Q3 2023), www.nso.go.th/nsoweb/storage/survey_detail/2023/20231113101421_58455.pdf

VIET NAM

As one of the most dynamic economies in ASEAN, Viet Nam is seen as a new "Asian Tiger".[349] Its digital economy is rapidly growing and diversifying, hitting USD 23 billion GMV in 2022 and expected to reach USD 50 billion by 2025—making it one of the fastest growing markets in Asia.[350]

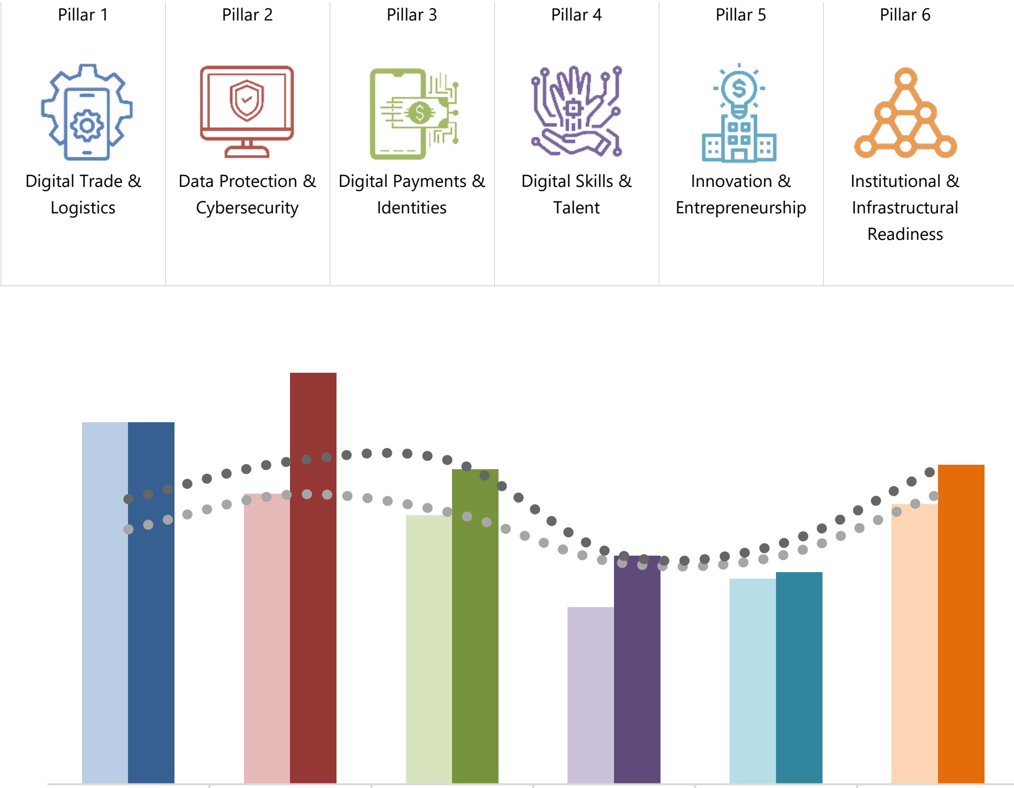
To realize its digital ambitions, Viet Nam will require a significant increase in its digital capabilities, supported by robust policies and legislative frameworks. Viet Nam has launched several national strategies and policies to guide its digital transformation in the years ahead, a process that was accelerated in 2019 by the COVID-19 pandemic. This acceleration culminated in 2020 with the launch of two key national strategies to achieve nationwide digitalization through selected sectors and technologies: the *National Digital Transformation Program*, which sets out the dual goal of (i) developing a digital government, a digital economy, and a digital society, and (ii) also establishing globally competitive Vietnamese digital technology enterprises;[351] and the *National Strategy for the Fourth Industrial Revolution Through 2030*, which expands on the digital transformation goals with defined domestic and international goals in e-government, connectivity development, and capabilities in emerging technologies. Specifically, the Strategy emphasizes the development of physical and digital infrastructure such as high-speed internet, and investing, researching, and developing key technologies to actively participate in the fourth industrial revolution such as AI, healthcare technology, IoT, big data, and blockchain.[352]

E-government has also emerged as a key priority for Viet Nam's central government. The digitalization of public service delivery, aimed at lowering transaction costs between government, businesses, and people, and leveraging cross-sectoral and cross-agency data, promises significant benefits for the Vietnamese.[353]

Together, these multiple initiatives should convergence into a solid foundation for further digital transformation processes that will steadily allow Viet Nam to future-proof its digital economy.

[349] Viet Nam Investment Review (2021) Viet Nam new Asian tiger: Business Times, https://vir.com.vn/vietnam-new-asian-tiger-business-times-91253.html
[350] Google, e-Conomy 2022 SEA Report: Vietnam, https://services.google.com/fh/files/misc/vietnam_e_conomy_sea_2022_report.pdf
[351] LuatViet Nam (2020) Decision No. 749/QD-TTg 2020 national digital transformation program through 2025, https://english.luatvietnam.vn/decision-no-749-qd-ttg-on-approving-the-national-digital-transformation-program-until-2025-with-a-vision-184241-Doc1.html
[352] Socialist Republic of Viet Nam Government News (2021) National strategy for 4th Industrial Revolution, https://en.baochinhphu.vn/national-strategy-for-4th-industrial-revolution-11140283.htm
[353] Ministry of Information and Communications (2021) Viet Nam issued its first e-government strategy towards digital government, https://english.mic.gov.vn/Pages/TinTuc/147615/Vietnam-issued-its-first-e-government-strategy-towards-digital-government.html

Figure 18: Viet Nam and ASEAN Scores – ADII 1.0 & ADII 2.0



	Pillar 1	Pillar 2	Pillar 3	Pillar 4	Pillar 5	Pillar 6
Viet Nam (ADII 1.0)	78.50	63.05	58.33	38.38	44.55	60.72
Viet Nam (ADII 2.0)	78.46	89.27	68.33	49.62	46.00	69.34
ASEAN (ADII 1.0)	55.27	62.81	58.84	48.21	49.32	62.85
ASEAN (ADII 2.0)	61.82	69.99	70.03	50.01	51.83	68.55

Figure 18 shows that Viet Nam performs relatively well in its ADII 2.0 Pillar scores, with three Pillars—Pillars 1, 2, and 6—that score above the ASEAN average, suggesting that Viet Nam is a frontrunner in these areas. At the same time, there are two Pillars—Pillars 3 and 4—that are very close to the ASEAN average, which may indicate that Viet Nam is keeping pace with other AMS in regard to these two dimensions. Lastly, Pillar 5 stands out as the weakest ADII 2.0 Pillar for Vietnam, despite a slight increase since ADII 1.0.

Viet Nam's Pillar 2 score is likely reflective of many recent initiatives on data protection and cybersecurity. In April 2023, the government promulgated the *Personal Data Protection Decree (PDPD)*, providing a comprehensive and uniform approach to personal data protection in Viet Nam. The PDPD takes effect in July 2023, with a two-year grace period for MSMEs.[354] In August 2022, the Ministry of Information and Communications (MIC) announced *Decision 964/QD-TTg* approving the *National Cybersecurity and Safety Strategy* to develop cybersecurity policies and laws, improve cybersecurity capacity, and regulate the collection, storage, and processing of personal data.[355]

[354] Tilleke & Gibbins (2023) A Closer Look at Vietnam's First-Ever Personal Data Protection Decree, https://www.tilleke.com/insights/a-closer-look-at-vietnams-first-ever-personal-data-protection-decree/
[355] Tech for Good Institute (2023) National-level priorities to grow the digital economy: Spotlight on Vietnam, https://techforgoodinstitute.org/blog/articles/advancing-digital-economy-through-national-level-priorities-spotlight-on-vietnam/

Meanwhile, the Ministry of Public Security (MPS) has also been working on a draft Cybersecurity Administrative Sanctions Decree (CASD) to enforce its Cybersecurity Law and PDPC.[356]

Regarding Pillar 1, Viet Nam boasts some progress in the growth and implementation of digital trade mechanisms, such as the *Viet Nam National Single Window (NSW)*, with reportedly nearly 66,300 participating enterprises as of October 2023. It is also reported that some 99.6% of customs clearance application forms are now handled online, and that it takes only 1-3 seconds for customs to clear cargo in the Green Line classification.[357] E-invoicing has also advanced digital trade and logistics for Viet Nam; under *Decree 123/2020/ND-CP*, Viet Nam has implemented and enforced the use of e-invoices since July 2022. Guidelines have also been provided through *Circular 68/2019/TT-BTC*, which outlines the specifications and parties that are obliged to comply with the regulation.[358]

On Pillar 6, Viet Nam has constructed robust 4G infrastructure, but has also added key 5G infrastructure, which has been earmarked as a priority by the government[359] and described as a 'pillar' of digitalization efforts that will underpin rapid development.[360] The *National Digital Transformation Until 2025, With a Vision to 2030, Towards the Fourth Industrial Revolution* includes investments in digital infrastructure as a key aspect of digital development, with ambitions of reaching 80% coverage of households with fiber optic Internet infrastructure.[361] Institutional coordination for digitalization is also apparent from the establishment of the National Committee on Digital Transformation in September 2021, which is tasked with directing and coordinating with ministries, agencies, and localities to advance national digital transformation.[362]

While Viet Nam scored the lowest in Pillar 5 on Innovation & Entrepreneurship, Viet Nam is well-positioned for progress on this front given ongoing efforts and initiatives. An example of government assistance is the *Supporting the National Innovation Start-Up Ecosystem to 2025* project, approved by the Prime Minister in *Decision No. 844/QD-TTg*. This project sets a goal of developing 600 start-ups by 2025, 100 of which will have capital of at least USD 85.5 million.[363] Vietnamese small businesses' swift recovery from the COVID-19 pandemic has also paved the way for growth, with a significant majority of local small businesses expecting to sustain their growth in 2023.[364]

Pillar 3 has also seen its share of initiatives. For example, the State Bank of Viet Nam (SBV) has been proactive in establishing an enabling regulatory environment, releasing a draft decree on a Fintech regulatory sandbox in June 2020,[365] and a second draft decree released in April 2022.[366] Rules are also in place for bank and non-bank financial institutions to offer intermediary payment services.[367] The Prime Minister has also signed and promulgated *Decision No. 1813/QD-TTg,* dated 28 October 2021, for the development of non-cash payments

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[356] Global Compliance News (2023) Vietnam: New draft Cybersecurity Administrative Sanctions Decree, www.globalcompliancenews.com/2023/06/03/https-insightplus-bakermckenzie-com-bm-technology-media-telecommunications_1-vietnam-new-draft-cybersecurity-administrative-sanctions-decree_05312023/
[357] VnEconomy (2023) Nearly 66,300 enterprises join National Single Window, https://vneconomy.vn/nearly-66-300-enterprises-join-national-single-window.htm
[358] Viet Nam Briefing (2022) Vietnam's E-invoice Implementation: Prepare for July 2022, www.vietnam-briefing.com/news/vietnam-e-invoice-implementation-prepare-for-july-2022.html
[359] Xinhua Net (2019) Viet Nam to launch 5G service soon: minister, www.xinhuanet.com/english/2019-03/22/c_137916442.htm
[360] Xinhua Net (2019) Viet Nam to launch 5G service soon: minister, www.xinhuanet.com/english/2019-03/22/c_137916442.htm
[361] MIC (2021) Viet Nam issued its first e-government strategy towards digital government, https://english.mic.gov.vn/Pages/TinTuc/tinchitiet.aspx?tintucid=147615
[362] Viet Nam Investment Review (2021) Viet Nam establishes National Committee on Digital Transformation, https://vir.com.vn/vietnam-establishes-national-committee-on-digital-transformation-87972.html
[363] VietnamPlus (2022) Asia's next start-up unicorns may spring from Vietnam: ADB report, https://en.vietnamplus.vn/asias-next-startup-unicorns-may-spring-from-vietnam-adb-report/237160.vnp
[364] VietnamPlus (2022) Digitalisation, innovation spur Vietnamese small businesses on expansion, https://en.vietnamplus.vn/digitalisation-innovation-spur-vietnamese-small-businesses-on-expansion/251936.vnp
[365] Lexology (2020) Vietnam: New draft decree on Fintech regulatory sandbox for Fintech services in Vietnam, www.lexology.com/library/detail.aspx?g=d7059851-c151-45c6-942b-17b80ec05b89
[366] Viet Nam Law Magazine (2022) Regulatory Fintech sandbox for banking activities proposed, https://vietnamlawmagazine.vn/regulatory-fintech-sandbox-for-banking-activities-proposed-48532.html
[367] These services include an electronic payment portal, cash collection and payment services, electronic money transfer services, and e-wallet services. LuatViet Nam (2014) Circular No. 39/2014/TT-NHNN dated December 11 2014, of the State Bank of Viet Nam guiding on payment intermediary service, https://english.luatvietnam.vn/circular-no-39-2014-tt-nhnn-dated-december-11-2014-of-the-state-bank-of-vietnam-guiding-on-payment-intermediary-service-91444-doc1.html

from 2021-2025. This serves as the basis for the development of relevant mechanisms and policies, the upgrading and developing of modern payments infrastructure, and the development of safe, secure, and innovative payment services to serve people and businesses.[368] In terms of digital identity, Viet Nam has made notable progress in establishing a comprehensive national digital identity system known as *VNeID* in July 2022, with the ambitious goal of 20 million users targeted by end 2023.[369] This initiative is underpinned by a legal framework, with the issuance of *Decree 59/2022/ND-CP* in September 2022, which outlines the use of electronic identity for both Vietnamese citizens and foreign residents.[370]

On Pillar 4, there has been an enhanced commitment to building digital capabilities. The MIC and local authorities have held training courses for community groups in 55 out of 63 provinces and centrally run cities. Between March and October 2022, more than 61,500 such groups were set up with nearly 284,000 members. In April 2022, the MIC launched an e-learning platform, *One Touch*, to provide training on digital skills.[371] The government has also partnered with private-sector players to grow digital talents such as with Google—which will provide certificate scholarships to student learners and mid-career professionals so they can pursue careers in tech and help enrich the Vietnamese start-up ecosystem.[372]

Looking at changes over time, Viet Nam has made positive progress for all pillars, except for Pillar 1, which has experienced a statistically negligible dip of 0.04 points between ADII 1.0 and ADII 2.0.

The biggest increase was for Pillar 2, where the score went from 63.05 in ADII 1.0 to 89.27 in ADII 2.0, placing it even higher above the ASEAN average compared to the ADII 1.0 Pillar score. This is likely attributed to the passing of the PDPD as discussed above, which represents a milestone for Viet Nam in enacting a comprehensive and robust law for effective and enforceable personal data protection.[373]

In relation to ASEAN averages, it is also notable that changes in Pillar scores from ADII 1.0 to ADII 2.0 have placed Viet Nam above the ASEAN average for Pillar 6, and significantly reduced its gap with the ASEAN average for Pillar 4. On the first observation, this progress may be explained by Viet Nam's policies and strategies to develop the digital infrastructure in the country, supported by greater institutional coordination mechanisms. On the second observation, Viet Nam has demonstrated strong progress to keep it at pace with other AMS, but there remains some unrealized potential for Viet Nam in this area. Looking ahead, it will have to join all other AMS in prioritizing the growth of its pool of digital talents, in step with the growth of its digital economy and the increasingly conducive environment for digital innovation.

[368] LuatViet Nam (2021) Decision 1813/QD-TTg 2021 approving the Scheme on development of non-cash payment in Viet Nam during 2021-2025, https://english.luatvietnam.vn/decision-no-1813-qd-ttg-approving-the-scheme-on-development-of-non-cash-payment-in-vietnam-during-2021-211698-doc1.html
[369] Socialist Republic of Viet Nam Government News (2023) Prime Minister asks for having at least 20 million VNeID app users by year's end, https://en.baochinhphu.vn/prime-minister-asks-for-having-at-least-20-million-vneid-app-users-by-years-end-111230713102204133.htm
[370] LuatViet Nam (2023) Formally issuing the Decree on electronic identifiers and authentications, https://english.luatvietnam.vn/legal-updates/decree-on-electronic-identifiers-and-authentications-892-91278-article.html
[371] One Touch (2020) About One Touch, https://onetouch.mic.gov.vn/gioi-thieu
[372] Vietnamplus (2023) Google offers 40,000 scholarships to support Vietnam's digital transformation, https://en.vietnamplus.vn/google-offers-40000-scholarships-to-support-vietnams-digital-transformation/270372.vnp#:~:text=Google%20on%20October%2028%20announced,the%20Google%20Career%20Certificates%20programme.
[373] Future of Privacy Forum (2023) Vietnam's Personal Data Protection Decree: Overview, Key Takeaways, And Context, https://fpf.org/blog/vietnams-personal-data-protection-decree-overview-key-takeaways-and-context

ANNEX B: INTERNATIONAL BEST PRACTICES

In addition to tracking how ASEAN fares and progresses in its digital integration efforts over time, it is important to benchmark the region's performance against that of bigger, more mature digital economies from around the Asia-Pacific.

Benchmarking ASEAN's Efforts Against Other Economies in APAC

Table 6 provides a detailed overview of the ADII 2.0 Pillar scores for ASEAN and for six non-ASEAN digital economies whose scores can be used to benchmark ASEAN's progress and performance (Australia, China, Japan, New Zealand, South Korea, and the United States).

Table 6: ASEAN vs. Benchmark Economies Scores – ADII 2.0

	Pillar 1 Digital Trade & Logistics	Pillar 2 Data Protection & Cybersecurity	Pillar 3 Digital Payments & Identities	Pillar 4 Digital Skills & Talent	Pillar 5 Innovation & Entrepreneurship	Pillar 6 Institutional & Infrastructural Readiness
ASEAN	**61.82**	**69.99**	**70.03**	**50.01**	**51.83**	**68.55**
Australia	92.00	97.75	95.50	53.86	65.80	80.87
China	89.77	94.60	83.87	**66.19**	70.56	72.65
Japan	**93.20**	**98.22**	85.00	47.52	76.40	80.53
New Zealand	90.00	90.73	**95.75**	58.97	65.45	80.94
South Korea	91.60	99.09	88.89	40.86	80.87	83.35
United States	91.20	98.33	89.00	62.14	**82.93**	**88.04**

Note: Some data for benchmark economies was not updated due to some third-party databases being unavailable or discontinued.

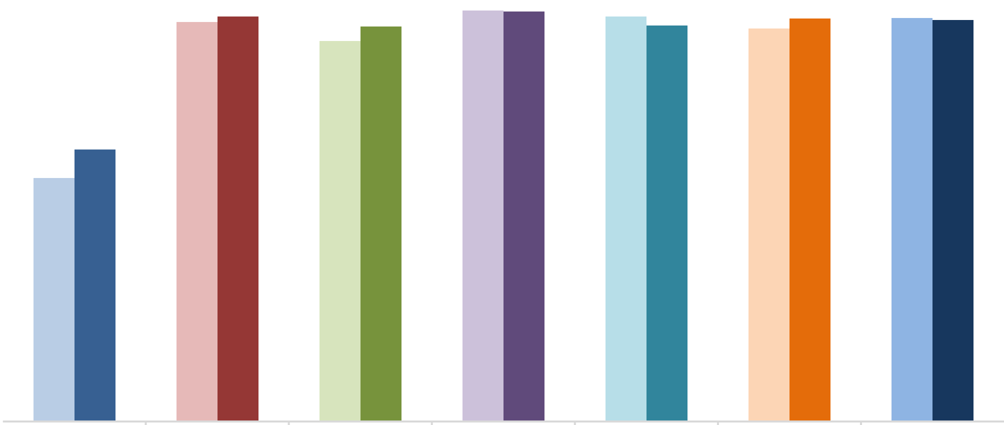
As noted throughout this report, the ADII 2.0 data shows that ASEAN has made steady progress across all six ADIII Pillars since the inaugural ADII 1.0 report in 2020-2021—though the breadth and pace of improvement has been largely uneven between AMS. Despite the marked upward trajectory observed for ASEAN, a comparison with the ADII 2.0 scores of more mature digital economies reveals that much remains to be done when it comes to consolidating digital development and achieving full digital integration in the region.

Indeed, ASEAN scores much lower than the six non-ASEAN benchmarking economies across all ADII Pillars, except in Pillar 4, where ASEAN outperforms Japan and South Korea. Pillar 4 stands out as the weakest Pillar across the board, both for ASEAN and non-ASEAN economies. Conversely, Pillars 2 and 3 stand out as the Pillars where the highest scores can be observed for non-ASEAN economies. Overall, Japan takes the lead on Pillars 1 and 2, while the United States dominates in Pillars 5 and 6. New Zealand takes the top spot on Pillar 3, and China leads the way on Pillar 4.

These findings suggest that ASEAN has much to gain from the methods and approaches used in these more mature digital economies; adopting best practices from Australia, China, Japan, New Zealand, South Korea, and the United States could support and accelerate AMS' journey towards becoming resilient, equitable, and sustainable digital economies.

Pillar 1: Digitally Enabled Trade & Logistics

Figure 19: Pillar 1 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	55.27	90.72	86.50	93.36	92.04	89.28	91.68
ADII 2.0	61.82	92.00	89.77	93.20	90.00	91.60	91.20

Pillar 1 is one of two Pillars (along with Pillar 3) for which ASEAN has experienced the highest point increase since ADII 1.0. Despite this progress, ASEAN's Pillar score of 61.82 remains the lowest compared to six non-ASEAN economies, where Pillar 6 scores range between 89 and 93. This discrepancy may stem from significant regional advancements—such as multiple trade agreements that call for a facilitation and dematerialization of cross-border trade—being weighed down by less conducive measures at the national level, such as stringent data localization rules hindering cross-border digital activities. This disparity underscores the need for alignment between regional initiatives and national policies to foster a more conducive environment for sustained growth and collaboration across borders.

Japan's leadership in this area, evident through its highest Pillar score of 93.20, presents a range of good practices for ASEAN to adopt and adapt. Japan's proactive approach in digital trade and logistics are marked by strategic policy frameworks. Notably, Japan has led global discussions on data governance and introduced the *Data Free Flow with Trust (DFFT)* concept, which promotes the free flow of data while protecting privacy, security, and intellectual property rights.[374] The *DFFT* was introduced at the G20 Osaka Summit in 2019 and subsequently endorsed at the G7 Hiroshima Summit in 2023 with the establishment of the *Institutional Arrangement for Partnership (IAP)* to operationalize DFFT.[375]

Japan's success extends to its strategic bilateral and plurilateral digital trade agreements. Most recently in June 2023, Japan and the European Union concluded the *European Union-Japan Digital Trade Principles*, aimed at establishing a shared understanding regarding crucial aspects of digital trade, including consumer and business trust, data governance, and the facilitation of digital trade processes.[376] Notably, Japan's bilateral digital trade agreements with the United States and the United Kingdom incorporate regulations addressing pertinent issues on digital platforms such as algorithms.[377]

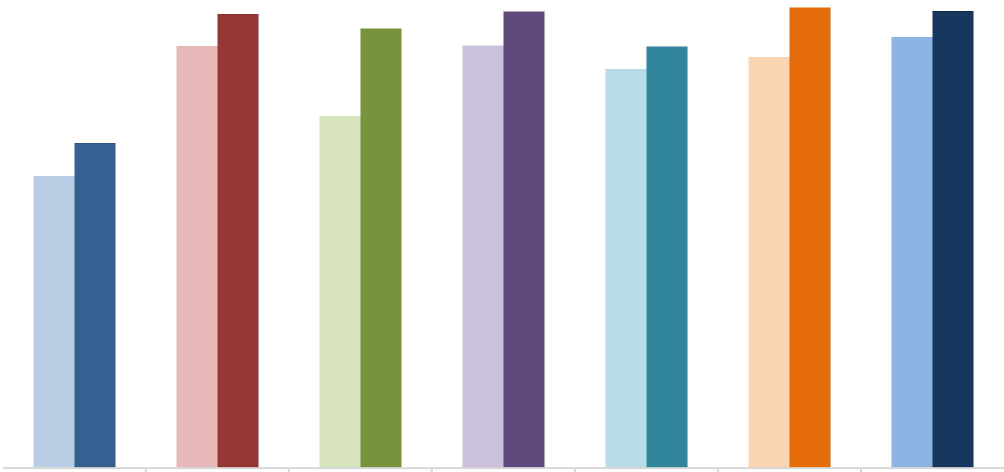
Moreover, Japan's enabling policies promote an environment conducive to digital trade and logistics growth. These include initiatives such as the *Comprehensive Logistics Policy Program (2021-2025)* prioritize digitalization and decarbonization in the logistics industry[378] and the *Priority Policy Program for Realizing Digital Society*, which charts a comprehensive roadmap for digital transformation and outlines initiatives such as electronic signatures, letters of attorney, business certificates, and digital IDs to enhance industry digitalization.[379] The country has also made efforts toward fostering interoperability among digital platforms, such as the *Connected Industries Open Framework*, which aims to facilitate data utilization across various digital platforms in the manufacturing sector.[380]

Japan's active participation in shaping global discussions on data governance, strategic trade agreements, and enabling policies offers a compelling model for ASEAN. By learning from Japan's approach, ASEAN can strengthen its digital trade frameworks and logistics systems, fostering enhanced cross-border growth and collaboration in the region.

[374] Japan Gov (2019) Abenomics, www.japan.go.jp/abenomics/_userdata/abenomics/pdf/1909_abenomics.pdf
[375] Digital Japan (n.d.) Exploring Together DFFT, www.digital.go.jp/assets/contents/node/basic_page/field_ref_resources/c82a2f37-7442-4e31-9ba9-15d3400f9cf4/b93f401a/20230816_dfft-en_outline_01.pdf
[376] SupplyChainbrain (2023) EU and Japan Agree to Cooperate on Digital and International Trade, www.supplychainbrain.com/articles/37587-eu-and-japan-agree-to-cooperate-on-digital-and-international-trade
[377] NBR (2023) Japan's Role and Strategy in the Formation of Digital Trade Rules in the Indo-Pacific, www.nbr.org/publication/japans-role-and-strategy-in-the-formation-of-digital-trade-rules-in-the-indo-pacific
[378] Smart CityKorea (2022) Japanese logistics industry strives to establish digitalization and decarbonization of supply chains, https://smartcity.go.kr/en/2022/03/10/%EC%9D%BC%EB%B3%B8-%EB%AC%BC%EB%A5%98%EC%97%85%EA%B3%84-%EA%B3%B5%EA%B8%89%EB%A7%9D%EC%9D%98-%EB%94%94%EC%A7%80%ED%84%B8%ED%99%94%C2%B7%ED%83%88%ED%83%84%EC%86%8C%ED%99%94-%EA%B5%AC%EC%B6%95/
[379] JETRO (2022) A Nation's Drive Towards a Data-first Digital Society Future, www.jetro.go.jp/en/invest/insights/japan-insight/nation-drive-datafirst-digital-society-future.html
[380] IVI (n.d.) The specifications of data exchange of "Connected Industries" has been released, https://iv-i.org/en/2019/11/02/english-the-specifications-of-data-exchange-of-connected-industries-has-been-released-public-review-started

Pillar 2: Data Protection & Cybersecurity

Figure 20: Pillar 2 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	62.81	90.77	75.73	90.93	85.91	88.42	92.73
ADII 2.0	69.99	97.75	94.60	98.22	90.73	99.09	98.33

Pillar 2 stands out as the ADII 2.0 Pillar for which the gap between ASEAN's average score (69.99) and the score of all other benchmarking economies is the widest. With a score of 99.09, South Korea dominates this Pillar and has an advantage of over 30 points over ASEAN. At 90.73, New Zealand is in next-to-last position but is still over 20 points ahead of ASEAN. Notably, Pillar 2, alongside Pillars 3 and 6, has seen consistent score improvement since ADII 1.0, which points to a general strengthening of all the enabling measures that allow digital investment and innovation to thrive.

South Korea's leadership in data protection and cybersecurity underscores its robust policies, offering best practices to inspire ASEAN policymakers. Central to South Korea's success is its comprehensive regulatory framework emphasizing data protection and cybersecurity. For instance, the *Personal Information Protection Act (PIPA)* mandates stringent guidelines for the collection, use, and safeguarding of personal data, ensuring stringent standards that enhance consumer trust and mitigate data breaches.

Moreover, South Korea's strategic initiatives, like the *National Cybersecurity Strategy*[381] and the *Strategic Plan to Foster the Data Protection Industry*,[382] underscore the country's commitment to nurturing a robust cybersecurity landscape and a thriving data protection industry. These efforts focus on securing critical infrastructure, fostering cybersecurity talent, and driving industry growth through strategic planning and innovation.

[381] Digwatch (2023) South Korean National Cybersecurity Strategy, https://dig.watch/resource/south-korean-national-cybersecurity-strategy
[382] Korea Net (2022) MSIT to announce 'Strategic Plan to Foster Data Protection Industry', www.korea.net/Government/Briefing-Room/Press-Releases/view?articleId=6296&type=O

South Korea's proactive international engagements in cybersecurity, exemplified by partnerships with France,[383] the United States,[384] and the United Kingdom,[385] highlight its collaborative approach in addressing global cyber-threats. These partnerships bolster information sharing and joint efforts in cyber-crisis management, contributing to a more coordinated response against cybersecurity threats.

Additionally, South Korea's commitment to cybersecurity education and training, demonstrated through national initiatives like the *Cyber Conflict Exercise (CCE)* [386] and hosting region-wide cybersecurity trainings such as the *ASEAN-Plus Cybersecurity Training* in November 2023[387], highlights its dedication to fostering cybersecurity readiness and talent development on a national and regional scale.

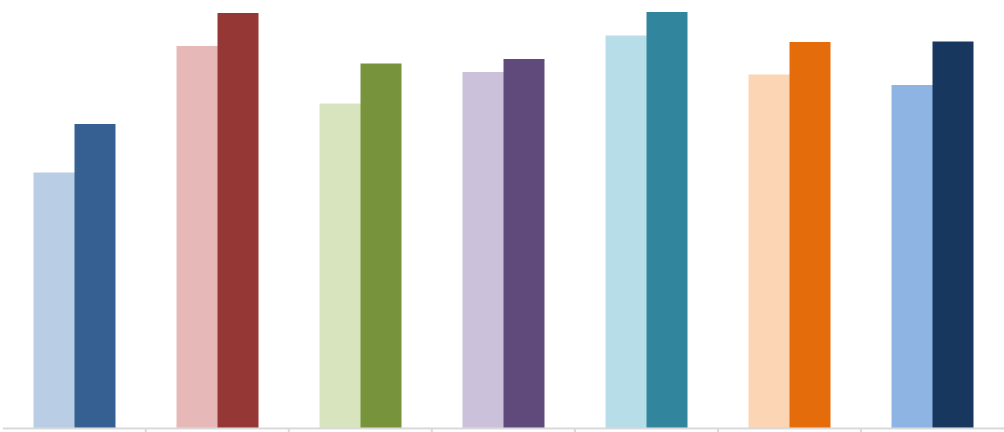
Furthermore, a notable aspect is South Korea's data-centric approach in informing policy formulation. The *Korea Global Cybersecurity Capability Assessment (GCCA)* tool enables the evaluation of national cybersecurity prowess by comparing and analyzing it against other countries using specified criteria.[388] This capability assessment serves the dual purpose of fortifying domestic cybersecurity measures, as well as establishing the groundwork for reinforcing cybersecurity through cross-border cyber-defense programs with international partners—thus enabling cooperative responses to global cyber-threats.[389]

South Korea's success in data protection and cybersecurity, demonstrated through robust regulatory frameworks, extensive education initiatives, strategic partnerships, and innovative assessment tools, provides a compelling blueprint for ASEAN. By leveraging and adopting similar strategies, ASEAN can fortify its cybersecurity measures and create a secure digital environment conducive to sustained growth and innovation.

[383] CNIL (2022) Declaration of cooperation between the South Korean data protection authority (PIPC) and the French data protection authority (CNIL), www.cnil.fr/en/declaration-cooperation-between-south-korean-data-protection-authority-pipc-and-french-data
[384] CISA (2023) CISA Signs Memorandum of Understanding with the Republic of Korea to Share Cyber Threat Information and Cybersecurity Best Practices, www.cisa.gov/news-events/news/cisa-signs-memorandum-understanding-republic-korea-share-cyber-threat-information-and-cybersecurity
[385] KBS World (2023) S. Korea, UK Sign Strategic Cyber Partnership, https://world.kbs.co.kr/service/news_view.htm?lang=e&Seq_Code=181956
[386] CCE (2023) Cyber Conflict Exercise 2023 Operational regulations, https://cce.cstec.kr/en/contents/documents/Cyber%20Conflict%20Exercise%202023%20Operational%20regulations.pdf
[387] Korea Times (2023) Korea hosts ASEAN-Plus cybersecurity training, www.koreatimes.co.kr/www/nation/2023/11/113_362921.html
[388] Carnegie Endowment for International Practice (2021) Korean Policies of Cybersecurity and Data Resilience, https://carnegieendowment.org/2021/08/17/korean-policies-of-cybersecurity-and-data-resilience-pub-85164
[389] Carnegie Endowment for International Practice (2021) Korean Policies of Cybersecurity and Data Resilience, https://carnegieendowment.org/2021/08/17/korean-policies-of-cybersecurity-and-data-resilience-pub-85164

Pillar 3: Digital Payments & Identities

Figure 21: Pillar 3 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	58.84	88.00	74.73	82.00	90.33	81.42	79.00
ADII 2.0	70.03	95.50	83.87	85.00	95.75	88.89	89.00

Pillar 3 is one of three Pillars (along with Pillars 2 and 6) for which no economies have seen their scores decrease since ADII 1.0. New Zealand leads in this pillar with an impressive score of 95.75, marking nearly a 25-point lead over the ASEAN average of 70.03. While Pillar 3, alongside Pillar 1, has observed the highest point increase for ASEAN since ADII 1.0 (a rise of 11.2 points), it remains at the tail-end of the rankings. This may be because the region started from a much lower point; as such, as remarkable and encouraging as ASEAN's progress has been so far, it is still insufficient to catch up to more mature digital ecosystems.

New Zealand's advancements in digital payments and identities serve as an instructive model for ASEAN's digital infrastructure development. The country has implemented numerous policy and regulatory measures that bolster and facilitate the digital payments landscape. For instance, the Reserve Bank of New Zealand (RBNZ) spearheads a *Future of Money* program, which includes redesigning the cash system and exploring the issuance of a central bank digital currency.[390]

Moreover, the *Retail Payment System Act 2022*, administered by the Commerce Commission, focuses on fostering competition and efficiency in the retail payment system for the long-term benefit of merchants and consumers in New Zealand. Additionally, collaborative efforts by RBNZ and FMA aim to implement the *Financial Market Infrastructures Act 2021*, which provides a comprehensive regulatory framework for the reliability of payment systems and other financial market transactions.[391]

New Zealand's success in digital identities also lies in its inclusive approach to digital identity management. Aside from prioritizing trust, security, and efficiency, the country emphasizes digital

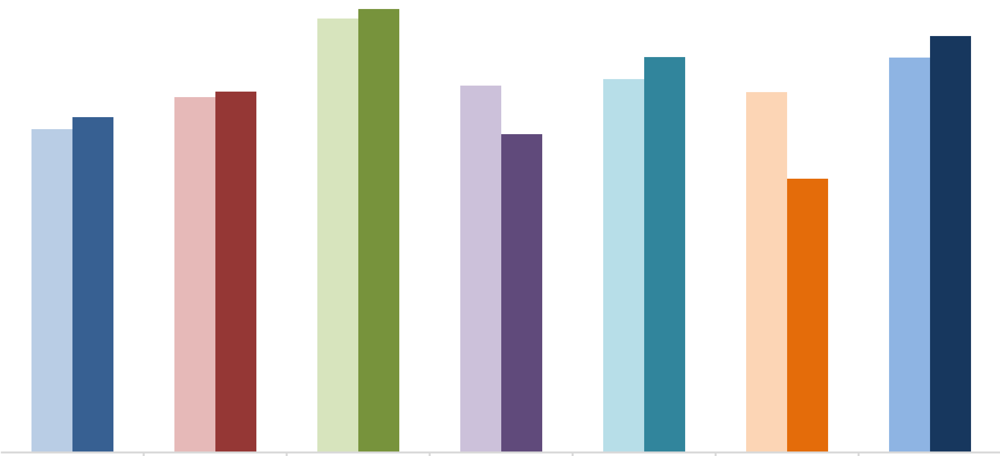
[390] RBNZ (2023) Future of money, www.rbnz.govt.nz/money-and-cash/future-of-money
[391] CoFR NZ (2023) CoFR vision for the future of New Zealand's payments, www.cofr.govt.nz/news-and-publications/payments-vision.html

inclusion. The government directs efforts to ensure equitable access to digital identity solutions for all citizens, taking into consideration *Māori tikanga* and *Māori* data sovereignty, and social inclusion for the disadvantaged.[392] Forthcoming initiatives like the *Digital Identity Services Trust Framework Act 2023* intend to promote the use of digital identities and cultivate a digital identity network, enabling businesses to use digital identities for access to their services, and allowing consumers to choose their preferred digital identity provider.[393]

New Zealand's leadership in digital payments and identities showcases robust policies that facilitate the digital payments landscape and an inclusive framework to digital identities that ASEAN can emulate given its characteristic diversity of cultures and languages. ASEAN can adopt these best practices to foster greater confidence and efficiency in digital transactions and identity management across the region.

Pillar 4: Digital Skills & Talent

Figure 22: Pillar 4 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	48.21	52.99	64.76	54.77	55.70	53.77	58.90
ADII 2.0	50.01	53.86	66.19	47.52	58.97	40.86	62.14

Pillar 4 is not only the Pillar for which ASEAN's average score has increased, but it is the only Pillar that has seen ASEAN move ahead of benchmarking economies. Indeed, Pillar 4 stands out as the only Pillar for which ASEAN no longer has the lowest score; supported by Japan's and South Korea's respective 7.2- and 12.9-point losses, ASEAN's modest 1.8-point increase allowed the region to move ahead in the benchmarking panel. With a score of 66.19, China takes top position, well ahead of the United States and New Zealand (interestingly, these same three economies already dominated the top 3 positions in ADII 1.0).

China's approach to nurturing digital skills and talent presents a compelling model for ASEAN. China has implemented a range of strategic policies and initiatives to cultivate a robust digital talent pool and

[392] Biometric Update (2023) NZ group calls for nationwide digital ID adoption, www.biometricupdate.com/202309/nz-group-calls-for-nationwide-digital-id-adoption
[393] Simpson Grierson (2023) New Digital Identity Framework – what does this mean for you?, www.simpsongrierson.com/insights-news/legal-updates/new-digital-identity-framework-what-does-this-mean-for-you

promote digital literacy. The *National Digital Talent Development Plan*, launched in 2019, stands as a comprehensive strategy addressing skill gaps and promoting talent development in emerging digital fields. This initiative emphasizes continuous upskilling, digital skills training, and vocational education to meet the evolving demands of the digital economy. Additionally, China has made efforts to build a strong foundation for digital education, notably through initiatives such as the *Internet Plus Education Action Plan*, which aims to integrate digital technologies into traditional educational frameworks to foster a digitally adept workforce.[394]

China has extended its efforts beyond national policy development by spearheading numerous programs and initiatives to advance digital literacy and skills. For instance, the Ministry of Education introduced the *Smart Education of China*, a public online platform recognized as the world's largest repository of educational materials.[395] Moreover, initiatives like *Chinadata.cn* have been pivotal in enhancing the digital capabilities of the general public by facilitating access to educational resources shared by prominent universities, research institutes, and social organizations.[396] Additionally, the Ministry of Commerce has rolled out a national e-commerce public service website which offers extensive resources including e-commerce news, information about exhibitions, and training courses across various sectors.[397] These initiatives have expanded access to digital educational resources, thereby fostering enhanced digital literacy.

China's dedication to fostering digital skills and talent extends to initiatives promoting entrepreneurship and innovation. Strategies such as the *Made in China 2025* prioritize advancements in key sectors like information technology and high-end equipment manufacturing.[398] These initiatives not only drive innovation, they also cultivate an environment that is conducive to talent development, encouraging individuals to explore and excel in digital innovation and entrepreneurship.

China's success is underpinned by robust public-private partnerships, notably exemplified by initiatives such as the new *Financial Talent Training Base* established in 2022 in Shanghai. The training base aims to train both local and international financial professionals, particularly college students within Pudong's financial institutions, while attracting top-tier global experts.[399]

By drawing from China's strategic initiatives and cultivating similar collaborative ecosystems, ASEAN can elevate its digital skills landscape. Emulating China's targeted approach in upskilling and fostering talent will empower the region's workforce to bridge skills gap and effectively meet the rising demand in digital talent and skills.

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[394] Mingfeng Wang, Tingting Liu and Wei Zhou (2023) Internet Plus, Industrial Transformation, and China's Evolving Urban System, Journal of Urban Technology, www.tandfonline.com/doi/full/10.1080/10630732.2023.2254203?src=

[395] People's Daily Online (2023) China takes measures to improve digital literacy and skills for general public, http://en.people.cn/n3/2023/0327/c90000-10227513.html

[396] People's Daily Online (2023) China takes measures to improve digital literacy and skills for general public, http://en.people.cn/n3/2023/0327/c90000-10227513.html

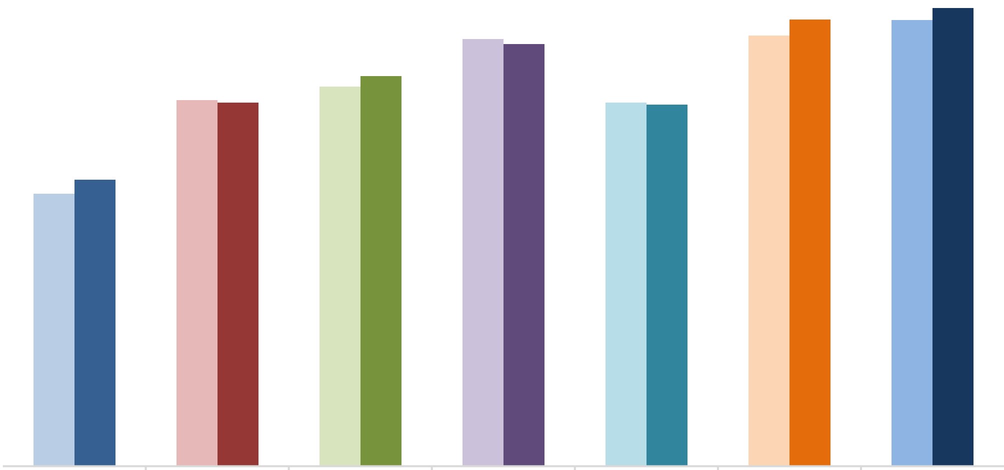
[397] People's Daily Online (2023) China takes measures to improve digital literacy and skills for general public, http://en.people.cn/n3/2023/0327/c90000-10227513.html

[398] Institute for Security and Development Policy (2018) Made in China 2025, https://isdp.eu/content/uploads/2018/06/Made-in-China-Backgrounder.pdf

[399] Shine (2022) New financial talent training base unveiled in Shanghai, www.shine.cn/biz/event/2212164123/

Pillar 5: Innovation & Entrepreneurship

Figure 23: Pillar 5 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	49.32	66.30	68.74	77.32	65.84	77.92	80.79
ADII 2.0	51.83	65.80	70.56	76.40	65.45	80.87	82.93

Unlike other Pillars, the scores for Pillar 5 have not made much progress across the board. ASEAN's average Pillar score has seen only a marginal increase from 49.32 to 51.83 (an increase of 2.5 points), mirroring the minimal fluctuations among non-ASEAN economies (no change above 3 points). This absence of noteworthy progress suggests a significant worldwide stalling of entrepreneurial activity, a trend that is reflected in the fact that the leaderboard remains unchanged since ADII 1.0. Indeed, the United States remains at the top, followed by South Korea and Japan; while ASEAN remains at the bottom, outpaced by New Zealand, Australia, and China.

The United States stands at the forefront of fostering innovation and entrepreneurship, setting a good example for ASEAN. The United States has championed various policy measures and initiatives that have propelled its innovation ecosystem. For instance, programs such as the *Small Business Innovation Research (SBIR)* and the *Small Business Technology Transfer (STTR)* allocate federal research funds to small businesses, encouraging technological innovation and commercialization.[400] Moreover, the STTR program drives partnerships between small businesses and non-profit research institutions, bridging the gap between performance of basic science and commercialization of resulting innovations.[401] Adapting to the evolving digital landscape, the United States introduced the *National Spectrum Strategy* in November 2023, outlining a roadmap to advance American innovation, competition, and security in wireless technologies.[402]

[400] SBIR (2023) The SBIR and STTR Programs, www.sbir.gov/about
[401] SBIR (2023) The SBIR and STTR Programs, www.sbir.gov/about
[402] NTIA (2023) Fact Sheet: Biden-Harris Administration Issues Landmark Blueprint to Advance American Innovation, Competition and Security in Wireless Technologies, www.ntia.gov/other-publication/2023/fact-sheet-biden-harris-administration-issues-landmark-blueprint-advance

Furthermore, the country's thriving venture capital (VC) ecosystem remains instrumental in driving innovation and fostering entrepreneurship. Policies like the *Jumpstart Our Business Start-ups (JOBS) Act* have eased regulations for fundraising and eased access to capital for start-ups and small businesses, creating over 100,000 jobs and over USD 1 billion in funding for American start-ups and small businesses since 2016.[403] Moreover, programs like the *Small Business Investment Company (SBIC)* initiative provide funding avenues for small businesses and start-ups, supporting their growth and innovation.[404] These initiatives have not only created job opportunities, they have also fostered a vibrant entrepreneurial environment with trickle-down effects.

The United States government actively engages in international science and technology collaborations to retain its global leadership in the innovation space. It holds multiple strategic energy dialogues with partners and allies to set out collaboration priorities on a bilateral basis and implements the initiatives through working groups and other mechanisms.

For example, on the clean energy innovation front, the United States has carried out bilateral engagements such as the *United States-India Partnership to Advance Clean Energy-Research (PACE-R)*, and the *United States -Israel Center of Excellence in Energy, Engineering and Water Technology (Energy Center)*.[405] It has also launched a new program called *Net Zero World Initiative* which works hand-in-hand with partner countries to co-create and implement tailored, actionable technical and investment roadmaps to increase the speed and scale of transition to net zero energy systems.[406] These partnerships facilitate the research, development, and commercialization of innovative technologies, accelerating innovation and creating an environment conducive to entrepreneurial growth.

Overall, the United States' comprehensive approach to nurturing innovation and entrepreneurship through strategic policy measures, initiatives, and global collaborations is a great example for ASEAN. Leveraging similar strategies and collaborative frameworks can enable ASEAN to drive a greater appetite for innovation and entrepreneurship in the region.

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[403] Yahoo Finance (2023) Obama Quietly Passed This 2012 Law That Has Helped Create Over 100,000 US Jobs And Over $1 Billion In Funding For US Small Businesses, https://finance.yahoo.com/news/obama-quietly-passed-2012-law-130313394.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAAqXm0f5WtgcghPTDWqBahGLFvwYYXJewBJcacQDyRnrh9gaVaJcZ5jrBCPKzPnzCSb8ANlMthP4LX9tL5ZLq59-yabGgVFcQGjtAMEKCI3d9mvf4piiLuEwY7p_V52hjK1UuUN8CfK5Ta7uy4iGvfe5WUPOfQBgNb0ggHxyniDb

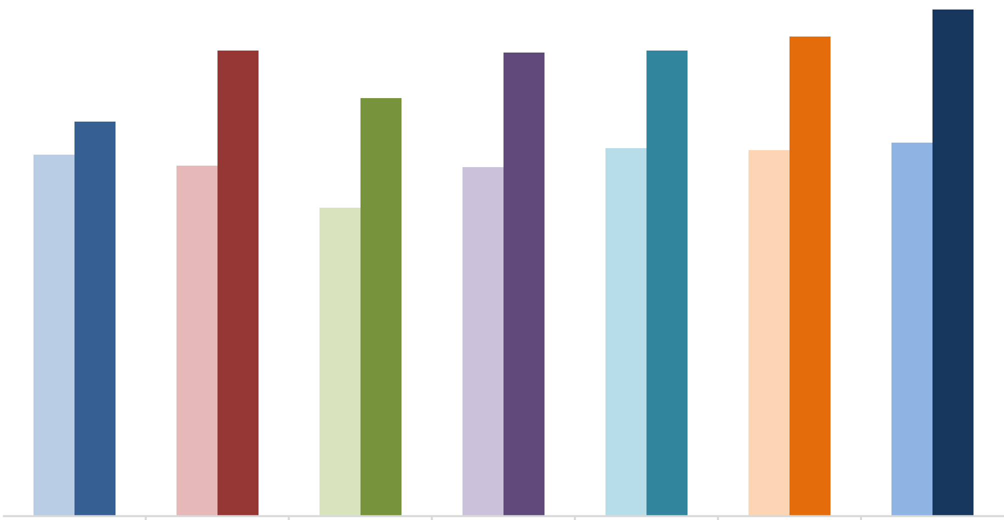
[404] SBA (2023) SBICs invest in small businesses, www.sba.gov/funding-programs/investment-capital#:~:text=An%20SBIC%20is%20a%20privately,in%20certain%20sectors%20or%20industries

[405] The White House (2023) National Innovation Pathway, www.whitehouse.gov/wp-content/uploads/2023/04/US-National-Innovation-Pathway.pdf

[406] The White House (2023) National Innovation Pathway, www.whitehouse.gov/wp-content/uploads/2023/04/US-National-Innovation-Pathway.pdf

Pillar 6: Institutional & Infrastructural Readiness

Figure 24: Pillar 6 ASEAN vs. Benchmark Economies Scores – ADII 2.0



	ASEAN	Australia	China	Japan	New Zealand	South Korea	United States
ADII 1.0	62.85	60.87	53.63	60.67	63.90	63.59	64.90
ADII 2.0	68.55	80.87	72.65	80.53	80.94	83.35	88.04

With an average score of 68.55, ASEAN saw its score advance by close to 6 points since ADII 1.0. Despite this progress, ASEAN scores the lowest compared to six non-ASEAN economies, whose Pillar 6 scores are comprised between 72 and 88. This is because unlike ASEAN, other economies have seen their Pillar 6 scores increase by 17 points or more, suggesting a faster, stronger, and more wide-reaching maturation of institutional and infrastructural capabilities. This is in stark contrast to ADII 1.0, where ASEAN was positioned right in the middle of the pack and ahead of China, Japan, and Australia. The United States leads this pillar with a score of 88.04, holding a nearly 20-point lead over the ASEAN average.

The United States' approach to institutional and infrastructural readiness offers valuable insights for ASEAN's development in these domains. The country has enacted various policies and initiatives to expand internet accessibility, notably introducing the *Broadband Equity Access and Deployment (BEAD)* program in June 2023. This initiative, part of President Biden's *Bipartisan Infrastructure Law*, allocates USD 42.45 billion to extend high-speed internet access, with the aim to make it affordable for all.[407] Additionally, under the same law, the *Affordable Connectivity Program (ACP)* was established to provide free and discounted high-speed internet to eligible households. The *ACP* has already aided over 21 million households, resulting in monthly savings of over USD 500 million on internet bills.[408]

[407] The White House (2023) Fact Sheet: Biden-Harris Administration Announces Over $40 Billion to Connect Everyone in America to Affordable, Reliable, High-Speed Internet, www.whitehouse.gov/briefing-room/statements-releases/2023/06/26/fact-sheet-biden-harris-administration-announces-over-40-billion-to-connect-everyone-in-america-to-affordable-reliable-high-speed-internet/

[408] The White House (2023) What they are saying: Wide Range of Community Advocates and Industry Applaud President Joe Biden's Efforts to Extend the Affordable Connectivity Program, www.whitehouse.gov/briefing-room/statements-releases/2023/10/30/what-they-are-saying-wide-range-of-community-advocates-and-industry-applaud-president-joe-bidens-efforts-to-extend-the-affordable-connectivity-program/

Furthermore, the United States has actively adapted its legal framework to support digital innovation. In response to the rise of emerging technologies, President Biden issued an *Executive Order on Ensuring Responsible Development of Digital Assets* in September 2022, addressing the risks and potential benefits of digital assets and their underlying technologies.[409]

Additionally, President Biden issued an *Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence (AI)* in October 2023, thereby establishing new standards for AI safety and security and safeguarding Americans' privacy while fostering innovation and competition.[410] Also on AI, a new *National Institute of Standards and Technology (NIST) Public Working Group on AI* was established in June 2023 to help NIST develop key guidance to help organizations address the unique risks associated with generative AI technologies.[411]

The United States government has also actively prioritized the provision of digital public services through several policy measures. Notably, the *21st Century Integrated Digital Experience Act (IDEA)*, launched in 2018, aims to enhance the digital experience for government customers and deliver improved digital services to users.[412]

In September 2023, the Office of Management and Budget (OMB) released new policy guidance to agencies on *Delivering a Digital-First Public Experience (M-23-22)*, a digital framework for Federal agencies to design, develop, and deliver modern websites and digital services.[413] These policy initiatives play a pivotal role in empowering the United States government to enhance the availability and accessibility of its public services and resources.

The United States' strategic policies and initiatives to enhance internet accessibility, support digital innovation, and improve digital public services have fortified its institutional and infrastructural readiness. Emulating these practices could greatly contribute to fortifying ASEAN's institutional and infrastructural capacities across the region.

[409] The White House (2022) Fact Sheet: White House Releases First-Ever Comprehensive Framework for Responsible Development of Digital Assets, www.whitehouse.gov/briefing-room/statements-releases/2022/09/16/fact-sheet-white-house-releases-first-ever-comprehensive-framework-for-responsible-development-of-digital-assets/

[410] The White House (2023) Fact Sheet: President Biden Issues Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence, www.whitehouse.gov/briefing-room/statements-releases/2023/10/30/fact-sheet-president-biden-issues-executive-order-on-safe-secure-and-trustworthy-artificial-intelligence/

[411] NIST (2023) Biden-Harris Administration Announces New NIST Public Working Group on AI, www.commerce.gov/news/press-releases/2023/06/biden-harris-administration-announces-new-nist-public-working-group-ai

[412] GSA (2020) 21st Century Integrated Digital Experience Act and GSA, www.gsa.gov/technology/government-it-initiatives/digital-experience/21st-century-idea-and-gsa

[413] The White House (2023) Why the American People Deserve a Digital Government, www.whitehouse.gov/omb/briefing-room/2023/09/22/why-the-american-people-deserve-a-digital-government/

Examples of Other Global Digital Integration Assessment Efforts

Globally, there are other efforts by governments, businesses, and academia that have approached the similar endeavor of assessing and tracking digital economy integration and development across different economies.

The three indexes listed below are intended to measure digital integration via slightly different lenses. These indexes are published regularly to offer reference points to track progress, which is key in assessing success and identifying gaps and bottlenecks.

European Commission – Digital Society and Economy Index (DESI)

The European Commission has published the *Digital Economy and Society Index (DESI)* since 2014. From 2023 onwards, the DESI will no longer be published as a standalone report but will be integrated into the new *State of the Digital Decade* report, which aims to monitor progress towards the region's digital targets.[414]

The DESI is intended to measure the progress made by European Union Member States towards developing digital economies and societies. It addresses four main policy areas, which group a total of 33 indicators.[415] The four areas are:

1. Digital skills
2. Digital infrastructure
3. Digital transformation of businesses
4. Digitalization of public services

The indicators from the DESI are compiled and collated by Eurostat as it coordinates across Union-level statistical activities and ensures they are produced according to established rules and statistical principles as defined in the *European Statistics Code of Practice*.[416]

BBVA Research – Digitization Index (DiGiX)

The research arm of Spanish financial services company Banco Bilbao Vizcaya Argentaria (BBVA) has published the *Digitization Index (DiGiX)* since 2016. The most recent report was published in 2022.[417]

The DiGiX draws data from multiple publicly available official data sources to capture the progress of digitization across 99 developed and developing countries. It assesses these countries using 20 indicators grouped into six dimensions, which are further grouped into three broad categories—supply conditions, demand conditions, and institutional environment.

[414] European Commission (2023) DESI 2023 dashboard for the Digital Decade, https://digital-decade-desi.digital-strategy.ec.europa.eu/datasets/desi/charts
[415] European Commission (2023) DESI 2023 Indicators, https://digital-decade-desi.digital-strategy.ec.europa.eu/datasets/desi/indicators#desi2023-1
[416] Eurostat (n.d.) European statistics code of practise, https://www.eustat.eus/about/codigo_buenas_practicas_europeas_i.html#:~:text=The%20European%20Statistics%20Code%20of%20Practice%20is%20the%20cornerstone%20of,statistical%20processes%20and%20statistical%20outputs.
[417] BBVA Research (2022) DiGiX 2022 Update: A Multidimensional Index of Digitization, www.bbvaresearch.com/en/publicaciones/digix-2022-update-a-multidimensional-index-of-digitization

The DiGiX's six dimensions are:

1. Infrastructure
2. Users Adoption
3. Enterprises Adoption
4. Government Adoption
5. Regulation
6. Affordability

Tufts University – Digital Intelligence Index (DII)

The Fletcher School at Tufts University collaborates with Mastercard to produce the *Digital Intelligence Index (DII)*, which assesses digital competitiveness, digital trust, and responsible data use across 90 economies, which are divided based on their performance.

The DII utilizes a "Digital Evolution scorecard" assessing 160 indicators across four key drivers. These drivers are:

1. Supply Conditions
2. Demand Conditions
3. Institutional Environment
4. Innovation and Change

Tufts works with Mastercard, Akamai, Globalwebindex, Blue Triangle, and PCRI as data partners to collate data for its indicators. The DII is in its third iteration, and there is no indication of a new publication being worked on after the 2020 edition.

ANNEX C: METHODOLOGY

ADII 2.0 INDICATORS

The ADII encompasses a wide range of indicators that measure the effective use of digital technologies to enable everything from digital business models to digital government services.

The indicators presented in this document were selected based on six key criteria:

Relevance
Indicators in each ADII pillar must have data that is relevant to each DIF priority area respectively. Data relevance provides a clear and accurate snapshot of the topic at hand and makes it easier to develop impactful strategies based on the information they convey.

Accessibility
The indicator must have data that is easily accessible. Whether it is a detailed report, a downloadable data file, or a dedicated micro-site, the data must be freely and/or openly available. This is to ensure that any other person or organization can access the same datasets.

Coverage
The indicator must have data for at least 8 of the 10 AMS. Any bigger gap in data coverage makes it difficult for overall scores to be accurate or representative. This is to ensure that the data effectively guides the policy recommendations that will be produced.

Timeliness
The indicator must have up-to-date data, or at least be published from 2018 onwards. Most of the databases retained use data from 2018, and only some very rare cases use 2017 data. This is to ensure that the data reflects current realities as closely as possible.

Consistency
The indicator must come from databases that are published regularly. Even if the methodology evolves over the years, it is important that the data used today is still available tomorrow. This is to ensure continuity and sustainability in future iterations of the ADII.

Transparency
The indicator must be from a reputable source that shares its approach and methodology. Whether it comes from an international organization or a private entity, the detailed methodology must be published to ensure that scoring mechanisms are impartial and objective.

PILLAR INDICATORS

PILLAR 1 – DIGITALLY ENABLED TRADE & LOGISTICS				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
1.1	Degree to which trade/customs processes are supported by digital technologies	Aggregated indicator that measures the extent to which customs and logistics are fully supported by digital and automated procedures.	• OECD Trade Facilitation Indicators (combination) • Data available for all AMS	Link
1.2	Degree to which digital certificates and signatures are in place	Measures the extent to which electronic signatures and digital certificates are available and used.	• OECD Trade Facilitation Indicators • Data available for all AMS	Link
1.3	Degree to which international standards for trade documents and procedures are followed	Measures the extent to which trade documents and procedures follow international standards.	• OECD Trade Facilitation Indicators • Data available for all AMS	Link
1.4	Level of quality of trade- and transport-related infrastructure	Measures the quality of connecting infrastructure and its ability to facilitate trade across borders.	• World Bank, Logistics Performance Index (LPI) • Does not include data for Brunei Darussalam	Link
1.5	Level of competence and quality of logistics services	Measures the competence and quality of logistics services (e.g., transport operators, customs brokers, etc.).	• World Bank, Logistics Performance Index (LPI) • Does not include data for Brunei Darussalam	Link

PILLAR 2 – DATA PROTECTION & CYBERSECURITY				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
2.1	Degree to which data protection measures are in place	Aggregated indicator that assesses the existence of a personal data protection law and the security safeguards it contains to appropriately protect against loss and unauthorized access.	• The TRPC Data Protection Index 2020 (combination) • Data available for all AMS • Privately owned database, but full dataset and methodology available • Data from a first-edition report, but will be published on a regular basis	Link
2.2	Level of legislative and regulatory cybersecurity capabilities	Aggregated indicator that assesses the existence of laws on cyber-crime and of regulations dealing with cybersecurity.	• United Nations, ITU – Global Cybersecurity Index (combination) • Data available for all AMS • Some data available within the report, detailed dataset available upon request	Link
2.3	Level of institutional cybersecurity capabilities	Aggregated indicator that assesses the ability, willingness, and commitment towards a national strategy for cybersecurity, including the existence of a government agency or body devoted to driving cybersecurity at a national level.	• United Nations, ITU – Global Cybersecurity Index (combination) • Data available for all AMS • Some data available within the report, detailed dataset available upon request	Link
2.4	Level of technical cybersecurity capabilities	Aggregated indicator that assesses the existence of a CIRT/CERT/CSIRT with national responsibility, as well as the ability, willingness, and commitment to applying international cybersecurity standards.	• United Nations, ITU – Global Cybersecurity Index (combination) • Data available for all AMS • Some data available within the report, detailed dataset available upon request	Link
2.5	Level of international cooperation on cybersecurity	Aggregated indicator that assesses the ability, willingness, and commitment to cooperate with foreign entities on cybersecurity.	• United Nations, ITU- Global Cybersecurity Index (combination) • Data available for all AMS • Some data available within the report, detailed dataset available upon request	Link

PILLAR 3 – DIGITAL PAYMENTS & IDENTITIES				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
3.1	Proportion of people who use digital platforms or devices for banking purposes only	Measures the share of adults (respondents aged 15 years and older) who have used a mobile phone or the internet to make a payment, to make a purchase, or to send or receive money through their financial institution account in the past 12 months.	• World Bank, Global Financial Development (Findex) database • Does not include data for Brunei Darussalam	Link
3.2	Proportion of people who use digital platforms or devices for any type of financial transaction	Measures the share of adults (respondents aged 15 years and older) who made or received digital payments in the past 12 months (who reported using mobile money, a debit or credit card, or a mobile phone to make a payment from an account, or reported using the internet to pay bills or to buy something online).	• World Bank, Global Financial Development (Findex) database • Does not include data for Brunei Darussalam	Link
3.3	Degree to which legal frameworks enable electronic transactions	Measures the extent to which a country has laws and regulations enabling domestic and cross-border electronic transactions (i.e. e-commerce or e-transaction laws that recognize electronic data/document from trading partners).	• UN Global Survey on Digital and Sustainable Trade Facilitation • Data available for all AMS	Link
3.4	Proportion of people who have a national identity card	Measures the share of adults (respondents aged 15 years and older) who possess a national identity card (electronic or otherwise).	• World Bank, Global Financial Development (Findex) database • Does not include data for Brunei Darussalam and the Philippines	Link
3.5	Degree to which a digitized ID system is in place	Measures the existence and usage of a national digital ID system.	• World Bank, Identification for Development (ID4D) global dataset • Data from 2018	Link

PILLAR 4 – DIGITAL SKILLS & TALENT				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
4.1	Proportion of graduates in science, technology, engineering, and mathematics	Measures the share of graduates from science, technology, engineering, and mathematics programs as a percentage of all tertiary-level graduates.	• UNESCO UIS database • Data available for all AMS • Some data before 2018	Link
4.2	Proportion of employment in knowledge-intensive services	Measures the share of managers, professionals, technicians, and associate professionals (including legislators and senior officials) as a percentage of total people employed.	• WIPO, Global Innovation Index (GII) database 2023 • Some data before 2018 • Does not include data for Myanmar	Link
4.3	Level of multi-stakeholder collaboration in R&D	Measures the extent to which companies collaborate in sharing ideas and innovating, and businesses and universities collaborate on R&D.	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link
4.4	Degree to which the active population has digital skills	Measures the extent to which extent the active population possesses sufficient digital skills (e.g. computer skills, basic coding, digital reading).	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link
4.5	Degree to which graduates have business-relevant skillsets	Measures the extent to which graduating students from secondary education and university possess the skills needed by businesses.	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link

PILLAR 5 – INNOVATION & ENTREPRENEURSHIP				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
5.1	Degree to which venture capital is available	Measures how easy it is for start-up and entrepreneurs with innovative but risky projects to obtain equity funding.	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link
5.2	Proportion of GDP expenditure on R&D	Measures the total domestic intramural expenditure on R&D during a given period, as a percentage share of GDP.	• WIPO, Global Innovation Index (GII) database 2023 • Does not include data for Lao PDR	Link
5.3	Degree to which innovative companies can grow	Measures the extent to which new companies with innovative ideas are able to grow rapidly.	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link
5.4	Degree to which it is easy to start a business	Measures the procedures officially required for an entrepreneur to start up and formally operate an industrial or commercial business, as well as the time and cost to complete these procedures and the paid-in minimum capital requirement.	• World Bank, Ease of Doing Business Index 2020 • Data available for all AMS	Link
5.5	Degree to which intellectual property protection frameworks are in place and are enforced	Aggregated indicator that measures the extent to which intellectual property is effectively protected through specific laws, bodies, and regulations.	• Global Innovation Policy Center (GIPC), International IP Index (IIPI) 2020 • Does not include data for Cambodia, Lao PDR, Myanmar[418]	Link

PILLAR 6 – INSTITUTIONAL & INFRASTRUCTURAL READINESS				
	INDICATOR	**DESCRIPTION**	**NOTES**	**SOURCE**
6.1	Proportion of Active mobile-broadband subscriptions	Measures the number of data and voice mobile-broadband subscriptions and data-only mobile-broadband subscription per 100 inhabitants.	• ITU, Measuring the Information Society Report 2018 • Data available for all AMS • 2018 data (2020 data available upon request)	Link
6.2	Proportion of Internet users	Measures the share of individuals accessing the Internet in the last three months, no matter the location or the device from which it is accessed.	• ITU, Country ICT Database • Data available for all AMS	Link
6.3	Degree to which government services are available and accessible digitally	Measures the quality, usability, and accessibility of governments' online services, resources, and portals.	• UN E-Government Survey 2020 • Data available for all AMS • A composite score comprising multiple scoring criteria (individual scoring data not made available)	Link
6.4	Degree to which a government is considered responsive to disruption and change	Measures a country's business community's perception of its ability to prepare for the future thanks to forward-looking policymakers and visionary institutions.	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link
6.5	Degree to which a legal framework is considered conducive for digital innovation	Measures a country's business community's perception of how fast the legal framework can adapt to digital business models (e.g., e-commerce, sharing economy, Fintech, etc.).	• World Economic Forum (WEF), Global Competitiveness Report 2019 • Does not include data for Myanmar	Link

[418] Indicator 5.5 is the only one that uses a database for which data is only available for seven AMS.

ADII 2.0 SCORING MECHANISM

INDICATOR SCORES

Each indicator value is normalized to be scored out of a scale of 20. A specific normalization method can be used to normalize each type of data being used.

- Percentages (i.e., The share of a given population that uses the internet)

In country "X", 59.6% of the population uses the internet, while in country "Y", the number is 89.9%. The following formula allows to normalize "59.6" and "89.9" from their original scale of 100 to a new scale of 20:

COUNTRY "X": (59.6 / 100) * 20 = 11.92
COUNTRY "Y": (89.9 / 100) * 20 = 17.98

In this scenario, Country "X" scores 11.9 out of 20 for this indicator, and Country "Y" scores 17.9.

- Raw values (i.e., The volume of mobile money transactions in a given country)

Raw, numerical indicators require more complex normalization. Internet speeds or numbers of transactions, for instance, are measured in absolute values with no maximum value limitations (Country "X" may have 450,547 internet transactions, while country "Y" may have 345).

In cases where the data range is very large, absolute values can be transformed into natural logarithmic values, and then normalized to a scale comprised between 0 and 20.

The logarithmic values can then be normalized to a new scale using the following formula (where the "maximum" and "minimum" values are defined based on the dataset's natural dispersion):

$$Normalized\ value = \left(\frac{actual\ value - minimum\ value}{maximum\ value - minimum\ value}\right) \times 20$$

Using the examples presented above, the two data points after natural log transformation are 13 (country "X") and 5.8 (country "Y"). Using "15" as the "maximum" value and "0" as the "minimum" value, the formula can be applied thusly:

Country "X":

$$\left(\frac{13 - 0}{15 - 0}\right) \times 20 = 17.3$$

Country "Y"

$$\left(\frac{5.8 - 0}{15 - 0}\right) \times 20 = 7.7$$

In this scenario, Country "X" scores 17.3 out of 20 for this indicator, and Country "Y" scores 7.7.

- Survey scores (i.e., The score attributed to a given country based on the existence of a type of regulation)

Survey scores can be normalized to a scale out of 20 regardless of their original scale. The World Economic Forum (WEF), for example, scores countries on a scale from 1 (very poor performance) to 7 (very strong performance), while some of the OECD's Trade Facilitation Indicators (TFIs) are scored from 0 ("no regulation in place") to 2 ("yes, a regulation exists").

WEF normalization: If Country "X" scores 5.4 out of 7 and Country "Y" scores 3.3 out of 7, the following formula can be used to normalize the WEF score:

Country "X": (5.4 / 7) * 20 = 15.42
Country "Y": (3.3 / 7) * 20 = 9.42

In this scenario, Country "X" scores 15.4 out of 20 for the WEF indicator, and Country "Y" scores 9.4.

OECD normalization: If Country "X" scores 2 out of 2 and Country "Y" scores 1 out of 2, the following formula can be used to normalize the OECD score:

Country "X": (2 / 2) * 20 = 20
Country "Y": (1 / 2) * 20 = 10

In this scenario, Country "X" scores 20 out of 20 for the OECD indicator, and Country "Y" scores 10.

PILLAR SCORES

Pillar scores are the simple sum of indicator scores, normalized to be scored out of a scale of 100. This allows a consistent scoring mechanism across pillars regardless of their total number of pillars.

If, for example, pillar 1 has 3 indicators and pillar 4 has 7 indicators, the normalization formula will keep their total scores comparable:

Pillar 1: Non-normalized score: 34.4 out of 60
Normalized score: (34.4 / 60) * 100 = 57.3 out of 100

Indicator 1.1: 15.5 / 20
Indicator 1.2: 12.3 / 20
Indicator 1.3: 6.6 / 20

Pillar 4: Non-normalized score: 79.9 out of 140
Normalized score: (79.9 / 140) * 100 = 57.07 out of 100

Indicator 4.1: 3.4 / 20
Indicator 4.2: 15.6 / 20
Indicator 4.3: 12.7 / 20
Indicator 4.4: 18.9 / 20
Indicator 4.5: 14.9 / 20

Indicator 4.6: 9.8 / 20
Indicator 4.7: 4.6 / 20

In this scenario, pillar 1 has a score of 57.3 out of 100, and pillar 4 scores 57.0.

COUNTRY SCORES

For each AMS, the Total ADII score will be the normalized sum of ADII components (normalized scores for both indicators and pillars) on a scale of 100.

REGION SCORES

Once the Total ADII score of each country has been calculated and normalized on a scale of 100, a simple average provides the regional ASEAN score.

ADII 2.0 DATA COMPLETION EXERCISE

Similar to ADII 1.0, ADII 2.0 uses data points from third-party data sources. Where indicator data is no longer updated or available, a data completion exercise was conducted between 2022 and 2023 to provide AMS with the opportunity to complete their respective ADII datasets, i.e., fill in missing values in the indicators that were used to construct the ADII.

Data which was provided through this exercise is marked by an * to indicate that these values come from the data completion exercise.

In cases where AMS did not have suitable domestic data to be used in place of the missing data, the missing values have been marked as Not Available ("N/A"). In such cases, the final pillar scores will be normalized to a total score of "100" where the N/A score will not be used to compute the final score.

DATA ENTRIES COMPLETED BY DATA COMPLETION COMMITTEES

The following tables provide a summary of how missing and outdated data entries were completed by each AMS' Data Completion Committees (DCCs) using alternative national data sources and/or custom-made qualitative assessments. While every effort has been taken to replicate the original methodology, additional information has been provided for situations where this has not been possible.

PILLAR 1 – DIGITALLY ENABLED TRADE & LOGISTICS	
INDICATOR 1.4 – LEVEL OF QUALITY OF TRADE- AND TRANSPORT-RELATED INFRASTRUCTURE	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
INDICATOR 1.5 – LEVEL OF COMPETENCE AND QUALITY OF LOGISTICS SERVICES	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.

PILLAR 3 – DIGITAL PAYMENTS & IDENTITIES	
INDICATOR 3.1 – PROPORTION OF PEOPLE WHO USE DIGITAL PLATFORMS OR DEVICES FOR BANKING PURPOSES ONLY	
COUNTRY	**NOTES**
Brunei Darussalam	• Data updated by Brunei DCC in Aug 2023 based on the results of national surveys with similar methodologies This survey is done through digital means as compared to the original survey which is done face-to-face or as a phone-based survey. • The timeframe of this survey is different from the original survey.
INDICATOR 3.2—PROPORTION OF PEOPLE WHO USE DIGITAL PLATFORMS OR DEVICES FOR ANY TYPE OF FINANCIAL TRANSACTION	
COUNTRY	**NOTES**

Brunei Darussalam	• Data updated by Brunei DCC in Aug 2023 based on the results of national surveys with similar methodologies. • This survey is done through digital means as compared to the original survey which is done face-to-face or as a phone-based survey.

INDICATOR 3.4 – PROPORTION OF PEOPLE WHO HAVE A NATIONAL IDENTITY CARD	
COUNTRY	**NOTES**
Brunei Darussalam	• Data updated by Brunei DCC in Aug 2023 based on the results of national surveys with similar methodologies. • Data provided considers the population 12 years old and over, as opposed to 15 years old and over in the original methodology.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on national data sources.

INDICATOR 3.5 – DEGREE TO WHICH A DIGITIZED ID SYSTEM IS IN PLACE	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on the presence of existing/new regulations, or third-party reports.
Indonesia	• Data updated by Indonesia DCC in Jul-Aug 2023 based on the presence of existing/new regulations, or third-party reports.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on the presence of existing/new regulations, or third-party reports.
Malaysia	• Data updated by Malaysia DCC in Sep 2023 based on the presence of existing/new regulations, or third-party reports.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on the presence of existing/new regulations, or third-party reports.
The Philippines	• Data updated by Philippines DCC in Nov 2023 based on national data sources.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the presence of existing/new regulations, or third-party reports.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on the presence of existing/new regulations, or third-party reports.

PILLAR 4 – DIGITAL SKILLS & TALENT

INDICATOR 4.1 – PROPORTION OF GRADUATES IN SCIENCE, TECHNOLOGY, ENGINEERING, AND MATHEMATICS	
COUNTRY	**NOTES**
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on national data sources.
Indonesia	• Data updated by Indonesia DCC in Jul-Aug 2023 based on national data sources.
Lao PDR	• In the absence of newer or alternative data sources, Lao PDR DCC has opted to retain the original score from ADII 1.0.

Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on national data sources. • The large increase from the ADII 1.0 score is due to a change in the number of non-STEM students, with the number of STEM students remaining largely the same.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on national data sources.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on third-party reports. • However, this proportion refers to the percentage of students pursuing a STEM-related degree rather than the percentage of students who have graduated with a STEM-related degree.

INDICATOR 4.3 – LEVEL OF MULTI-STAKEHOLDER COLLABORATION IN R&D

COUNTRY	NOTES
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on a qualitative assessment with input from private sector and government sources.
Indonesia	• Data updated by Indonesia DCC in Jul-Aug 2023 based on the results of national surveys with similar methodologies.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment with input from private sector and government sources.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on the results of national surveys with similar methodologies.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on a qualitative assessment with input from private sector and government sources.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• In the absence of newer or alternative data sources, Thailand DCC has opted to retain the original score from ADII 1.0.
Viet Nam	• In the absence of newer or alternative data sources, Viet Nam DCC has opted to retain the original score from ADII 1.0.

INDICATOR 4.4 – DEGREE TO WHICH THE ACTIVE POPULATION HAS DIGITAL SKILLS

COUNTRY	NOTES
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• In the absence of newer or alternative data sources, Cambodia DCC has opted to retain the original score from ADII 1.0.

Indonesia	• Data updated by Indonesia DCC in Jul-Aug 2023 based on the results of national surveys with similar methodologies.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account the level of internet/mobile penetration, and the usage of digital technology in daily life.
Malaysia	• Data updated by Malaysia DCC in Sep 2023 based on the results of national surveys with similar methodologies • This survey measures access to ICT services and equipment rather than digital skills.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 by taking a survey conducted for one segment of the population as a starting point, and making a qualitative assessment on how that figure can be more representative of the active population as a whole.
The Philippines	• In the absence of newer or alternative data sources, the Philippines DCC has opted to retain the original score from ADII 1.0.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• In the absence of newer or alternative data sources, Thailand DCC has opted to retain the original score from ADII 1.0.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on a qualitative assessment taking into account local initiatives, training programs and implementation efforts targeted at digital upskilling, and based on third-party reports.

INDICATOR 4.5 – DEGREE TO WHICH GRADUATES HAVE BUSINESS-RELEVANT SKILLSETS

COUNTRY	NOTES
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on the results of national surveys with similar methodologies.
Indonesia	• Indonesia DCC has opted to leave this data entry as “Not Available (N/A)”. It will not be taken into account when calculating ADII scores and averages (a missing value does not count as “0”, it is simply not used in the calculation of the average).
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account their own experiences with hiring graduates.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 by taking a survey conducted for one segment of the population as a starting point, and making a qualitative assessment on how that figure can be more representative of the active population as a whole.
The Philippines	• In the absence of newer or alternative data sources, The Philippines DCC has opted to retain the original score from ADII 1.0.

Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the results of national surveys with similar methodologies.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on a qualitative assessment and based on third-party reports.

PILLAR 5 – INNOVATION & ENTREPRENEURSHIP	
INDICATOR 5.1 – DEGREE TO WHICH VENTURE CAPITAL IS AVAILABLE	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on a qualitative assessment taking into account the level of development of the venture capital and private equity ecosystems or associations.
Indonesia	• In the absence of newer or alternative data sources, Indonesia DCC has opted to retain the original score from ADII 1.0.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account the level of development of the venture capital and private equity ecosystems or associations.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on a qualitative assessment taking into account the level of development of the venture capital and private equity ecosystems or associations, and based on third-party reports.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on a qualitative assessment with input from private sector and government sources, and presence of existing regulations.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the results of national surveys with similar methodologies.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on a qualitative assessment taking into account initiatives to support innovation and start-ups, and based on third-party reports.

INDICATOR 5.2 – GROSS EXPENDITURE ON R&D, PROPORTION OF GDP	
COUNTRY	**NOTES**
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 (based on the target set for gross expenditure on R&D).

INDICATOR 5.3 – DEGREE TO WHICH INNOVATIVE COMPANIES CAN GROW	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on a qualitative assessment taking into account the number of successful start-ups in their countries, and considerations such as the presence of industry- and government-led initiatives to support such companies.
Indonesia	• Data updated by Indonesia DCC in Jul-Aug 2023 based on the results of national surveys with similar methodologies.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account the number of successful start-ups in their countries, and considerations such as the presence of industry- and government-led initiatives to support such companies.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on a qualitative assessment taking into account the number of successful start-ups in their countries, and considerations such as the presence of industry- and government-led initiatives to support such companies, and based on third-party reports.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on a qualitative assessment taking into account existing policy, institutional, and operational support.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the results of national surveys with similar methodologies.
Viet Nam	• In the absence of newer or alternative data sources, Viet Nam DCC has opted to retain the original score from ADII 1.0.

INDICATOR 5.4 – DEGREE TO WHICH IT IS EASY TO START A BUSINESS	
COUNTRY	**NOTES**
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• In the absence of newer or alternative data sources, Cambodia DCC has opted to retain the original score from ADII 1.0.

Indonesia	• In the absence of newer or alternative data sources, Indonesia DCC has opted to retain the original score from ADII 1.0.
Lao PDR	• In the absence of newer or alternative data sources, Lao PDR DCC has opted to retain the original score from ADII 1.0.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• In the absence of newer or alternative data sources, Myanmar DCC has opted to retain the original score from ADII 1.0.
The Philippines	• Data updated by Philippines DCC in Nov 2023 based on a qualitative assessment taking into account implementation of laws, policies, and provision of institutional support.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• In the absence of newer or alternative data sources, Thailand DCC has opted to retain the original score from ADII 1.0.
Viet Nam	• In the absence of newer or alternative data sources, Viet Nam DCC has opted to retain the original score from ADII 1.0.

INDICATOR 5.5 – DEGREE TO WHICH INTELLECTUAL PROPERTY PROTECTION FRAMEWORKS ARE IN PLACE AND ARE ENFORCED

COUNTRY	NOTES
Cambodia	• In the absence of newer or alternative data sources, Cambodia DCC has opted to retain the original score from ADII 1.0.
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account national data sources and legal frameworks.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on a qualitative assessment taking into account national data sources and legal frameworks.

PILLAR 6 – INSTITUTIONAL & INFRASTRUCTURAL READINESS

INDICATOR 6.4 – DEGREE TO WHICH A GOVERNMENT IS CONSIDERED RESPONSIVE TO DISRUPTION AND CHANGE

COUNTRY	NOTES
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on a qualitative assessment taking into account the government's response to COVID-19, the number of currently established plans, and how often the government reviews them.

Indonesia	• Indonesia DCC has opted to leave this data entry as "Not Available (N/A)". It will not be taken into account when calculating ADII scores and averages (a missing value does not count as "0", it is simply not used in the calculation of the average).
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account the number of currently established plans and how often the government reviews them.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.
Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on the results of national surveys with similar methodologies.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on a qualitative assessment with input from private sector and government sources, and presence of existing regulations.
Singapore	• Data updated by Singapore DCC in Oct 2023 based on the results of national surveys with similar methodologies.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the results of national surveys with similar methodologies.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on a qualitative assessment taking into account the government's response to COVID-19, the implementation of laws and regulations relating to the digital economy, and based on third-party reports.

INDICATOR 6.5 – DEGREE TO WHICH A LEGAL FRAMEWORK IS CONSIDERED CONDUCIVE FOR DIGITAL INNOVATION

COUNTRY	NOTES
Brunei Darussalam	• In the absence of newer or alternative data sources, Brunei Darussalam DCC has opted to retain the original score from ADII 1.0.
Cambodia	• Data updated by Cambodia DCC in Sep 2023 based on a qualitative assessment taking into account the presence of existing regulations and steps taken to support implementation.
Indonesia	• Indonesia DCC has opted to leave this data entry as "Not Available (N/A)". It will not be taken into account when calculating ADII scores and averages (a missing value does not count as "0", it is simply not used in the calculation of the average).
Lao PDR	• Data updated by Lao PDR DCC in Jul-Aug 2023 based on a qualitative assessment taking into account the presence of existing regulations and steps taken to support implementation.
Malaysia	• In the absence of newer or alternative data sources, Malaysia DCC has opted to retain the original score from ADII 1.0.

Myanmar	• Data updated by Myanmar DCC in Aug 2023 based on a qualitative assessment taking into account the presence of existing regulations and steps taken to support implementation.
The Philippines	• Data updated by the Philippines DCC in Nov 2023 based on a qualitative assessment with input from private sector and government sources, and presence of existing regulations.
Singapore	• In the absence of newer or alternative data sources, Singapore DCC has opted to retain the original score from ADII 1.0.
Thailand	• Data updated by Thailand DCC in Jul 2023 based on the results of national surveys with similar methodologies.
Viet Nam	• Data updated by Viet Nam DCC in Oct 2023 based on a qualitative assessment taking into account national plans, and the outcomes of workshops with businesses.

SUMMARY OF DATA ENTRIES COMPLETED BY DATA COMPLETION COMMITTEES

Out of the ADII's 30 indicators, nine were no longer collected/published (3.5, 4.3, 4.4, 4.5, 5.1, 5.3, 5.4, 6.4, and 6.5) and eight were updated only for some AMS (1.4, 1.5, 3.1, 3.2, 3.4, 4.1, 5.2, and 5.5).

Table 7: Overview of Missing and Outdated Data Entries Completed by DCCs

	PILLAR 1 DIGITALLY ENABLED TRADE & LOGISTICS					PILLAR 2 DATA PROTECTION & CYBERSECURITY					PILLAR 3 DIGITAL PAYMENTS & IDENTITIES				
	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR
Brunei Darussalam	Updated	Updated	Updated	Outdated	Outdated	Updated	Updated	Updated	Updated	Updated	Missing	Missing	Updated	Outdated	Missing
Cambodia	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Indonesia	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Lao PDR	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Malaysia	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Myanmar	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
The Philippines	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing	Missing
Singapore	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Thailand	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing
Viet Nam	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Updated	Missing

	PILLAR 4 DIGITAL SKILLS & TALENT					PILLAR 5 INNOVATION & ENTREPRENEURSHIP					PILLAR INSTITUTIONAL & INFRASTRUCTURAL READINESS				
	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR	INDICATOR
Brunei Darussalam	Updated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Cambodia	Outdated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Missing	Updated	Updated	Updated	Missing	Missing
Indonesia	Outdated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Lao PDR	Outdated	Updated	Missing	Missing	Missing	Missing	Missing	Missing	Missing	Missing	Updated	Updated	Updated	Missing	Missing
Malaysia	Updated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Myanmar	Outdated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Missing	Updated	Updated	Updated	Missing	Missing
The Philippines	Updated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Singapore	Updated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Thailand	Outdated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing
Viet Nam	Outdated	Updated	Missing	Missing	Missing	Missing	Updated	Missing	Missing	Updated	Updated	Updated	Updated	Missing	Missing

Legend: *Updated* *Missing (not updated or no longer published, completed by DCCs)* *Available but outdated (completed by DCCs)*